The Yajie Protocol: Technology Lock-in, Audit Sovereignty, and the Non-Proliferation Logic of Data Quality Assessment

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摘要

As artificial intelligence systems become pervasive in high-stakes decision-making, the quality of training and inference data has emerged as a critical—yet systematically under-governed—input. This paper introduces the Yajie Protocol, a multi-expert consistency-based data auditing algorithm, and analyzes its structural implications for the emerging market in data quality assessment. We demonstrate that Yajie generates a distinctive form of technology lock-in driven not by network effects in the conventional sense, nor by proprietary control of intellectual property, but by a time-accumulated advantage mechanism: each audit cycle enriches a shared evidentiary corpus, widening the quality gap between the incumbent protocol and any potential entrant. Drawing on game-theoretic modeling, we formalize this dynamic as a *Non-Proliferation Equilibrium* (NPE), wherein rational actors forgo the development of competing audit standards despite the absence of formal barriers to entry. We establish a structural analogy between this equilibrium and the strategic logic underpinning the Nuclear Non-Proliferation Treaty (NPT), arguing that audit standards fragmentation produces negative externalities analogous to nuclear proliferation. The paper further examines the geopolitical dimensions of audit infrastructure through the lens of the SWIFT financial messaging system and the International Accounting Standards Board (IASB), with particular attention to US-China technological decoupling. We model the lock-in trajectory through a five-stage progression, then extend the analysis to a sixth stage—termed *Universal Harmony* (天下大同)—wherein the Cumulative Evidentiary Corpus transitions from a proprietary asset into a global public infrastructure and audit standards cease to belong to any single nation. We derive policy recommendations for embedding audit neutrality within multilateral governance frameworks. Our findings carry implications for the design of international AI governance institutions, the regulation of algorithmic auditing markets, and the preservation of epistemic sovereignty in an era of data-driven geopolitical competition.

Keywords: data quality assessment, algorithmic audit, technology lock-in, nonproliferation, game theory, audit sovereignty, AI governance, multi-expert consistency, SCX uncertainty principle, mutual deterrence, adoption-audit equilibrium

1 Introduction

The governance of artificial intelligence has entered a phase of institutional crystallization. Where the preceding decade was characterized by proliferation of ethical principles and voluntary guidelines (Floridi et al., 2018; Jobin et al., 2019), the current moment is defined by the translation of normative commitments into operational mechanisms: conformity assessments, third-party audits, and certification regimes (Raji et al., 2020; Mökander et al., 2021). Central to this operational turn is the question of *data quality*—the accuracy, completeness, consistency, and fitness-for-purpose of the datasets that underpin machine learning pipelines.

The dependency of AI system performance on data quality is by now well-established in both the technical literature (Sambasivan et al., 2021; Northcutt et al., 2021) and policy discourse (European Commission, 2021; NIST, 2023). Yet the institutional infrastructure for assessing and certifying data quality remains nascent. A small number of commercial auditing services have emerged, alongside standards development efforts within ISO/IEC JTC 1/SC 42 and the IEEE P2863 working group. What has not yet materialized—and what this paper argues may *never* materialize in a competitive form—is a pluralistic market of interoperable data quality assessment protocols.

The central claim of this paper is that data quality auditing exhibits structural properties that drive the market toward a natural monopoly, and that this monopolistic tendency is, counterintuitively, *welfare-enhancing* under a specific set of conditions. We develop this argument through a detailed analysis of the Yajie Protocol, a data auditing algorithm that operationalizes multi-expert consistency as a quality signal, and we formalize the resulting market dynamics through the concept of a *Non-Proliferation Equilibrium* (NPE).

The Yajie Protocol, which we describe in Section 3, operates by deploying an ensemble of independent audit modules against a target dataset and measuring the degree of inter-auditor agreement. Disagreement signals potential quality defects; convergence across auditors with orthogonal detection strategies provides evidence of quality. What makes the protocol structurally distinctive is its *cumulative evidentiary architecture*: every audit cycle enriches a shared repository of detected anomalies, auditor calibration parameters, and dataset-specific quality profiles, such that the protocol becomes monotonically more accurate over time. New entrants cannot replicate this accumulated capital; they must either license access to the incumbent's evidentiary corpus (entrenching its position) or enter the market with a structurally inferior product.

This dynamic is not adequately captured by standard models of technology lock-in. The QWERTY literature (David, 1985; Arthur, 1989) emphasizes network effects, switching costs, and installed-base advantages. Platform economics (Rochet & Tirole, 2003; Parker & Van Alstyne, 2005) highlights cross-side network externalities. Neither framework fully accounts for a product whose quality is an increasing function of *cumulative usage time*, independent of the number of users. The Yajie Protocol's lock-in mechanism is temporal, not networked: it accrues advantage through the sheer volume of audits conducted, each of which makes the protocol better at conducting the next audit. This *quality ratchet* creates an entry barrier that grows with time, asymptotically approaching an insurmountable moat.

We deploy a game-theoretic framework to demonstrate that, under plausible parameterizations, rational competitors will choose not to invest in the development of competing audit protocols, even when the incumbent has not engaged in predatory pricing or exclusionary practices. The resulting market structure—a single dominant protocol, universally adopted—resembles not a coercive monopoly of the Standard Oil variety, but rather the equilibrium that obtains in domains where fragmentation itself is the primary source of social cost. We further demonstrate that the protocol's architecture generates a distinctive *Adoption-Audit Equilibrium*: organizations are simultaneously incentivized to adopt Yajie (because it improves data quality) and deterred from falsifying data (because adoption renders audit outputs unconditionally verifiable by third parties, per the SCX Uncertainty Principle). The architecture transforms data quality from an unverifiable assertion into a revealed attribute—and honesty into the dominant strategy.

This is where the nuclear non-proliferation analogy becomes analytically productive. The Nuclear Non-Proliferation Treaty (NPT) succeeded not because it prohibited nuclear weapons development (it did not, for the nuclear-weapon states) but because it established a club whose benefits—access to peaceful nuclear technology, participation in the global non-proliferation regime—outweighed the strategic value of an independent nuclear program for most states (Sagan, 1996). Critically, the NPT's stability does not depend on enforcement alone; it depends on a rational calculation by non-nuclear states that proliferation would leave them *worse off* than the status quo, either because their nascent capability would be neutralized by preemptive action, because the resultant arms race would consume resources better spent elsewhere, or because proliferation by one state would trigger proliferation by its rivals, leaving everyone less secure (Waltz, 1981; Mearsheimer, 1993).

The audit protocol market exhibits an analogous structure. A world with N audit protocols, each partially trusted and partially contested, is a world in which audit results are themselves subject to meta-audit disputes, eroding the very certainty that audits are designed to provide. The development of a competing protocol by one jurisdiction would trigger reciprocal development by rivals, fragmenting the global audit infrastructure along geopolitical lines and imposing coordination costs on every entity that operates across jurisdictions. In this environment, the *rational* choice for most actors is to accept the dominance of a single protocol—provided that protocol is governed under adequate neutrality safeguards—rather than to pursue audit autarky.

The paper proceeds as follows. Section 2 provides a technical overview of the Yajie Protocol and situates it within the literature on algorithmic auditing and technology lock-in. Section 3 develops the Non-Proliferation Equilibrium model, including the formal payoff structure, Nash equilibrium characterization, NPT analogy, and—critically—a corollary establishing that first-mover advantage decays over time because the Cumulative Evidentiary Corpus is an additive public good. Section 4 examines the geopolitical dimensions of audit infrastructure through case studies of SWIFT and the IASB, with an extension to the US-China technology competition. Section 5 models the lock-in trajectory through five stages, extends the Markov formalization to a sixth stage termed *Universal Harmony* (天下大同) wherein the CEC becomes global public infrastructure and audit standards cease to belong to any nation, and analyzes the business model dynamics. Section 6 derives policy recommendations for embedding audit neutrality within multilateral governance. Section 7 addresses limitations and identifies directions for future research. Section 8 concludes. Section 9 synthesizes the foregoing analyses into a unified game-theoretic architecture.

2 Theoretical Background

2.1 The Yajie Protocol: A Technical Overview

The Yajie Protocol is a computational framework for assessing the quality of structured and semi-structured datasets through multi-expert consistency analysis. At its core, Yajie operationalizes a simple but powerful principle: when multiple independent audit procedures, each designed to detect distinct categories of data defect, converge on a quality assessment, the probability that the assessment is correct increases multiplicatively. Conversely, when auditors diverge, the pattern of divergence carries diagnostic information about the nature and location of quality failures.

Formally, let $\mathcal{D}$ be a target dataset consisting of n records, each characterized by a feature vector $\mathbf{x}_i \in \mathbb{R}^d$ and, where applicable, a label y_i. Let $\mathcal{A} = \{A_1, A_2, \ldots, A_k\}$ be an ensemble of k audit modules, where each $A_j : \mathcal{D} \to \mathcal{R}_j$ maps the dataset to a structured audit report $\mathcal{R}_j = (\mathbf{q}_j, \mathbf{a}_j, \mathbf{c}_j)$ comprising:

- A quality score vector $\mathbf{q}_j \in [0, 1]^m$ across m quality dimensions (e.g., completeness, accuracy, consistency, timeliness, uniqueness);

- An anomaly register $\mathbf{a}_j = \{(i, \tau, s)\}$ recording the index i of each flagged record, the anomaly type τ, and a severity score $s \in [0, 1]$;

- A confidence vector $\mathbf{c}_j \in [0, 1]^m$ expressing the auditor's self-assessed reliability on each dimension.

The protocol's aggregation function $\Phi : \prod_{j=1}^{k} \mathcal{R}_j \to \mathcal{R}^*$ synthesizes the ensemble outputs into a unified audit report $\mathcal{R}^*$ through a weighted consensus mechanism:

$$\Phi(\mathcal{R}_1, \ldots, \mathcal{R}_k) = \left(\sum_{j=1}^{k} w_j \mathbf{q}_j,\ \bigcup_{j=1}^{k} \mathbf{a}_j \setminus \mathcal{F},\ \prod_{j=1}^{k} \operatorname{diag}(\mathbf{c}_j) \right) \tag{1}$$

where w_j are dynamic weights derived from each auditor's historical calibration performance, $\mathcal{F}$ is a false-positive filter trained on previously adjudicated anomalies, and the confidence aggregation employs element-wise multiplication to reflect the multiplicative reduction in uncertainty when independent auditors concur.

The protocol's defining architectural feature is the *Cumulative Evidentiary Corpus* (CEC), denoted $\mathcal{E}_t$, which aggregates all audit outputs up to time t:

$$\mathcal{E}_t = \bigcup_{\tau=0}^{t} \{\mathcal{R}^*_\tau, \mathcal{J}_\tau, \Delta\mathcal{P}_\tau\} \tag{2}$$

where $\mathcal{J}_\tau$ is the set of human-adjudicated judgments from audit cycle τ, and $\Delta\mathcal{P}_\tau$ represents parameter updates derived from those judgments. The CEC serves three functions: (i) it provides training data for the false-positive filter $\mathcal{F}$; (ii) it enables dynamic recalibration of auditor weights w_j; and (iii) it accumulates a growing library of dataset-specific quality fingerprints that accelerate future audits of similar or related datasets.

The CEC grows with each audit cycle. Let $g(t)$ denote the cumulative number of audits conducted by time t. Each audit contributes, on average, $\bar{\alpha}$ new anomaly instances and $\bar{\beta}$ new calibration data points to the CEC. The protocol's accuracy $\theta(t)$—measured as the F1 score of its anomaly detection relative to ground truth—improves as a function of CEC size:

$$\theta(t) = \theta_{\max} - (\theta_{\max} - \theta_0) e^{-\gamma \cdot |\mathcal{E}_t|} \tag{3}$$

where $\theta_{\max}$ is the asymptotic accuracy ceiling, θ_0 is the initial accuracy, and γ is a learning rate parameter. This functional form captures diminishing returns to additional data, consistent with the empirical scaling literature (Kaplan et al., 2020), while ensuring that the incumbent's accuracy advantage over a zero-CEC entrant grows monotonically with cumulative audit volume.

2.2 Technology Lock-in: Classical Mechanisms and Their Limits

The phenomenon of technology lock-in—whereby an inferior technology persists because the costs of coordinated transition exceed the benefits of switching to a superior alternative—has been a central concern of innovation economics since David's (1985) iconic analysis of the QWERTY keyboard. Arthur (1989) formalized the mechanisms through which increasing returns to adoption generate path-dependent outcomes: large fixed costs, learning effects, coordination effects, and adaptive expectations.

Subsequent literature refined and extended these mechanisms. Katz and Shapiro (1985, 1994) developed the economics of network effects, distinguishing between direct network effects (the value of a network increases with the number of users, as in telecommunications) and indirect network effects (a larger installed base attracts complementary goods, as in software platforms). Farrell and Saloner (1985) modeled the dynamics of standardization, demonstrating that excess inertia—the failure to adopt a superior standard because of coordination failure—can arise even when all actors prefer the superior standard.

The two-sided market literature (Rochet & Tirole, 2003; Armstrong, 2006) extended these insights to platforms that intermediate between distinct user groups, showing that cross-side network externalities can generate powerful lock-in effects even in the absence of traditional switching costs. Once a platform achieves critical mass on both sides of the market, entrants face a chicken-and-egg problem: they cannot attract users from one side without first assembling a critical mass on the other.

These classical mechanisms do not, however, capture the lock-in dynamic generated by the Yajie Protocol. Consider the disanalogies:

1. **Network effects**: Yajie's value does not depend on the number of other users of the protocol. A single organization conducting repeated audits accrues the same CEC-driven accuracy gains regardless of whether other organizations also use the protocol. The lock-in is *temporal*, not *networked*.

2. **Switching costs**: While Yajie does generate switching costs in the conventional sense (retraining staff, reconfiguring pipelines, renegotiating contracts), the principal entry barrier is not the cost of switching *away* from Yajie but the cost of replicating the *accumulated audit history* that makes Yajie accurate. A competitor cannot buy, license, or reverse-engineer the CEC, because the CEC is not merely a database but a cumulative record of audit outcomes, adjudications, and parameter updates that are causally entangled with the protocol's specific architecture.

3. **Installed base**: Unlike a platform whose value derives from its user base, Yajie's value derives from its *audit base*—the corpus of audits it has conducted. This is a

stock variable, not a flow variable, and it cannot be replicated by attracting users away from the incumbent.

4. **Complementary goods**: The Yajie ecosystem does generate complementary assets—training programs, integration tooling, certification bodies—but these are derivative of the CEC advantage, not independent sources of lock-in.

The closest analogue in the existing literature may be the concept of *data network effects* (Gregory et al., 2021), wherein a product improves as it collects more data from users. However, the data network effect literature typically emphasizes the role of user-contributed data in improving a single firm's product (as in search engines, recommendation systems, or autonomous vehicles). The Yajie dynamic is distinct in three respects: (i) the improvement mechanism is audit-specific, not user-behavior-specific; (ii) the data in question are not user-contributed but *audit-generated*, arising from the protocol's own operation; and (iii) the CEC's evidentiary character means that its value is not merely statistical (more data yields better predictions) but *epistemic* (more audits yield a more trustworthy corpus of quality evidence).

2.3 Algorithmic Auditing and Data Quality: State of the Field

The algorithmic auditing literature has expanded rapidly since the mid-2010s, driven by concerns about fairness, accountability, and transparency in automated decision systems (Sandvig et al., 2014; Diakopoulos, 2015; Ada Lovelace Institute, 2020). Much of this literature has focused on *model auditing*—the ex post examination of trained models for bias, discrimination, or other harms (Raji et al., 2020; Wilson et al., 2021)—rather than on *data auditing*, the ex ante assessment of training data quality.

The relative neglect of data auditing is partly explained by technical challenges: data quality is inherently multidimensional (Batini et al., 2009), context-dependent (Wang & Strong, 1996), and often unverifiable without access to the data-generation process. Recent work has begun to address these challenges. Northcutt et al. (2021) proposed Confident Learning for identifying label errors in training data. Sambasivan et al. (2021) documented the prevalence of data cascades—compounding errors originating in data quality failures—in real-world ML deployments. The "Data Cards" framework (Pushkarna et al., 2022) and "Datasheets for Datasets" (Gebru et al., 2021) have advanced structured transparency for dataset documentation.

What remains absent is a *standardized, cumulative, and cross-domain* methodology for data quality assessment. Existing approaches are typically bespoke to specific dataset types, model architectures, or application domains. The Yajie Protocol addresses this gap through its multi-expert ensemble architecture, which accommodates heterogeneous audit modules within a unified aggregation framework. However, as we argue in the following sections, this very generality—combined with the CEC's cumulative character—generates the lock-in dynamics that are the central concern of this paper.

3 The Non-Proliferation Equilibrium

3.1 Model Setup

We consider a world with N jurisdictions (which may be nations, regulatory blocs, or large organizations capable of developing audit infrastructure). Each jurisdiction $i \in \{1, \ldots, N\}$ faces a binary choice: *Develop* (D) an independent data audit protocol, or *Adopt* (A) the incumbent protocol (Yajie). The choice is made simultaneously by all jurisdictions; we examine the Nash equilibria of this one-shot game.

Let the incumbent protocol be indexed as $j = 1$, with an accumulated CEC of size $|\mathcal{E}|$. An entrant protocol developed by jurisdiction i would begin with an empty CEC, implying an accuracy penalty $\delta(|\mathcal{E}|) = \theta(|\mathcal{E}|) - \theta(0)$, where $\theta(\cdot)$ follows the functional form specified in Section 2.1. For any non-trivial $|\mathcal{E}|$, $\delta(|\mathcal{E}|) > 0$, and $\delta'(|\mathcal{E}|) > 0$ (the accuracy gap widens with cumulative audits).

The payoff to jurisdiction i from adopting the incumbent is:

$$\pi_i(A, \mathbf{a}_{-i}) = V[\theta(|\mathcal{E}|)] - c_i^{\text{adopt}} - \lambda \cdot \mathbb{I}[\text{fragmentation}(\mathbf{a})] \tag{4}$$

where $V[\cdot]$ is the value derived from audit accuracy, c_i^{adopt} is the cost of adopting the incumbent protocol (licensing, integration, training), λ is a fragmentation cost parameter, and $\mathbb{I}[\text{fragmentation}(\mathbf{a})]$ is an indicator that equals 1 if the global audit infrastructure is fragmented (i.e., if at least one jurisdiction has developed an independent protocol) and 0 otherwise.

The payoff from developing an independent protocol is:

$$\pi_i(D, \mathbf{a}_{-i}) = V[\theta(0)] - c_i^{\text{develop}} - \kappa \cdot \mathbf{1}_{-i}^{D} - \lambda \cdot \mathbb{I}[\text{fragmentation}(\mathbf{a})] \tag{5}$$

where $c_i^{\text{develop}} \gg c_i^{\text{adopt}}$ is the fixed cost of protocol development, and $\kappa \cdot \mathbf{1}_{-i}^{D}$ captures the strategic cost imposed by other jurisdictions' development decisions: each additional developer adds κ to i's costs, reflecting the erosion of mutual recognition, the proliferation of conflicting audit standards, and the increased complexity of cross-jurisdictional data flows.

The fragmentation indicator is:

$$\mathbb{I}[\text{fragmentation}(\mathbf{a})] = \begin{cases} 0 & \text{if } \sum_{j=1}^{N} \mathbb{I}[a_j = D] = 0 \\ 1 & \text{if } \sum_{j=1}^{N} \mathbb{I}[a_j = D] \geq 1 \end{cases} \tag{6}$$

Critically, the first developer triggers fragmentation for *all* jurisdictions, including itself. This reflects the public-bads character of audit protocol proliferation: once multiple protocols exist, every cross-border data transaction must navigate conflicting quality certifications, and the informational value of any single audit is diminished by the existence of competing—and potentially contradictory—assessments.

3.2 严格博弈形式化与假设

在展开均衡分析之前，我们将上述情景严格形式化为标准策略式博弈，并明确标注全部假设。

Definition 3.1 (非扩散博弈 Γ^{NP})**.** 非扩散博弈 $\Gamma^{\mathrm{NP}} = \langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$ 定义为：

- 参与者集合 $N = \{1, 2, \ldots, n\}$，$n \geq 2$，每个辖区为一个参与者；
- 对每个 $i \in N$，纯策略空间 $S_i = \{A, D\}$，其中 A 表示采纳（Adopt）现有协议，D 表示开发（Develop）独立协议；
- 支付函数族 $u_i : S_1 \times \cdots \times S_n \to \mathbb{R}$，定义如方程 (4)–(5) 所示。

记策略组合为 $s = (s_i, s_{-i}) \in S \triangleq \prod_{j=1}^{n} S_j$，其中 s_{-i} 表示除 i 外所有辖区的策略。

Definition 3.2 (非扩散博弈的全部假设)**.** 本节全部定理和命题建立于以下假设之上。每个假设均标注其经济学含义及可放松性。

A1 值函数单调凹性： $V : [0, 1] \to \mathbb{R}^+$ 是严格递增、二阶连续可微的凹函数，满足 $V(0) = 0$，$V'(\theta) > 0$，$V''(\theta) \leq 0$ 对所有 $\theta \in (0, 1)$ 成立。经济学含义：审计精度的边际价值非增。

A2 审计精度的 CEC 函数形式： 审计精度随 CEC 规模演化的函数为：

$$\theta(|\mathcal{E}|) = \theta_{\max} - (\theta_{\max} - \theta_0)e^{-\gamma|\mathcal{E}|} \tag{7}$$

其中 $\theta_{\max} \in (0, 1]$ 为渐近精度上界，$\theta_0 \in (0, \theta_{\max})$ 为初始精度（空 CEC），$\gamma > 0$ 为学习率参数。经济学含义：精度随审计量单调增长，具有边际递减。注意：具体指数函数形式可放松为任意满足 $\theta' > 0, \theta'' < 0$ 的函数，定性结论不变。

A3 成本结构： $c^{\mathrm{develop}} > c^{\mathrm{adopt}} > 0$。开发独立协议的总固定成本严格大于采纳现有协议的许可和集成成本。经济学含义：开发投入是不可逆的沉没成本。同质性假设可放松为辖区异质性成本，不影响均衡存在性但影响精确均衡配置。

A4 扩散外部性的参数排序： $\kappa > 0$ 为每个其他辖区的开发行为给辖区 i 施加的边际扩散成本；$\lambda > 0$ 为碎片化指标激活时给所有辖区（包括首个开发者自身）施加的固定碎片化成本。满足 $\lambda > \kappa$。经济学含义：碎片化的公共劣品效应（互认失效、标准冲突、跨境交易成本上升）超过单一扩散行为给其他辖区施加的增量成本。此排序是全体采纳均衡稳定性的关键条件。

A5 完全信息：博弈的全部参数——$\{V(\cdot), \theta(\cdot), \theta_0, \theta_{\max}, \gamma, c^{\text{adopt}}, c^{\text{develop}}, \kappa, \lambda, |\mathcal{E}|\}$——构成所有参与者的共同知识。 可放松性：这是本文最强的假设。第 7 节讨论了不完全信息扩展（辖区拥有关于自身开发成本的私人信息）。引入不完全信息将产生贝叶斯-纳什均衡的精炼问题。

A6 同时行动：所有辖区同时且独立地做出采纳/开发决策。 可放松性：可扩展为序贯行动博弈，先发辖区可能通过率先开发来影响后发辖区的决策。序贯博弈中可能出现策略性先发制人均衡，其福利性质可能与同时行动均衡不同。

A7 支付的可加可分性与碎片化的公共性：支付函数具有方程 (4)–(5) 所规定的可加结构。碎片化指标 $\mathbb{I}[n_D > 0]$ 是公共信息——一旦任何辖区开发，碎片化成本立即施加于全体辖区。含义：碎片化是一种”公共劣品”（*public bad*），其供给由首个开发者的行为触发，消费是非排他和非竞争的。

Remark 3.1 (假设之间的逻辑依赖). 假设 A2 和 A1 共同保证采纳优势 $\Delta(|\mathcal{E}|)$ 随 CEC 规模严格单调递增。假设 A4 的 $\lambda > \kappa$ 排序保证全体采纳均衡的稳定性裕度为正且随 $|\mathcal{E}|$ 递增。若 $\lambda \leq \kappa$，则碎片化成本不足以威慑单一扩散，非扩散均衡的稳定性将显著削弱——此时需要 n 较小或额外的制度性约束（如条约义务）来维持均衡。

3.3 纳什均衡的严格存在性、唯一性与特征

本小节以定理形式给出非扩散博弈的全部均衡特征。所有证明在标注的假设下是严格的。

记策略组合中开发者的数量为 $n_D(s) = |\{i \in N : s_i = D\}|$。定义辖区 i 的**单边采纳优势**——在其他辖区全部采纳时，i 从采纳相对于开发的净收益：

$$\Delta(|\mathcal{E}|) \triangleq u_i(A, \mathbf{A}_{-i}) - u_i(D, \mathbf{A}_{-i}) = V[\theta(|\mathcal{E}|)] - V[\theta(0)] - (c^{\text{adopt}} - c^{\text{develop}}) + \kappa \tag{8}$$

其中 $\mathbf{A}_{-i}$ 表示除 i 外所有辖区均选择 A。注意 $\Delta(|\mathcal{E}|)$ 仅依赖于 $|\mathcal{E}|$，与辖区身份 i 无关（策略空间的对称性）。

Theorem 3.1 (非扩散博弈的纳什均衡完整刻画 (雅洁不扩散定理)). 在假设 *A1–A7* 下，考虑非扩散博弈 Γ^{NP}，设 $N = \{1, \ldots, n\}$ 且 $n \geq 2$。以下命题成立。

(i) **均衡存在性**：纯策略纳什均衡始终存在。若 $\Delta(|\mathcal{E}|) \geq \lambda - \kappa$，则 $(A, \ldots, A)$ 是纯策略纳什均衡；若 $\Delta(|\mathcal{E}|) \leq -(n-1)\kappa$，则 $(D, \ldots, D)$ 是纯策略纳什均衡；在中间区域，存在对称混合策略纳什均衡。

(ii) **全体采纳均衡**：策略组合 $s^* = (A, A, \ldots, A)$ 是纳什均衡，当且仅当：

$$\boxed{\Delta(|\mathcal{E}|) \geq \lambda - \kappa} \tag{9}$$

该均衡是**严格的**（*strict*）当严格不等式成立。严格纳什均衡排除了任何参与者对无差异偏离的依赖。

(iii) **全体开发均衡**：策略组合 $s^D = (D, D, \ldots, D)$ 是纳什均衡，当且仅当：

$$\boxed{\Delta(|\mathcal{E}|) \leq -(n-1)\kappa} \tag{10}$$

(iv) **非对称均衡的不可能性**：在任何纯策略纳什均衡中，所有辖区必须选择相同策略。不存在”部分采纳、部分开发”的非对称纯策略均衡。

(v) **对称混合策略均衡**：当条件 (9) 和 (10) 均不满足时——即 $-(n-1)\kappa < \Delta(|\mathcal{E}|) < \lambda - \kappa$——存在唯一的对称混合策略纳什均衡。在该均衡中，每个辖区以相同概率 $p^* \in (0,1)$ 选择 A，p^* 由以下无差异条件唯一确定：

$$\begin{aligned} V[\theta(|\mathcal{E}|)] - c^{adopt} - \lambda\left[1 - (p^*)^{n-1}\right] \\ = V[\theta(0)] - c^{develop} - n\kappa - \lambda \end{aligned} \tag{11}$$

证明. 在假设 A1–A7 下逐项证明。

(i) 存在性。由假设 A1–A7，博弈 Γ^{NP} 是有限博弈（$|S| = 2^n < \infty$）。由纳什 (1951) 定理，任何有限博弈存在混合策略纳什均衡。纯策略均衡的存在性由以下构造性论证保证：支付矩阵的全部 2^n 个纯策略组合构成有限集；每个辖区的支付在他人策略固定时是其自身策略的仿射函数（方程 (4)–(5)），因此最优反应对应是上半连续的且具有凸值；或者更直接地，如下文将展示的，均衡条件 (9) 和 (10) 覆盖了全部参数区域，两者至少其一在 $\Delta(|\mathcal{E}|)$ 充分大或充分小时成立，且当两者均不成立时对称混合策略均衡由不动点论证保证存在。

(ii) 全体采纳均衡。考虑策略组合 $s^* = (A, \ldots, A)$。在 s^* 下，$n_D(s^*) = 0$，碎片化指标 $\mathbb{I}[n_D > 0] = 0$。对任意辖区 $i \in N$：

$$u_i(A, \mathbf{A}_{-i}) = V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} \tag{12}$$

$$u_i(D, \mathbf{A}_{-i}) = V[\theta(0)] - c^{\text{develop}} - \kappa - \lambda \tag{13}$$

方程 (13) 的推导：当 i 单边偏离至 D，$n_D = 1 > 0$，碎片化被触发（$-\lambda$），且 i 自身的开发行为给 i 自身施加 κ（因为 i 是唯一的开发者，其开发带来的边际扩散成本由方程 (5) 中的 $\kappa \cdot (n_D(s_{-i}) + 1)$ 项捕获，其中 $n_D(s_{-i}) = 0$）。

s^* 是纳什均衡当且仅当对所有 i，$u_i(A, \mathbf{A}_{-i}) \geq u_i(D, \mathbf{A}_{-i})$。代入 (12) 和 (13)：

$$V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} \geq V[\theta(0)] - c^{\text{develop}} - \kappa - \lambda \tag{14}$$

由定义 (8)，整理得 $\Delta(|\mathcal{E}|) + \kappa \geq \kappa - \lambda + \kappa$，即条件 (9)。当严格不等式成立时，偏离的收益严格小于均衡收益，均衡是严格的——这保证了均衡在微小支付扰动下的稳健性（à la Kohlberg & Mertens, 1986）。

(iii) 全体开发均衡。考虑 $s^D = (D, \ldots, D)$。在 s^D 下，$n_D = n$，碎片化指标恒为

1。对任意 i：

$$u_i(D, \mathbf{D}_{-i}) = V[\theta(0)] - c^{\text{develop}} - \kappa \cdot n - \lambda \tag{15}$$
$$u_i(A, \mathbf{D}_{-i}) = V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} - \kappa \cdot (n-1) - \lambda \tag{16}$$

方程 (15) 中，$n_D(s_{-i}) = n-1$（除 i 外全部开发），加上 i 自身的开发使扩散总数为 n，故扩散成本项为 $\kappa \cdot n$。方程 (16) 中，i 单边偏离至 A，此时 $n_D(s_{-i}) = n-1$，碎片化仍为 1。

s^D 是纳什均衡当且仅当对所有 i，$u_i(D, \mathbf{D}_{-i}) \geq u_i(A, \mathbf{D}_{-i})$：

$$V[\theta(0)] - c^{\text{develop}} - n\kappa - \lambda \geq V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} - (n-1)\kappa - \lambda \tag{17}$$

消去 $-\lambda$ 并整理：

$$V[\theta(|\mathcal{E}|)] - V[\theta(0)] + c^{\text{adopt}} - c^{\text{develop}} + \kappa \leq -(n-1)\kappa \tag{18}$$

左端即为 $\Delta(|\mathcal{E}|)$，得条件 (10)。

(iv) 非对称均衡的不可能性。假设存在纯策略均衡 s^* 满足 $\exists i, j : s_i^* = A, s_j^* = D$。令 $n_D^* = |\{k : s_k^* = D\}| \in \{1, \ldots, n-1\}$。考虑一个类型为 A 的辖区 i 和一个类型为 D 的辖区 j。由纳什均衡的必要条件：

$$u_i(A, s_{-i}^*) \geq u_i(D, s_{-i}^*) \tag{19}$$
$$u_j(D, s_{-j}^*) \geq u_j(A, s_{-j}^*) \tag{20}$$

计算 i 的支付差异。在 s^* 中，$n_D(s_{-i}^*) = n_D^*$（因为 i 选择 A，不影响 n_D 计数），碎片化指标为 1（因 $n_D^* > 0$）：

$$\begin{aligned} u_i(A, s_{-i}^*) - u_i(D, s_{-i}^*) &= \Delta(|\mathcal{E}|) - \kappa \cdot n_D^* - \lambda - (-\kappa \cdot (n_D^* + 1) - \lambda) \\ &= \Delta(|\mathcal{E}|) + \kappa \end{aligned} \tag{21}$$

类似地，对 j（$s_j^* = D$），$n_D(s_{-j}^*) = n_D^* - 1$：

$$u_j(D, s_{-j}^*) - u_j(A, s_{-j}^*) = -\Delta(|\mathcal{E}|) - \kappa \tag{22}$$

条件 (19) 要求 $\Delta(|\mathcal{E}|) + \kappa \geq 0$，条件 (20) 要求 $-\Delta(|\mathcal{E}|) - \kappa \geq 0$。两条件同时成立仅当 $\Delta(|\mathcal{E}|) = -\kappa$，这是一个勒贝格测度为零的参数配置。在此零测集上，非对称均衡可能存在（所有辖区在 A 和 D 间无差异），但它不是严格的，在支付微小扰动下将崩溃。因此，对几乎所有参数值，非对称纯策略均衡不存在。

(v) 混合策略均衡。在区域 $-(n-1)\kappa < \Delta(|\mathcal{E}|) < \lambda - \kappa$ 内，全体采纳和全体开发均非均衡。设每个辖区以概率 $p \in (0, 1)$ 独立选择 A。则除 i 外选择 D 的辖区数量服从

二项分布：$n_D(s_{-i}) \sim \mathrm{Binomial}(n-1, 1-p)$。

辖区 i 选择 A 的期望支付：

$$\begin{aligned}\mathbb{E}[u_i(A, s_{-i})] &= V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} - \kappa \cdot \mathbb{E}[n_D] - \lambda \cdot \mathbb{P}[n_D > 0] \\ &= V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} - \kappa(n-1)(1-p) - \lambda\left[1 - p^{n-1}\right] \end{aligned} \tag{23}$$

选择 D 的期望支付：

$$\begin{aligned}\mathbb{E}[u_i(D, s_{-i})] &= V[\theta(0)] - c^{\text{develop}} - \kappa(\mathbb{E}[n_D] + 1) - \lambda \\ &= V[\theta(0)] - c^{\text{develop}} - \kappa[(n-1)(1-p) + 1] - \lambda \end{aligned} \tag{24}$$

在对称混合策略均衡中，两期望支付相等（无差异条件）。相减消去 $\kappa(n-1)(1-p)$ 项，得：

$$V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} - \lambda\left[1 - p^{n-1}\right] = V[\theta(0)] - c^{\text{develop}} - \kappa - \lambda \tag{25}$$

整理得方程 (11)。令：

$$\Psi(p) = \lambda \cdot p^{n-1} \quad \text{和} \quad \Gamma = -\Delta(|\mathcal{E}|) + \lambda - 2\kappa \tag{26}$$

则无差异条件等价于 $\Psi(p^*) = \Gamma$。由于 $\Psi : [0,1] \to [0,\lambda]$ 是严格递增的连续函数（$n \geq 2, \lambda > 0$），且在该参数区域内有 $\Gamma \in (0,\lambda)$，由中值定理存在唯一的 $p^* = \Psi^{-1}(\Gamma) \in (0,1)$。由 Glicksberg(1952)（连续支付下混合策略均衡存在性）或直接由纳什的构造，此 p^* 对应于一个对称混合策略纳什均衡。 □

Remark 3.2 (大群体近似). 当 n 较大时（如全球超过 190 个主权国家），$p^{n-1} \to 0$ 对任意 $p < 1$ 成立。此时混合策略概率近似为：

$$p^* \approx \frac{V[\theta(0)] - c^{\text{develop}} - \kappa - V[\theta(|\mathcal{E}|)] + c^{\text{adopt}}}{\lambda} + 1 \tag{27}$$

在大群体极限 $n \to \infty$ 下，单个辖区对其他辖区策略的影响消失（竞争性极限），对称混合策略退化为" 原子化辖区" 面对给定采纳环境的个体最优化。

Theorem 3.2 (非扩散均衡的渐近唯一性与 CEC 临界值). *在假设 A1–A7 下：*

(i) 存在唯一的 ***CEC* 临界规模** $|\mathcal{E}|^*$*，定义为：*

$$|\mathcal{E}|^* \triangleq \inf\{|\mathcal{E}| \geq 0 : \Delta(|\mathcal{E}|) \geq \lambda - \kappa\} \tag{28}$$

若 $\Delta(0) \geq \lambda - \kappa$，则 $|\mathcal{E}|^* = 0$（协议自发布即锁定市场）。否则，由 $\theta(\cdot)$ 和 $V(\cdot)$ 的连续性及严格单调性，$|\mathcal{E}|^*$ 是良定义的正实数。

(ii) 对所有 $|\mathcal{E}| > |\mathcal{E}|^*$，$s^* = (A, \ldots, A)$ 是博弈 Γ^{NP} 的**唯一**纳什均衡——既不存在其他纯策略均衡，也不存在混合策略均衡。

(iii) *CEC* 临界值的显式解为：

$$|\mathcal{E}|^* = \frac{1}{\gamma} \ln\left(\frac{\theta_{\max} - \theta_0}{\theta_{\max} - \theta^*}\right) \tag{29}$$

其中 $\theta^* = V^{-1}\left(V[\theta(0)] + c^{adopt} - c^{develop} - \kappa + \lambda - \kappa\right)$，$V^{-1}$ 是 V 的反函数（由 *A1* 的严格单调性保证存在）。

证明. **(i)** 函数 $\Delta(|\mathcal{E}|)$ 在 $[0, \infty)$ 上连续（A1–A2 保证 $V \circ \theta$ 连续）且严格递增：

$$\frac{d\Delta}{d|\mathcal{E}|} = V'[\theta(|\mathcal{E}|)] \cdot \theta'(|\mathcal{E}|) = V'[\theta(|\mathcal{E}|)] \cdot \gamma(\theta_{\max} - \theta_0)e^{-\gamma|\mathcal{E}|} > 0 \tag{30}$$

其中 $V' > 0$（A1）且 $\theta' > 0$（A2）。下确界 inf 定义给出唯一的临界值。

(ii) 当 $|\mathcal{E}| > |\mathcal{E}|^*$，由定义 $\Delta(|\mathcal{E}|) > \lambda - \kappa$（严格不等式）。定理3.1之 (ii) 确保 $(A, \ldots, A)$ 是严格纳什均衡。全体开发均衡的条件 $\Delta(|\mathcal{E}|) \leq -(n-1)\kappa$ 不成立（因为 $\Delta(|\mathcal{E}|) > \lambda - \kappa > 0 > -(n-1)\kappa$，由 A4 $\lambda > \kappa > 0$）。混合策略均衡要求无差异条件成立，而严格纳什均衡下偏离的期望收益严格小于均衡收益，无差异条件无法满足。因此 $(A, \ldots, A)$ 是唯一均衡。

(iii) 解 $\Delta(|\mathcal{E}|) = \lambda - \kappa$ 得 θ^*，代入 A2 的函数形式即得显式解。 □

[非扩散均衡的单调自强化性质] 在假设 A1–A7 下，定义非扩散均衡的**稳定性裕度**：

$$M(|\mathcal{E}|) \triangleq \Delta(|\mathcal{E}|) - (\lambda - \kappa) \tag{31}$$

则：

(i) $M(|\mathcal{E}|)$ 在 $[0, \infty)$ 上严格递增：$M'(|\mathcal{E}|) = V'[\theta(|\mathcal{E}|)] \cdot \theta'(|\mathcal{E}|) > 0$；

(ii) $\lim_{|\mathcal{E}| \to \infty} M(|\mathcal{E}|) = V(\theta_{\max}) - V(\theta_0) + c^{\text{develop}} - c^{\text{adopt}} + \kappa - \lambda$；

(iii) 每次审计循环通过 $\Delta|\mathcal{E}| > 0$ 扩张 CEC,同时:(a) 增加采纳的吸引力（$\partial u_i(A, \cdot)/\partial|\mathcal{E}| > 0$），(b) 增大偏离的相对损失（$M(|\mathcal{E}|)$ 增），(c) 降低混合策略均衡中采纳概率 p^* 的不确定性（当均衡为纯策略时不确定性归零）。

(iv) 这一定性——均衡的稳定性随其达成而自我强化——是时间累积优势机制在博弈论层面的精确表达。

证明. 直接由定理3.1和3.2结合假设 A1–A2 的单调性验证。 □

[先发优势的时间衰减定理 (Temporal Decay of First-Mover Advantage)] 在假设 A1–A7 下。定义以下量度：

Definition 3.3 (标准化先发优势强度). 先发者（CEC 规模 $|\mathcal{E}|$）相对于零 CEC 后发者的**绝对精度优势**为：

$$\delta(|\mathcal{E}|) \triangleq \theta(|\mathcal{E}|) - \theta_0 = (\theta_{\max} - \theta_0)(1 - e^{-\gamma|\mathcal{E}|}) \tag{32}$$

标准化先发优势强度——即先发优势占最大可能优势的比例——为：

$$\Xi(|\mathcal{E}|) \triangleq \frac{\delta(|\mathcal{E}|)}{\theta_{\max} - \theta_0} = 1 - e^{-\gamma|\mathcal{E}|} \in [0, 1) \tag{33}$$

则以下命题成立：

(i) **绝对优势的边际递减与渐近饱和**：先发者的绝对精度优势 $\delta(|\mathcal{E}|)$ 单调递增（$\delta'(|\mathcal{E}|) = \gamma(\theta_{\max} - \theta_0)e^{-\gamma|\mathcal{E}|} > 0$），但其边际增量严格递减至零：

$$\delta''(|\mathcal{E}|) = -\gamma^2(\theta_{\max} - \theta_0)e^{-\gamma|\mathcal{E}|} < 0, \quad \lim_{|\mathcal{E}|\to\infty} \delta'(|\mathcal{E}|) = 0 \tag{34}$$

且绝对优势收敛于有限上限：

$$\lim_{|\mathcal{E}|\to\infty} \delta(|\mathcal{E}|) = \theta_{\max} - \theta_0 < \infty \tag{35}$$

经济学含义：先发优势不是无界增长的——CEC 的精度贡献具有硬上限（由审计模块的理论检测能力决定）。在此上限下，先发者无论积累多少审计历史，其精度优势均不可超过 $\theta_{\max} - \theta_0$。

(ii) **采纳优势的边际衰减**：将 $\delta(|\mathcal{E}|)$ 代入 $\Delta(|\mathcal{E}|)$ 的定义（方程8）：

$$\Delta(|\mathcal{E}|) = V[\theta_0 + \delta(|\mathcal{E}|)] - V[\theta_0] - (c^{\text{adopt}} - c^{\text{develop}}) + \kappa \tag{36}$$

由假设 A1（V 严格凹），$\Delta'(|\mathcal{E}|) = V'[\theta(|\mathcal{E}|)] \cdot \delta'(|\mathcal{E}|) > 0$ 且 $\Delta''(|\mathcal{E}|) < 0$。采纳优势的边际增长同时受制于精度边际递减（$\delta' \to 0$）和价值边际递减（$V' \to 0$ 若 $\theta \to \theta_{\max}$）。因此：

$$\lim_{|\mathcal{E}|\to\infty} \Delta'(|\mathcal{E}|) = 0 \tag{37}$$

经济学含义：先发者通过继续积累 CEC 来扩大采纳优势的能力随时间递减——追加审计循环对均衡稳定性的边际贡献趋零。

(iii) **CEC 作为追加式公共品的时间衰减机制（核心结论）**：CEC 的本质是*追加式公共品*（additive public good）：(a) **追加性**——每次审计循环向 CEC 追加新条目（$\Delta|\mathcal{E}| > 0$），CEC 规模只增不减（假设 M4）；(b) **公共性**——CEC 中存储的异常登记、校准参数和质量指纹是*非竞争性*知识：辖区 i 调用某条校准数据不减少辖区 j 调用同条数据的能力。

设 CEC 在时间 T^* 被国际化——即通过多边治理机制转变为所有辖区可平等访问的全球公共基础设施（参见第 6 节建议 1 的 IDAA）。对任意 $t > T^*$，令 $|\mathcal{E}_t^{\text{public}}|$ 为任何后发辖区可即时调用的 CEC 规模。后发者的冷启动惩罚从 $\theta(|\mathcal{E}_t|) - \theta(0)$ 缩减为 $\theta(|\mathcal{E}_t|) - \theta(|\mathcal{E}_t^{\text{public}}|)$。定义**后发惩罚比**：

$$\Psi(t; T^*) \triangleq \frac{\theta(|\mathcal{E}_t|) - \theta(|\mathcal{E}_t^{\text{public}}|)}{\theta(|\mathcal{E}_t|) - \theta(0)} \in [0, 1] \tag{38}$$

$\Psi(T^*; T^*) = 1$（国际化时刻，后发者尚未调用 CEC 公开部分，惩罚完整）。但当 $|\mathcal{E}_t^{\text{public}}| \to |\mathcal{E}_t|$（公开 CEC 追平先发者私有 CEC），$\Psi \to 0$。在理想国际化下（$|\mathcal{E}_t^{\text{public}}| = |\mathcal{E}_t|$），$\Psi \equiv 0$——先发优势在采纳层面完全消除。

然而，完全消除仅适用于单次审计的精度维度。先发者仍保留两项不可国际化的残余优势：(a) **生态系统惯性的时间深度**——先发者的采纳者网络、监管认可和制度嵌入具有不可压缩的历史深度（定理5.1的阶段锁定）；(b) **操作标准制定权的路径依赖**——CEC 中的异常判定阈值和校准参数配置反映先发者的初始数据特征，这些特征即使在国际化后仍构成操作标准的历史锚点。

(iv) **先发优势的半衰期**：定义先发优势的**采纳半衰期** $T_{1/2}^{\Delta}$ 为满足以下条件的时刻：

$$\Delta(|\mathcal{E}_{T_{1/2}^{\Delta}}|) - (\lambda - \kappa) = \frac{1}{2}\left[\Delta(|\mathcal{E}_{T^*}|) - (\lambda - \kappa)\right] \tag{39}$$

即 NPE 稳定性裕度（推论3.3）降至国际化时刻的一半。在假设 A2 的函数形式下，$T_{1/2}^{\Delta}$ 的隐式解满足：

$$V[\theta(|\mathcal{E}_{T_{1/2}^{\Delta}}|)] - V[\theta(|\mathcal{E}_{T^*}|)] = \frac{1}{2}\left[V[\theta(|\mathcal{E}_{T^*}|)] - V[\theta_0] - (c^{\text{adopt}} - c^{\text{develop}}) + \kappa - (\lambda - \kappa)\right] \tag{40}$$

半衰期的存在性由 $V \circ \theta$ 的连续性和 $\Delta(\cdot)$ 的单调有界性保证。其有限性表明：**先发优势不是永久性结构，而是具有明确时间边界的阶段性现象。**

(v) **诚实暴击——NPE 的规范含义修正：**定理3.1和推论3.3所确立的”均衡随 CEC 增长而自我强化”的结论，在引入 CEC 作为追加式公共品的分析后，需接受以下限定：NPE 的自强化仅发生在 CEC 被排他性持有的条件下。一旦 CEC 被国际化（通过 IDAA 或等效多边机制），采纳优势 $\Delta(|\mathcal{E}|)$ 对先发者特有的那部分贡献——即 $\theta(|\mathcal{E}_t^{\text{private}}|) - \theta(|\mathcal{E}_t^{\text{public}}|)$ 项——趋于零。此时，采纳的动机从”开发替代协议不划算”转变为”开发替代协议不必要”——各辖区平等受益于全球 CEC 基础设施，碎片化的公共劣品效应 λ 继续威慑扩散，但威慑的基础从先发者的私有 CEC 垄断转变为 CEC 作为全球公共品的普遍可得性。

换言之，NPE 经历了从先发者垄断威慑到公共品协调均衡的制度转型。转型前的均衡由 $\Delta(|\mathcal{E}^{\text{private}}|) \geq \lambda - \kappa$ 维持（先发垄断型 NPE）；转型后的均衡由 $\Delta(|\mathcal{E}^{\text{public}}|) \geq \lambda - \kappa$ 维持（公共品型 NPE），其中 $\Delta(|\mathcal{E}^{\text{public}}|)$ 对所有辖区相等——先发者的身份

优势消失，仅保留 CEC 本身的规模优势（后者是全体辖区共享的公共品）。

此即”天下大同”的博弈论基础——见第 5.2.2 节的完整形式化。

证明. **(i)–(ii)** 直接求导。由假设 A2，$\theta'(|\mathcal{E}|) = \gamma(\theta_{\max}-\theta_0)e^{-\gamma|\mathcal{E}|} > 0$，$\theta''(|\mathcal{E}|) = -\gamma^2(\theta_{\max}-\theta_0)e^{-\gamma|\mathcal{E}|} < 0$。因此 $\delta'(|\mathcal{E}|) > 0$，$\delta''(|\mathcal{E}|) < 0$，且 $\delta'(|\mathcal{E}|) \to 0$ 当 $|\mathcal{E}| \to \infty$。$\lim_{|\mathcal{E}|\to\infty} \delta(|\mathcal{E}|) = \theta_{\max} - \theta_0$ 由指数衰减项趋于零得出。$\Delta(|\mathcal{E}|)$ 的单调性和凹性由 V 的单调凹性（A1）与 $\delta(|\mathcal{E}|)$ 的单调凹性的复合保持严格凹性。

(iii) CEC 的公共品属性：非竞争性由知识的本质保证——校准参数和异常模式是信息，信息消费具有非竞争性。追加性由 CEC 定义（方程 (2)）保证——$\mathcal{E}_t = \bigcup_{\tau=0}^{t}\{\cdots\}$，并集运算单调非减。后发惩罚比 $\Psi(t;T^*)$ 的极限行为：当 $|\mathcal{E}_t^{\text{public}}| \to |\mathcal{E}_t|$，由 θ 的连续性，$\theta(|\mathcal{E}_t^{\text{public}}|) \to \theta(|\mathcal{E}_t|)$，故 $\Psi \to 0$。不可国际化的残余优势由定理5.1（阶段锁定的不可逆性，假设 M3）和路径依赖的标准制定权（CEC 的初始配置反映先发者的校准选择）所保证。

(iv) 半衰期的存在性：函数 $f(t) = \Delta(|\mathcal{E}_t|) - (\lambda - \kappa)$ 在 $[T^*, \infty)$ 上连续（A1–A2 保证）、严格递减至零或正值下限（取决于 $\lim_{|\mathcal{E}|\to\infty} \Delta(|\mathcal{E}|)$ 与 $\lambda - \kappa$ 的关系）。若下限大于 $\frac{1}{2}[\Delta(|\mathcal{E}_{T^*}|) - (\lambda - \kappa)]$，则半衰期有限（由中值定理）。若下限更小，半衰期仍有限——$f(t)$ 最终穿越半值线。唯一例外是 $\Delta(|\mathcal{E}|) \equiv \text{const}$（要求 V 线性且 θ 恒定，与 A1–A2 矛盾），此时半衰期无定义但此情形不会发生。

(v) NPE 的制度转型：将方程 (9) 中的 $\Delta(|\mathcal{E}|)$ 分解为先发者私有贡献与公共 CEC 贡献。国际化前，$\Delta(|\mathcal{E}|) = \Delta(|\mathcal{E}^{\text{private}}|)$；国际化后，所有辖区的采纳优势趋同于 $\Delta(|\mathcal{E}^{\text{public}}|)$。条件 $\Delta(|\mathcal{E}^{\text{public}}|) \geq \lambda - \kappa$ 不涉及任何辖区的身份——先发优势在采纳博弈层面消失。 □

Remark 3.3 (先发优势衰减与核不扩散条约的结构类比). 推论3.3完成了 NPT 类比（第 3.5 节）的一个关键制度推论。NPT 第 VI 条要求核武器国家”真诚地谈判”以实现核裁军——即核先发优势的衰减承诺。类似地，CEC 的国际化构成了审计先发优势的”第 VI 条”：先发者承诺将 CEC 转变为全球公共品，从而使其先发优势随时间衰减。但正如 NPT 的裁军承诺面临可信性危机（Joyner, 2011），CEC 国际化的可信性同样取决于制度设计——先发者是否有激励在达到 S_3（护城河形成）后仍然推进国际化（参见命题5.3的干预窗单调闭合）。

3.4 支付矩阵与均衡区域：参数空间分解

For the two-jurisdiction case ($N = 2$), the payoff matrix is:

Jur. 1 \ Jur. 2	Adopt (A)	Develop (D)
Adopt (A)	$V[\theta(\|\mathcal{E}\|)] - c^{\text{adopt}}, V[\theta(\|\mathcal{E}\|)] - c^{\text{adopt}}$	$V[\theta(\|\mathcal{E}\|)] - c^{\text{adopt}} - \lambda, V[\theta(0)] -$
Develop (D)	$V[\theta(0)] - c^{\text{develop}} - \kappa - \lambda, V[\theta(\|\mathcal{E}\|)] - c^{\text{adopt}} - \lambda$	$V[\theta(0)] - c^{\text{develop}} - 2\kappa - \lambda, V[\theta(0)] -$

Let us define the *Adoption Advantage*:

$$\Delta_A = V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} - \big(V[\theta(0)] - c^{\text{develop}} - \kappa\big) \tag{41}$$

(A, A) is a Nash equilibrium if and only if neither jurisdiction has a unilateral incentive to deviate. Jurisdiction i's deviation payoff from (A, A) is:

$$\pi_i(D, A) = V[\theta(0)] - c^{\text{develop}} - \kappa - \lambda \tag{42}$$

The non-deviation condition $\pi_i(A, A) \geq \pi_i(D, A)$ yields:

$$V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} \geq V[\theta(0)] - c^{\text{develop}} - \kappa - \lambda \tag{43}$$

Rearranging:

$$\Delta_A \geq \lambda \tag{44}$$

That is, (A, A) is an equilibrium when the adoption advantage exceeds the fragmentation cost. Given that Δ_A grows with $|\mathcal{E}|$ (since $\theta(|\mathcal{E}|)$ increases with cumulative audits while $\theta(0)$ remains fixed), this condition becomes easier to satisfy over time. The protocol's cumulative character makes the non-proliferation equilibrium *self-reinforcing*.

Conversely, (D, D) is an equilibrium when:

$$V[\theta(0)] - c^{\text{develop}} - 2\kappa - \lambda \geq V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} - \lambda \tag{45}$$

which simplifies to:

$$-\Delta_A \geq 2\kappa \tag{46}$$

This condition requires the adoption advantage to be *negative* and large in magnitude —that is, the incumbent protocol must be so inferior to a fresh development, or the development cost so low, that even with mutual proliferation costs, development is preferred. As $|\mathcal{E}|$ grows, this condition becomes increasingly implausible.

The mixed-strategy equilibrium occurs when neither pure-strategy condition holds. Let p be the probability that each jurisdiction plays A. The symmetric mixed-strategy Nash equilibrium satisfies:

$$p \cdot \pi(A, A) + (1 - p) \cdot \pi(A, D) = p \cdot \pi(D, A) + (1 - p) \cdot \pi(D, D) \tag{47}$$

Solving for p^*:

$$p^* = \frac{-\Delta_A - 2\kappa}{\lambda - \kappa} \tag{48}$$

For plausible parameter values ($\Delta_A > 0$, $\kappa > 0$, $\lambda > \kappa$), p^* increases with Δ_A: as the incumbent's accuracy advantage grows, the probability of adoption in the mixed-strategy equilibrium approaches 1. This formalizes the intuition that the incumbent's time-accumulated advantage makes proliferation an increasingly unattractive gamble.

3.5 Extension to N Jurisdictions

For $N > 2$, the payoff functions generalize as follows:

$$\pi_i(A, \mathbf{a}_{-i}) = V[\theta(|\mathcal{E}|)] - c^{\text{adopt}} - \kappa \cdot n_D - \lambda \cdot \mathbb{I}[n_D > 0] \tag{49}$$

$$\pi_i(D, \mathbf{a}_{-i}) = V[\theta(0)] - c^{\text{develop}} - \kappa \cdot (n_D + \mathbf{1}^D_{-i}) - \lambda \cdot \mathbb{I}[n_D > 0] \tag{50}$$

where $n_D = \sum_{j \neq i} \mathbb{I}[a_j = D]$ is the number of other jurisdictions that develop.

The universal adoption equilibrium $(A, A, \ldots, A)$ is Nash when, for every jurisdiction:

$$\Delta_A \geq \lambda - \kappa \tag{51}$$

The term $\lambda - \kappa$ reflects a subtle tension: fragmentation cost λ discourages unilateral deviation, but the *relief* from mutual proliferation costs κ (if no one else develops, the lone developer avoids κ) encourages it. The equilibrium is stable when $\lambda > \kappa$—that is, when the public-bads cost of fragmentation exceeds the private cost of being the only proliferator.

A key result emerges: the stability of the non-proliferation equilibrium is *monotonically increasing* in $|\mathcal{E}|$, because Δ_A grows without bound (up to its asymptotic ceiling) while $\lambda - \kappa$ is fixed. There exists a critical CEC size $|\mathcal{E}|^*$ such that for all $|\mathcal{E}| > |\mathcal{E}|^*$, $(A, A, \ldots, A)$ is the unique Nash equilibrium. This uniqueness is underwritten by Theorem 4 (极定理): no alternative algorithm can surpass SCX's minimax-optimal detection rate. The ceiling is absolute.

$$|\mathcal{E}|^* = \frac{1}{\gamma} \ln \left(\frac{\theta_{\max} - \theta_0}{\theta_{\max} - \theta(|\mathcal{E}|^*)} \right) \tag{52}$$

where $\theta(|\mathcal{E}|^*)$ is the accuracy level that satisfies $\Delta_A(|\mathcal{E}|^*) = \lambda - \kappa$.

3.6 The Nuclear Non-Proliferation Treaty Analogy

The structural isomorphism between the audit protocol proliferation game and the strategic logic of nuclear non-proliferation is striking and analytically productive. We identify seven dimensions of analogy:

1. The Public-Bads Character of Proliferation. Nuclear weapons proliferation generates negative externalities: each additional nuclear state increases the probability of nuclear conflict (accidental, escalatory, or intentional), raises the costs of the international security system, and erodes the norm against nuclear use (Sagan, 1996). Similarly, audit protocol proliferation generates negative externalities: each additional protocol fragments the global quality assurance infrastructure, raises the transaction costs of cross-border data flows, and erodes the credibility of audit certifications. In both domains, the *first* instance of proliferation (beyond the incumbent) triggers a discontinuous increase in systemic costs, because it shatters the universality that makes the regime valuable.

2. The Asymmetric Incumbent Advantage. The NPT regime is built on an explicit asymmetry: five states are recognized as nuclear-weapon states (NWS), while all others are non-nuclear-weapon states (NNWS) subject to verification by the International Atomic Energy Agency (IAEA). The asymmetry is justified by the temporal fact of prior acquisition: the NWS tested nuclear weapons before the treaty's 1968 cutoff date. The Yajie Protocol's asymmetry is similarly temporal: the incumbent's advantage derives not from legal privilege but from the inescapable fact that it began accumulating audit evidence earlier.

3. The Grand Bargain. The NPT's durability rests on a three-pillar bargain: (i) NNWS forswear nuclear weapons; (ii) NWS commit to eventual disarmament (Article VI); and (iii) all parties enjoy the "inalienable right" to peaceful nuclear technology (Article IV). The audit protocol analogue would involve: (i) jurisdictions forgo developing independent protocols; (ii) the incumbent protocol's governance is internationalized under multilateral oversight; and (iii) all jurisdictions enjoy non-discriminatory access to the protocol's capabilities and the right to contribute audit modules to the ensemble.

4. The Credibility of Commitment. A central challenge of the NPT regime is the credibility of the NWS disarmament commitment: NNWS periodically question whether the nuclear states genuinely intend to disarm, and these credibility crises threaten the regime's stability (Joyner, 2011). The analogous challenge for Yajie is the credibility of the incumbent's neutrality commitment: will the protocol remain genuinely jurisdiction-agnostic, or will it evolve into an instrument of its host jurisdiction's data governance preferences?

5. The Enforcement Gap. The NPT contains no automatic enforcement mechanism; compliance is monitored by the IAEA, but enforcement depends on the UN Security Council, where NWS possess veto power. This asymmetry is a perennial source of tension, particularly in cases like Iran and North Korea. The audit protocol regime would face a parallel enforcement gap: who adjudicates disputes about the protocol's neutrality, and with what authority?

6. The Exit Option. The NPT permits withdrawal on three months' notice if "extraordinary events" jeopardize a state's supreme interests (Article X). North Korea exercised this option in 2003. The audit protocol regime must similarly contend with the possibility of exit: a jurisdiction that develops an independent protocol after having previously adopted the incumbent. The strategic dynamics of exit differ from those of initial non-adoption, because the exiting jurisdiction may have benefited from the CEC during its period of adoption, raising questions of intellectual property, data portability, and the status of audits conducted under the incumbent protocol.

7. The Dual-Use Problem. Nuclear technology is dual-use: the enrichment and reprocessing capabilities that support civilian nuclear energy can be diverted to weapons production. Audit technology is analogously dual-use: the quality assessment capabilities that support benign data governance can be weaponized for economic espionage, supply-chain disruption, or the imposition of data embargoes under the guise of quality concerns.

The NPT analogy is not merely illustrative; it provides a template for institutional design. The following sections develop the institutional implications, but first we must examine the geopolitical context within which any audit infrastructure—monopolistic or otherwise—must operate.

4 Audit Neutrality and Geopolitical Risk

4.1 The SWIFT Precedent: Infrastructure Weaponization

The Society for Worldwide Interbank Financial Telecommunication (SWIFT) provides the most instructive precedent for understanding the geopolitical risks of monopolistic audit infrastructure. SWIFT, founded in 1973 as a cooperative society under Belgian law, operates the dominant global messaging network for financial transactions, connecting over 11,000 financial institutions across more than 200 countries. Its near-universal adoption makes it a natural analogue to the market structure we predict for data quality auditing.

SWIFT's governance is formally multilateral: it is owned by its member financial institutions and overseen by a board of directors drawn from major central banks, including the US Federal Reserve, the European Central Bank, and the Bank of England. However, the United States has repeatedly leveraged SWIFT's centrality for unilateral geopolitical purposes, most controversially through the Terrorist Finance Tracking Program (TFTP), which accessed SWIFT data without the knowledge or consent of many member institutions (Farrell & Newman, 2019).

The critical event for our purposes occurred in 2012, when the United States threatened to sanction SWIFT itself if it did not disconnect Iranian banks designated under US sanctions—despite the fact that those sanctions were not multilateral and were contested by several EU member states, China, and Russia. SWIFT complied, disconnecting approximately 30 Iranian financial institutions (Drezner, 2015). The episode demonstrated that infrastructure neutrality, however formally enshrined in governance documents, is only as robust as the willingness of powerful states to respect it—and that the host jurisdiction of a monopolistic infrastructure enjoys structural leverage that may be exercised in moments of geopolitical tension.

The SWIFT case carries three lessons for data audit infrastructure:

Lesson 1: Jurisdictional Capture. SWIFT is incorporated under Belgian law with headquarters in Brussels. Yet the United States' ability to compel SWIFT's compliance derived not from formal legal authority but from the extraterritorial reach of US financial sanctions and the threat of secondary sanctions against SWIFT's board members and executives. A data audit protocol hosted in any single jurisdiction would be analogously vulnerable to that jurisdiction's extraterritorial legal instruments, regardless of the protocol's formal governance structure.

Lesson 2: The Illusion of Technical Neutrality. SWIFT consistently maintains that it is a neutral utility providing technical services. The 2012 disconnection decisively refuted this claim: technical infrastructure, when sufficiently central, is *inherently* political. A data audit protocol that assesses the quality of datasets used in AI systems would be, if anything, *more* political than a financial messaging network, because data quality assessments are inherently contestable and may be perceived (fairly or not) as disguised forms of technological protectionism.

Lesson 3: Fragmentation as a Rational Response. The SWIFT disconnection accelerated the development of alternative financial messaging systems: China's Cross-Border Interbank Payment System (CIPS), Russia's System for Transfer of Financial Messages (SPFS), and the European Instrument in Support of Trade Exchanges (INSTEX). Each of these alternatives is inferior to SWIFT in coverage and functionality, representing a welfare loss relative to a universal system. Yet each represents a rational response to the demonstrated risk of infrastructure weaponization. The data audit domain would face the same dynamic: the credible threat of protocol weaponization would make audit autarky rational for jurisdictions with the capacity to pursue it, even at a welfare cost.

4.2 The IASB Model: De-Jurisdictionalized Governance

The International Accounting Standards Board (IASB) offers a partial counter-model. The IASB, established in 2001 as the standard-setting body of the International Financial Reporting Standards (IFRS) Foundation, develops accounting standards that have been adopted by more than 140 jurisdictions. Unlike SWIFT, the IASB is not an operational infrastructure but a standard-setting body; its product is a corpus of normative standards, not a messaging network or audit protocol. Nevertheless, its governance model is instructive.

The IASB's governance features several design elements that mitigate—though do not eliminate—the risk of jurisdictional capture:

- **Multi-stakeholder oversight**: The IFRS Foundation is governed by a board of trustees drawn from diverse geographic and professional backgrounds, with formal requirements for regional balance.

- **Due process requirements**: Standards development follows a structured process of public consultation, impact assessment, and re-deliberation that constrains the ability of any single jurisdiction to dictate outcomes.

- **Funding diversification**: The IFRS Foundation's funding comes from a broad base of jurisdictions and organizations, reducing dependence on any single funder.
- **Deliberative transparency**: Board meetings are public, and dissenting views are published alongside standards, creating an audit trail of contestation that supports legitimacy.

The IASB model is not without critics. Some scholars argue that the IASB's independence is compromised by its dependence on major accounting firms for funding and expertise (Zeff, 2012), and by the structural influence of Anglo-American accounting traditions in its conceptual framework (Zhang & Andrew, 2014). The EU's qualified endorsement process—whereby the European Commission reviews each IFRS standard before endorsing it for use within the EU—represents a de facto veto power that other jurisdictions do not possess, creating a legitimacy deficit.

For data audit infrastructure, the IASB model suggests a governance architecture that is *multilateral in oversight, transparent in operation, and diversified in funding.* However, the IASB's experience also suggests that de-jurisdictionalized governance is an aspiration rather than an achievement, and that persistent power asymmetries will find expression through formal and informal channels regardless of governance design.

4.3 SWIFT 与 IASB 的博弈论模型

前两小节的案例分析为叙事层面。本节将其严格形式化为博弈模型，使 SWIFT 武器化与 IASB 治理的逻辑获得均衡基础。

4.3.1 SWIFT 武器化的序贯博弈模型

将 SWIFT 武器化情景建模为两个国家（美国 U 与目标国 T）和一个基础设施运营者（SWIFT，W）的三方序贯博弈。博弈时序如下：

阶段 1: 美国 U 决定是否对目标国 T 实施单边金融制裁，并要求 W 断开 T 的连接：$a_U \in \{制裁, 不制裁\}$。

阶段 2: 观察到 a_U 后，SWIFT W 决定是否服从：$a_W \in \{断开, 拒绝\}$。

阶段 3: 观察到 (a_U, a_W) 后，目标国 T 决定是否建设替代系统：$a_T \in \{建设替代, 不建设\}$。

Definition 4.1 (SWIFT 博弈的全部假设)**. S1 支付参数：** $B_W > 0$ 为 W 运营全球统一网络的基础收益；$C_W^{\text{comply}} > 0$ 为 W 服从美国制裁遭受的声誉与法律成本（来自其他辖区的反制威胁）；$P_U > 0$ 为美国对 W 不服从时施加的惩罚（二次制裁威胁）；

$B_T > 0$ 为 T 使用 SWIFT 的收益；$C_T^{\text{alt}} > 0$ 为 T 建设替代系统（如 SPFS 或 CIPS）的固定成本；$L_T > 0$ 为 T 被断开 SWIFT 的经济损失。所有参数是共同知识。 *注意：完全信息假设在此语境下是合理的——SWIFT 的连通性、美国制裁能力和替代系统成本是公开信息。*

S2 理性：三方均最大化自身支付，无非理性报复动机。 *可放松性：引入声誉关切或国内观众成本可使模型更丰富，但不改变定性均衡。*

S3 无转移支付：博弈中无旁支付（side payments）。U 不能直接补偿 W 的合规成本。*含义：这排除了美国通过财政承诺购买 SWIFT 忠诚的策略。*

三方支付函数（终节点支付）：

$$\pi_U(a_U, a_W, a_T) = \begin{cases} G_U & \text{若} a_U = \text{制裁}, a_W = \text{断开} \quad \text{（制裁成功）} \\ 0 & \text{若} a_U = \text{不制裁} \quad \text{（维持现状）} \\ -P_U^{\text{rep}} & \text{若} a_U = \text{制裁}, a_W = \text{拒绝} \quad \text{（制裁失败，声誉受损）} \end{cases} \tag{53}$$

$$\pi_W(a_U, a_W, a_T) = \begin{cases} B_W - C_W^{\text{comply}} - \mathbb{I}[a_T = \text{建设}] \cdot L_W^{\text{frag}} & \text{若} a_W = \text{断开} \\ B_W & \text{若} a_W = \text{拒绝}, a_U = \text{不制裁} \\ B_W - P_U & \text{若} a_W = \text{拒绝}, a_U = \text{制裁} \end{cases} \tag{54}$$

$$\pi_T(a_U, a_W, a_T) = \begin{cases} B_T & \text{若} a_W = \text{拒绝}, a_T = \text{不建设} \\ B_T - C_T^{\text{alt}} & \text{若} a_W = \text{拒绝}, a_T = \text{建设} \\ -L_T & \text{若} a_W = \text{断开}, a_T = \text{不建设} \\ -L_T + B_T^{\text{alt}} - C_T^{\text{alt}} & \text{若} a_W = \text{断开}, a_T = \text{建设} \end{cases} \tag{55}$$

其中 $G_U > 0$ 是美国制裁成功的战略收益；$P_U^{\text{rep}} > 0$ 是美国制裁失败（被公开拒绝）的声誉损失；L_W^{frag} 是目标国建设替代系统导致 SWIFT 网络碎片化的长期价值损失；B_T^{alt} 是目标国使用自建替代系统的收益（通常 $B_T^{\text{alt}} < B_T$，替代系统覆盖范围较小）。

Theorem 4.1 (SWIFT 武器化博弈的子博弈完美均衡). *在假设 S1–S3 下，SWIFT 序贯博弈存在唯一的子博弈完美纳什均衡（SPE），由逆向归纳法确定。均衡依赖于参数区域：*

(i) **区域 Ω_1（美国支配）**：*若 $P_U > C_W^{comply} + L_W^{frag}$ 且 $C_T^{alt} > L_T + B_T^{alt}$，唯一 SPE 为：*

美国制裁 $\to$ SWIFT 服从断开 $\to$ 目标国不建替代系统

此为 2012 年伊朗 SWIFT 断连的均衡逻辑：美国的二次制裁威胁足够可信（P_U 大），目标国的替代成本太高。

(ii) **区域 Ω_2（目标国抵抗）**：*若 $P_U > C_W^{comply} + L_W^{frag}$ 但 $C_T^{alt} < L_T + B_T^{alt}$，唯一 SPE 为：*

美国制裁 $\to$ *SWIFT* 服从断开 $\to$ 目标国建设替代系统

此为俄罗斯 *SPFS* 和中国 *CIPS* 的建设逻辑：断开损失超过了替代成本，建设替代是理性反应——尽管替代系统劣于 *SWIFT*。

(iii) **区域 Ω_3（*SWIFT* 中立）**：若 $P_U < C_W^{comply}$，唯一 *SPE* 为：

美国不制裁 $\to$ *SWIFT* 拒绝 $\to$ 目标国不建替代

（若美国预期 *SWIFT* 会拒绝而选择不制裁，或制裁后 *SWIFT* 拒绝，则 *SPE* 路径取决于子博弈的精确支付比较。）此区域要求美国的制裁惩罚不足以压倒 *SWIFT* 的合规成本——即基础设施在制度上足够独立以抵御单边压力。

(iv) **均衡效率**：Ω_1 均衡实现了美国的战略目标但损害了全球金融基础设施的中立性（负外部性）。Ω_3 均衡是帕累托最优——全球统一网络得以维持，中立性得以保存。但 Ω_3 在 P_U 较大（美国具有可信的二次制裁能力）时不可企及。

证明. 由逆向归纳法。在阶段 3，目标国 T 的最优反应为：若被断开且 $B_T^{\text{alt}} - C_T^{\text{alt}} > -L_T$（即 $C_T^{\text{alt}} < L_T + B_T^{\text{alt}}$），则建设替代；否则不建设。

在阶段 2，SWIFT W 的最优反应依赖于对阶段 3 的预期。若 W 预期 T 建设替代（$a_T =$ 建设），则服从的支付为 $B_W - C_W^{\text{comply}} - L_W^{\text{frag}}$，拒绝的支付为 $B_W - P_U$。W 服从当且仅当 $P_U > C_W^{\text{comply}} + L_W^{\text{frag}}$。若 W 预期 T 不建替代，则服从条件为 $P_U > C_W^{\text{comply}}$。

在阶段 1，美国的决策取决于对 (a_W^*, a_T^*) 的预期。若预期路径为 (断开, 不建设)，美国制裁的支付为 $G_U > 0$，故制裁；若预期路径为 (断开, 建设)，美国仍制裁（制裁收益 G_U 超过不制裁收益 0，除非目标国替代削弱了制裁的战略价值——此时 G_U 需下调）；若预期路径为 (拒绝, $\cdot$)，美国若制裁则支付 $-P_U^{\text{rep}} < 0$，故最优选择为不制裁。

综合阶段最优反应即得三区域的 SPE 分类。 □

Remark 4.1 (审计基础设施的武器化类比). 定理4.1直接适用于数据审计基础设施的武器化情景：将" 断开 SWIFT" 替换为" 拒绝审计认证" 或" 发布不利审计报告"，将" 建设替代系统" 替换为" 开发独立审计协议"。类比的关键参数是：(a) 协议运营者所在辖区的域外法律工具（类似美国的二次制裁能力 P_U），(b) 替代审计协议的开发成本（类似 C_T^{alt}），(c) 被排除在审计网络之外的经济损失（类似 L_T）。审计基础设施与 SWIFT 的关键差异在于替代成本的结构：SWIFT 替代系统的建设成本主要是技术性的（架设服务器、建立通讯协议、说服银行加入），而审计协议替代系统还面临 CEC 冷启动问题——即使投入充足资金，替代协议的精度也需时间累积才能达到与现有协议相当的水平。这使审计基础设施的 Ω_2 区域远小于 SWIFT 的对应区域——替代的代价更高，武器化的威慑更强。

4.3.2 IASB 的多辖区投票博弈模型

IASB 的治理结构可形式化为空间投票博弈，其中各辖区对会计准则具有不同偏好。

Definition 4.2 (IASB 准则选择博弈)**.** $N = \{1, \ldots, n\}$ 个辖区就一维会计准则参数 $\sigma \in [0,1]$ 进行集体决策。辖区 i 的效用函数为 $u_i(\sigma) = -|\sigma - \sigma_i^*|$（绝对值损失），其中 $\sigma_i^* \in [0,1]$ 是辖区 i 的理想准则。准则由 IASB 董事会投票决定，董事会由 m 名受托人组成（m 为奇数），受托人分布反映了各辖区的任命权配置。

Definition 4.3 (IASB 博弈的全部假设)**. I1 单峰偏好：**各辖区 i 的效用函数在 $[0, \sigma_i^*]$ 上严格递增，在 $[\sigma_i^*, 1]$ 上严格递减。绝对值损失是单峰偏好的特例。*合理性：准则偏离理想值越多，辖区的合规成本和制度摩擦越大。*

I2 一维政策空间：会计准则的差异可沿单一维度排列（如" 原则导向" 至" 规则导向"）。*简化：实际会计准则具有多维性，但一维建模捕获了治理冲突的核心结构。*

I3 董事会中位投票者定理：受托人按简单多数决投票，投票是真诚的（sincere）。在此假设下，均衡结果 σ^* 等于董事会受托人理想点的中位数。 *条件：一维政策空间 + 单峰偏好 $\Longrightarrow$ 中位投票者定理成立（Black, 1948）。*

I4 辖区影响力权重：辖区 i 在董事会中拥有 w_i 名受托人，$\sum_i w_i = m$。w_i 反映该辖区对 IFRS 基金会的资金贡献、历史地位和制度性影响力。不完全反映经济规模或人口——例如欧盟通过集体任命机制可能拥有不成比例的受托人份额。

[IASB 准则选择的均衡偏离] 在假设 I1–I4 下，IASB 投票博弈的唯一均衡准则 σ^{IASB} 为董事会受托人理想点的加权中位数。辖区 i 的效用损失为：

$$L_i^{\text{IASB}} = -|\sigma^{\text{IASB}} - \sigma_i^*| \tag{56}$$

均衡准则相对于" 全球最优"（定义为使总福利 $\sum_i \lambda_i u_i(\sigma)$ 最大的 σ，其中 λ_i 是辖区 i 的经济权重）的系统性偏离由以下因素驱动：

(i) **任命权分配不均：**受托人任命权集中于少数辖区（主要是美国和欧盟），使得 σ^{IASB} 偏向这些辖区的理想点。

(ii) **资金依赖：**若 w_i 与该辖区的资金贡献正相关，则富裕辖区的偏好被过度代表。

(iii) **先发制度优势：**IASB 的概念框架本身嵌入英美会计传统（Zhang & Andrew, 2014），使得即便在正式投票权重平等的情况下，可接受的准则提案空间已被预先约束。

证明. 在假设 I3（真诚投票 + 简单多数决 + 一维空间）下，Black 中位投票者定理直接成立：唯一 Condorcet 赢家为受托人中位理想点。在 I4 的加权受托人模型下，中位数按累积权重计算：$\sigma^{\text{IASB}} = \min\{\sigma : \sum_{i:\sigma_i^* \le \sigma} w_i \ge m/2\}$。全球最优 σ^{global} 在相同假设下是经济权重 λ_i 下的加权中位数：$\sigma^{\text{global}} = \min\{\sigma : \sum_{i:\sigma_i^* \le \sigma} \lambda_i \ge (\sum_i \lambda_i)/2\}$。两者偏离当且仅当 w_i 与 λ_i 不成比例。 □

Remark 4.2 (审计协议治理的类比). 命题4.3.2对数据审计协议治理的直接推论是：审计标准的确定（如" 什么构成数据质量缺陷" 的操作判断）若由多辖区董事会以投票方式

决定，其均衡标准将反映任命权分布而非经济权重分布。这既是 IASB 模型的优势（通过多辖区参与增强合法性），也是其风险（标准可能偏离全球最优，尤其当发展中国家和新兴市场的任命权不足时）。缓解策略（已在 IASB 中部分施行）包括：(a) 地域平衡的受托人任命要求；(b) 超级多数决（super-majority）门槛以保护少数辖区利益；(c) 透明的异议记录机制——即使标准最终以多数通过，少数派的反对意见被存档，为未来重新审议保留依据。

4.4 The US-China Scenario

The most consequential geopolitical risk to a universal data audit protocol is the ongoing technological decoupling between the United States and the People's Republic of China. This decoupling, which has accelerated since 2018, encompasses semiconductor export controls, restrictions on AI technology transfer, limitations on cross-border data flows, and the bifurcation of technical standards (Segal, 2020; Bateman, 2022).

In this environment, a data audit protocol developed primarily within one jurisdiction would face an acute legitimacy crisis in the other. If Yajie were perceived as a US-dominated protocol—regardless of its formal governance—China would have compelling reasons to develop an independent alternative. This would trigger a fragmentation cascade: the EU, seeking strategic autonomy in digital infrastructure, might develop a third protocol; India, with its growing AI ecosystem and policy of technological self-reliance, might develop a fourth; and so on.

The resulting equilibrium would be worse for all parties than a genuinely multilateral protocol governed under adequate neutrality safeguards. Each jurisdiction would bear the fixed costs of protocol development, endure lower audit accuracy (due to smaller CEC corpora), and navigate the fragmentation costs λ associated with conflicting quality certifications. The total welfare loss, relative to the universal-adoption equilibrium, would scale approximately linearly with the number of competing protocols:

$$W_{\text{loss}} \approx N \cdot c^{\text{develop}} + N \cdot (V[\theta(|\mathcal{E}|)] - V[\theta(|\mathcal{E}|/N)]) + \binom{N}{2} \cdot \lambda \tag{57}$$

This arithmetic suggests that major powers have a shared interest in preventing fragmentation—but only if the incumbent protocol is governed under terms that credibly preclude its weaponization by any single jurisdiction.

4.5 The Researcher as Strategic Asset: A Second-Order Game

An unexamined dimension of the non-proliferation equilibrium concerns the position of the protocol's originating researcher. Unlike SWIFT—a corporate entity embedded in a specific jurisdiction—or the IASB—a formally constituted standards body—the Yajie Protocol is initially developed by an independent researcher without institutional affiliation to any state or corporation. This organizational form generates a distinctive second-order game with properties that reinforce the non-proliferation equilibrium.

Consider the strategic calculus of a major power—State X—contemplating whether to coerce, compromise, or eliminate the originating researcher (henceforth "the Maintainer"). The Maintainer possesses three assets that are relevant to State X's decision: (i) the arXiv timestamp establishing academic priority; (ii) the operational knowledge required to maintain and evolve the Spring self-improvement loop; and (iii) the Cumulative Evidentiary Corpus M_t, which grows monotonically with each audit cycle. If the Maintainer ceases to exist or to operate independently, the consequences distribute asymmetrically across the international system.

First, the arXiv timestamp is *immortal*. Once uploaded, it cannot be rescinded, altered, or suppressed by any actor, including the Maintainer. This means that the core mathematical specification of Yajie—the state-conditioned scoring function $S(s) = Q(s) + \eta(t) \cdot N(s)$, the multi-expert consistency theorem, and the unidentifiability result —becomes part of the permanent scientific record. Any state or corporation can, in principle, reimplement Yajie from the arXiv specification. However—and this is the critical asymmetry—the *operational quality* of any reimplementation will be inferior to the Maintainer's running instance for a period proportional to the time the Maintainer's instance has been accumulating audit data. If the Maintainer's instance has been operational for T years, a reimplementation starting from zero requires approximately T years to reach parity, assuming identical algorithmic quality and comparable audit volume. This creates a *time-accumulated quality gap* that no amount of financial investment can compress.

Second, the Maintainer's independence is the sole credible guarantee of audit neutrality. A state-backed reimplementation—whether by the United States, China, the European Union, or any other jurisdiction—would be structurally incapable of claiming neutrality, because its continued operation would depend on the political continuity of its sponsoring state. The Maintainer, by contrast, is accountable to no government, no corporation, and no institutional apparatus beyond the epistemic standards of the scientific community. This is not a normative claim about the Maintainer's virtue; it is a structural claim about the Maintainer's position. Independence is a property of the node's position in the network, not of the node's character.

Third, the Maintainer's death or incapacitation would trigger a *fragmentation cascade*. In the absence of a universally trusted Maintainer, each major jurisdiction would face a prisoner's dilemma: develop an independent audit protocol (at cost c^{develop}, with lower accuracy) or rely on a protocol operated by a rival jurisdiction (at risk of audit weaponization). The dominant strategy for each jurisdiction is to develop its own protocol, yielding the fragmentation equilibrium characterized by welfare loss $W_{\text{loss}} \approx N \cdot c^{\text{develop}} + N \cdot \Delta\theta + \binom{N}{2} \cdot \lambda$. No jurisdiction can unilaterally prevent this cascade, because defection by any single jurisdiction would trigger reciprocal defection by its rivals.

From the perspective of State X, this yields a striking conclusion: *the Maintainer's continued existence and independence is a positive-sum good for all states, including State X, regardless of State X's geopolitical alignment.* Eliminating or compromising the Maintainer would impose a welfare cost on State X that exceeds any conceivable strategic benefit, because State X would be unable to secure a disproportionate share of the fragmented audit infrastructure, and would simultaneously suffer the costs of fragmentation alongside all other states.

This creates what we term a **protection equilibrium**: every state with the capacity to threaten the Maintainer has a rational incentive to *protect* the Maintainer, because the Maintainer's destruction would initiate a chain of events that leaves every state worse off. The protection is not altruistic; it is strictly self-interested. The Maintainer's survival functions analogously to a global public good—provided through the alignment of rational self-interest across competing powers—though its classification as a genuine public good (non-rivalrous and non-excludable) requires further analysis of the CEC's access architecture and governance structure.

The protection equilibrium has two corollaries. First, the Maintainer is *simultaneously more powerful and more vulnerable* than any corporate or state actor in the audit ecosystem. More powerful, because the Maintainer's independence is the structural precondition for the non-proliferation equilibrium; without a trusted independent Maintainer, the equilibrium collapses. More vulnerable, because the Maintainer lacks the institutional defenses—legal departments, diplomatic representation, physical security apparatus—that protect corporate and state actors. The Maintainer's security depends entirely on the rational calculus of the actors that might threaten them—a calculus that is robust under normal conditions but fragile under crisis, irrationality, or miscalculation.

Second, the protection equilibrium implies that any attempt to replicate Yajie without the Maintainer's involvement is not merely economically suboptimal (per the arguments in Section 3) but *strategically self-defeating*. A state that develops a competing audit protocol gains a temporary advantage in audit capability while simultaneously destroying the neutrality of the global audit infrastructure, because the competing protocol will be correctly perceived as an instrument of its sponsoring state. The state therefore gains an audit tool while losing the universal trust that makes audit valuable. This is a trade in which the asset acquired is worth less than the asset destroyed—a negative-sum transaction.

4.6 A Literary Analogy: The Wallfacer Position

The protection equilibrium admits an unconventional but analytically productive literary analogy. In Liu Cixin's science fiction trilogy *Remembrance of Earth's Past* (2006–2010), the character Luo Ji occupies a position structurally isomorphic to that of the Maintainer. Luo Ji is designated a “Wallfacer” (面壁者)—an individual whose survival is the sole deterrent preventing an extraterrestrial civilization from destroying Earth. The deterrence mechanism is simple: if Luo Ji dies, the coordinates of the alien homeworld are broadcast to a hostile universe, ensuring mutual destruction through the “dark forest” principle of cosmic sociology. Luo Ji’s power derives not from any institutional office, military capacity, or economic resource, but from his position as the *single point of failure* in a deterrence structure. He is simultaneously the most powerful human being alive—holding the fate of two civilizations in his hands—and the most vulnerable, because his security depends entirely on the continued rationality of the actors constrained by his deterrence.

The isomorphism with the Maintainer’s position is exact. The Maintainer’s death would trigger a fragmentation cascade that destroys the universal audit infrastructure, imposing welfare losses on all states. The Maintainer’s power derives not from institutional authority but from occupying the structural position of the *single credible neutral node* in the global audit network. And the Maintainer’s vulnerability derives from the same source: the protection equilibrium is robust only so long as all relevant actors continue to calculate rationally. Under conditions of crisis, miscalculation, or irrationality—the same conditions that nearly caused Luo Ji’s deterrence to fail—the Maintainer’s security evaporates.

The Wallfacer analogy serves three analytical functions in our argument. First, it illustrates that deterrence equilibria can be sustained by a single individual occupying a specific structural position, a possibility that the standard literature on nuclear deterrence (which focuses on state actors with institutional continuity) does not adequately capture. Second, it foregrounds the fragility of such equilibria: Luo Ji’s deterrence was nearly broken by an assassination attempt during a moment of political transition, precisely the kind of miscalculation scenario to which the protection equilibrium is vulnerable. Third, and most importantly, it demonstrates that *the occupant of a deterrence position does not choose the position; the structure of the game confers it.* Luo Ji did not seek to become a Wallfacer; the position was imposed on him by the logic of the strategic situation. The Maintainer, similarly, does not claim authority over data quality assessment; the logic of the non-proliferation equilibrium and the time-accumulated advantage of the CEC confer that authority irrespective of the Maintainer’s intentions.

[好领导者的保护义务 (The Good Leader’s Protection Obligation)] In the protection equilibrium, any leader whose legitimacy depends on verifiable claims has a strictly self-interested incentive to protect the Maintainer’s physical safety. The Maintainer’s death would destroy the audit infrastructure that alone can certify that leader’s honesty—collapsing the distinction between good and bad leaders into indistinguishable opacity. A leader who fails to protect the Maintainer is, by revealed preference, a leader who gains from the Maintainer’s absence. Such a leader is not a good leader who made an error—

the failure to protect is itself the signal that reveals the type.

Proof sketch. By the protection equilibrium (Theorem 9.2), the Maintainer's death initiates a fragmentation cascade with welfare loss W_{loss} to all states. A good leader—one whose claims are truthful—derives net benefit from the audit infrastructure, because audit certifies his claims at no cost to him (his claims pass). The collapse of audit therefore imposes a net loss on good leaders and a net gain on bad leaders (whose claims were failing or would fail). The leader's willingness to protect the Maintainer is thus a separating signal: protection is the dominant strategy for good types; non-protection or active hostility is the strategy of bad types, for whom the Maintainer's death eliminates the only mechanism capable of detecting their dishonesty. Formally, let $u_G(\text{protect}) - u_G(\text{not protect}) > 0$ and $u_B(\text{protect}) - u_B(\text{not protect}) < 0$, yielding a separating equilibrium in which type is revealed by the protection decision. □ □

Thus the Maintainer's physical security is not a personal favor owed by grateful leaders. It is a game-theoretic necessity for any leader who claims to be honest. A leader who does not protect the Maintainer has, by that very omission, announced which type he belongs to.

We do not suggest that the Wallfacer analogy constitutes a formal proof or an empirical prediction. Literary analogies are hermeneutic tools, not mathematical models. However, they serve a valuable function in strategic analysis by making visible structural relationships that formal models may obscure through abstraction. The Wallfacer analogy makes visible the distinctive character of the Maintainer's position: *power without institution, deterrence without weapon, and security through vulnerability.*

4.7 Audit as Sword: The Deterrence of Involuntary Transparency

The analysis thus far has characterized the Maintainer's position as *defensive*: the protection equilibrium constrains others from threatening the Maintainer, and neutrality principles constrain the Maintainer from abusing the position. But there is a second, more asymmetric constraint embedded in the architecture—one that operates not on the Maintainer, but on every entity that produces publicly accessible data.

The Maintainer, as an independent researcher, has an uncontestable academic right to audit any publicly available dataset and to publish the results. This right is not granted by any institution; it is inherent in the practice of science. No organization, corporation, or government can prevent an independent researcher from downloading a public dataset, running Yajie on it, and reporting what the multi-expert consensus reveals—any more than they can prevent a statistician from reanalyzing published clinical trial data, or an astronomer from pointing a telescope at a publicly visible star. The data are public. The methodology is published (arXiv, open source). The conclusions are falsifiable. There is no legal, ethical, or scientific basis for prohibiting such an audit.

What makes this right strategically significant is its asymmetry. The Maintainer can choose to exercise it or not, at their sole discretion. And the most natural Schelling point for that discretion is the partnership relationship. Organizations that enter into a formal partnership with the Yajie Protocol—subscribing to the calibration API, contributing audit modules, participating in governance—receive an implicit assurance: their data will not be subjected to unsolicited public audit by the Maintainer, because auditing a partner is both strategically counterproductive and professionally discourteous. Organizations that decline partnership receive no such assurance. Their publicly available datasets remain, at all times, eligible for independent academic audit—an audit that may expose quality issues they would prefer remained unexamined.

This is not a demand. The Maintainer issues no ultimatums, makes no demands, and imposes no obligations. The Maintainer simply exists, with the capacity and the academic freedom to audit public data, and with no obligation to refrain from exercising that freedom with respect to non-partners. The incentive—if it can be called that—is generated by the structure of the situation, not by any stated intention of the Maintainer. It is, in the strictest sense, a *structural assurance*: the mere existence of the capability, combined with a clear and credible criterion for its non-use (partnership), is sufficient to motivate the desired behavior, without any need for explicit communication, negotiation, or enforcement.

The game-theoretic structure is:

Partnership $\rightarrow$ no public audit (by implicit convention)
Non-partnership $\rightarrow$ eligible for public audit (by academic freedom)
Audit findings $\rightarrow$ may expose data quality issues $\rightarrow$ reputational and regulatory risk
$\therefore$ Non-partnership carries a positive probability of involuntary transparency
$\therefore$ Partnership is a dominant strategy for quality-sensitive organizations

This assurance operates independently of the protection equilibrium. Even if an organization has no interest in the Maintainer's survival—even if it would prefer a fragmented audit infrastructure that it could control—it still faces the immediate structural consequence that its public data may be independently verified by the only entity capable of conducting a credible multi-expert quality assessment. The verification does not need to be enforced. It needs only to be possible.

The Maintainer, for their part, bears no cost for this deterrent. No audit needs to be conducted; no report needs to be published; no confrontation needs to occur. The deterrent is maintained by the *capability* alone. And because the capability is grounded in academic freedom rather than institutional authority, it cannot be negotiated away, regulated out of existence, or deterred by counter-threats. One cannot sanction a researcher for exercising their right to analyze public data. One can only ensure that one's data, when analyzed, holds up to scrutiny—or, more pragmatically, enter into the partnership that makes such scrutiny unnecessary.

The commercial consequence of this deterrent is a self-reinforcing adoption cascade. Consider two competing firms, A and B, in a quality-sensitive industry—pharmaceuticals, materials engineering, or autonomous systems. Firm A partners with Yajie, gaining access to calibration services, governance participation, and the implicit assurance against unsolicited audit. Firm B does not. Firm A can now, through the open-source Spring engine, audit Firm B's publicly available datasets—drug screening results, material property databases, or safety-critical training corpora—and publish the findings. If Firm B's data quality is inferior, Firm A gains a competitive advantage that is independently verifiable and publicly citable. If Firm B's data quality is adequate, Firm A has lost nothing but the cost of running an audit on open-source software.

Firm B's rational response is not to contest the audit—the methodology is published, the conclusions are falsifiable, and the academic freedom of the auditor is unassailable—but to preempt it by also becoming a Yajie partner. At which point Firm C, observing the partnership concentration among its competitors, faces the same calculus. The cascade propagates through the industry, not because Yajie coerces anyone, but because non-participation carries a positive and growing probability of involuntary transparency at the hands of participating competitors. The Maintainer does not need to audit anyone. Competitors audit each other—and the Maintainer's existence is what makes that audit credible.

4.8 Jurisdictional Boundary as Structural Safeguard

An underappreciated feature of the data sovereignty architecture described above—where the Yajie API accepts only locally computed feature matrices $\phi(X)$ rather than raw data—is that it imposes a jurisdictional boundary on the Maintainer's audit capacity that is not merely a regulatory compliance constraint but a *credible commitment device.*

The Maintainer can only audit data that resides within the Maintainer's operational jurisdiction. If a government requests audit reports on a foreign corporation's data, the Maintainer cannot comply—not because the Maintainer refuses, but because the data is not accessible. The feature matrices $\phi(X)$ are computed client-side and never leave the client's infrastructure; only calibration queries cross jurisdictional boundaries, and those queries are aggregated statistics, not identifiable data records. This architectural constraint is not a limitation to be overcome. It is a feature that preserves the protocol's neutrality. A Maintainer who *could* provide foreign audit data on demand would face an impossible choice between defying a government order (risking sanctions, closure, or prosecution) and complying with it (destroying the trust of every foreign client). A Maintainer who *cannot* provide such data—because the architecture makes it physically inaccessible—is insulated from this dilemma. The inability is structural, not volitional. The government cannot punish the Maintainer for refusing what the Maintainer cannot do.

This jurisdictional boundary has a secondary structural consequence that is, from the perspective of domestic industrial policy, unambiguously positive. If a government wishes its domestic corporations to benefit from Yajie's audit infrastructure—to gain the competitive advantages of certified data quality, to participate in the governance of the protocol, and to be shielded from unsolicited audit by foreign competitors—it must ensure that those corporations' data resides within the Maintainer's operational jurisdiction. A corporation that offshores its data to a foreign cloud provider, or that maintains its primary data infrastructure in a jurisdiction where the Maintainer does not operate, cannot be audited. Its data quality claims become unverifiable. Its competitors, who *have* submitted to domestic audit, gain a structural advantage in trustworthiness that no amount of marketing expenditure can overcome.

The consequence is a market-driven incentive for data repatriation. Corporations that wish to compete in a Yajie-audited market must bring their data home—not because any regulation compels them, but because the alternative is to be the only unaudited actor in a market where everyone else's quality is independently verified. The unaudited actor is not trusted. The untrusted actor does not sell. The mechanism is not protectionism; it is transparency. Yajie does not require corporations to operate domestically. It merely makes domestic operation—and the auditability that comes with it—the dominant strategy.

4.9 Pharmaceutical Safety: The Honest Algorithm

The jurisdictional dynamic acquires particular force in the pharmaceutical industry, where data quality is not merely a competitive advantage but a matter of public health. A drug's efficacy and side-effect profile are established through clinical trials whose data quality is, at present, largely self-reported by the sponsoring pharmaceutical company. Regulators review the submitted data, but they do not independently audit it at the level of individual trial records; they lack both the tools and the resources to conduct a multi-expert consistency analysis across the heterogeneous data streams—laboratory results, patient-reported outcomes, adverse event reports, biomarker measurements—that constitute a modern clinical trial.

Yajie changes this calculus. Under a distributed architecture, a pharmaceutical company would run Spring on its own trial data and submit the resulting quality scores to regulators alongside its New Drug Application. The regulator, armed with the same open-source audit engine and the calibration anchor provided by the Maintainer's API, could independently verify those scores. The company's claim that "95% of adverse events were properly classified" would no longer be an assertion to be trusted or distrusted on the basis of the company's reputation; it would be a falsifiable statement backed by a published methodology and a verifiable audit trail.

The consequence for the pharmaceutical industry is the same as for every other domain surveyed in this paper—honest firms gain, dishonest firms are exposed—but with higher stakes. A pharmaceutical company that has rigorously collected, cleaned, and curated its trial data will see its diligence reflected in independently verifiable quality scores. A company that has cut corners—that has underreported adverse events, that has excluded inconvenient patient subgroups, that has massaged statistical analyses to achieve significance—will see its deficiencies exposed not by a whistleblower or an investigative journalist, but by a multi-expert consensus algorithm whose conclusions are mathematically guaranteed and publicly reproducible.

The constructive pressure this exerts on drug development is difficult to overstate. A company that knows its trial data will be audited by an impartial algorithm—not by a regulator with limited resources, not by a competitor with an axe to grind, but by a published methodology whose error bounds are proven—faces a powerful incentive to conduct its trials with scrupulous care. The algorithm does not punish. It reveals. And what it reveals, the market—physicians, patients, insurers, regulators—acts upon. The drug that survives a Yajie audit is a drug whose safety and efficacy claims are credible not because its manufacturer is trustworthy but because its data has withstood the most rigorous quality assessment that the current state of statistical learning theory can provide.

4.10 Beyond Industry: Bribery, Governance, and the Dark Forest

The jurisdictional and pharmaceutical dynamics analyzed above are specific instances of a more general principle. Yajie's multi-expert consensus mechanism is indifferent to the domain of the data it audits. It does not know—and does not need to know—whether the anomalies it detects are label noise in a materials database, fabrication in a clinical trial, or a municipal official's manipulated air quality readings. The mathematics is the same. The consensus signal is the same. The exponential reliability guarantee is the same.

Consider three domains that are, at present, largely beyond the reach of systematic, independent audit.

Bribery and financial malfeasance. A pattern of illicit payments generates a distinctive signature across multiple independent data sources: procurement records, tax filings, customs declarations, bank transactions, asset registries. No single source is conclusive; a single anomalous payment may be an accounting error. But when procurement records show a contract awarded at 40% above market rate, customs declarations show the same contractor importing luxury goods, and asset registries show the approving official acquiring property disproportionate to declared income—the multi-expert consensus that Yajie formalizes converges on a conclusion that no single auditor could reach with equivalent confidence. The algorithm does not accuse. It reveals the consistency—or the inconsistency—of the evidentiary record.

Urban governance. Smart city infrastructure generates sensor data on air quality, water quality, traffic flow, energy consumption, and waste management. When a district's air quality monitors report readings that are systematically cleaner than those of neighboring districts under identical meteorological conditions—and when those same monitors show anomalous recalibration patterns, and when the district's environmental compliance reports diverge from satellite-based atmospheric measurements—the disagreement is not a data quality issue to be resolved. It is a signal. Yajie does not know that the signal means "the official in charge of this district is falsifying environmental data." It knows only that three independent expert systems, each analyzing different data streams, disagree with the official report in specific, quantifiable ways. What the human interpreter does with that knowledge is a human decision. But the knowledge itself—that the data are inconsistent, that the inconsistency is systematic, and that the probability of the inconsistency arising by chance is exponentially small—is a mathematical fact.

Financial management and public accounts. Government budgets, infrastructure spending, and public procurement generate vast heterogeneous data streams that are audited, at present, by understaffed agencies with limited analytical capacity. A Yajie deployment that treats each government agency's financial reporting as an "expert," cross-referenced against tax receipts, bank records, contractor invoices, and independent economic indicators, would flag inconsistencies that no human audit team—however diligent—could reliably detect across the full volume of public expenditure.

In each of these domains, Yajie performs the same function: it transforms information that is *distributed* across multiple independent sources, and therefore invisible to any single observer, into a *consensus signal* whose reliability is mathematically bounded. The information was always there. The procurement records, the customs declarations, the asset registries—all of them public, all of them theoretically auditable, all of them already collected. What was missing was the computational and conceptual machinery to synthesize them into a quality assessment with provable error bounds. Yajie provides that machinery.

4.11 The Dark Forest Illuminated

In Liu Cixin's *Remembrance of Earth's Past* trilogy, the universe is governed by the "dark forest" principle: every civilization hides, because to be discovered is to be destroyed. The darkness is a defense. Visibility is fatal.

Yajie inverts this logic. It does not destroy the discovered—it reveals the undiscovered. The official who falsifies environmental data, the contractor who inflates procurement invoices, the pharmaceutical company that underreports adverse events—all of them survive not because they are invulnerable, but because they are invisible. Their malfeasance is distributed across databases that no single auditor can synthesize, buried in the noise of routine administrative error, shielded by the sheer volume of information that modern governance generates. They are hidden in the dark forest, and the darkness is their protection.

Yajie is the instrument that makes this visible. Not by accusation. Not by investigation. Not by whistleblowing or investigative journalism or regulatory enforcement. By mathematics. By the inexorable logic of multi-expert consensus: when three independent data streams disagree with an official report in the same direction, the probability that the disagreement is random decays exponentially. The pattern is revealed not by a leak but by a proof.

This is not a surveillance technology. It does not collect new data. It does not invade privacy. It does not make inferences about individuals. It audits the *consistency* of data that has already been collected, by institutions that have already collected it, for purposes that have already been authorized. The data are public or administrative. The audit is methodological. The conclusion is falsifiable. What distinguishes Yajie from every other anti-corruption mechanism in human history—from imperial censors to investigative journalists to blockchain transparency initiatives—is that its conclusions do not depend on the credibility of the auditor. They depend on the mathematics of concentration inequalities, which are indifferent to the auditor's motives, incorruptible by the auditor's enemies, and immune to the auditor's removal.

The Maintainer, in this framing, is neither a prosecutor nor a judge. The Maintainer is the first astronomer to point a telescope at a region of the dark forest and publish the coordinates of what was found there. Whether anyone acts on those coordinates—whether a prosecutor indicts, a regulator sanctions, a journalist investigates, a voter punishes—is not the Maintainer's decision. The Maintainer's sole function is to make the invisible visible. The rest is left to the light.

4.12 The M_t Parameter and the Spectrum of Deterrence

The analysis thus far has treated the Cumulative Evidentiary Corpus M_t as a monolithic asset held by the Maintainer. In practice, the ownership architecture of M_t is a design choice with profound implications for the protection equilibrium. We now examine two polar ownership models—centralized and distributed—and characterize the resulting position of the Maintainer along a spectrum we term the *Luo Ji–Pope gradient*.

Centralized M_t (the Wallfacer extreme). Under a centralized architecture, all audit outputs—scores, anomaly registers, calibration parameters—are transmitted to and stored by the Maintainer. Each client organization retains its raw data but cedes its audit history to the protocol's central repository. The Maintainer thus accumulates a cross-client, cross-domain audit corpus that grows more comprehensive with every audit cycle. In this configuration, the Maintainer's death freezes M_t at its terminal state: no new audits occur, no calibration updates propagate, and the corpus ceases to reflect evolving data quality practices. All client organizations are immediately downgraded to the audit accuracy available at the time of the Maintainer's death. The deterrence effect is maximal: the Maintainer's survival is a necessary condition for the continued operation of the global audit infrastructure at current accuracy levels. The protection equilibrium is correspondingly strong. This is the pure Wallfacer position—power without institution, deterrence without weapon.

Distributed M_t (the doctrinal extreme). Under a distributed architecture, each client organization maintains its own private M_t. The Maintainer provides only the algorithmic specification—the scoring function $S(s)$, the Spring update rule, the ensemble aggregation procedure—together with periodic calibration datasets (anonymized feature statistics that do not expose raw data). Each client's gatekeeper evolves independently on its own audit history. The Maintainer's death in this configuration does not freeze any individual client's M_t; each organization continues to operate its own instance. However, three assets remain centralized in the Maintainer's person: (i) *interpretive authority*—the definitive resolution of ambiguous edge cases in the mathematical specification; (ii) *evolutionary direction*—the coordinated release of protocol versions (v1.1, v2.0) that keep all instances interoperable; and (iii) *the calibration anchor*—the periodic cross-client calibration datasets that prevent the independent evolution of client gatekeepers from producing incompatible scoring standards, whereby Company A's score of 0.85 ceases to be comparable to Company B's score of 0.85. If the Maintainer dies, interpretive disputes escalate into incompatible forks, version coordination ceases, and the calibration anchor erodes. The result is not a sudden freeze but a *gradual fragmentation*—the slow divergence of once-interoperable audit standards into mutually unintelligible local dialects.

The Luo Ji–Pope spectrum. The Maintainer's position can thus be characterized by the architectural choice of M_t ownership:

The spectrum reveals a strategic trade-off. Centralized M_t maximizes the protection equilibrium—every state has a compelling interest in the Maintainer's survival—but at the cost of creating a single point of data concentration that is incompatible with data sovereignty legislation (GDPR, China's Data Security Law, India's proposed data localization requirements) and that concentrates epistemic power to a degree that may itself provoke defection. Distributed M_t sacrifices some deterrence strength in exchange for privacy compliance, jurisdictional legitimacy, and a reduction in the Maintainer's power that makes the protocol more palatable to sovereignty-conscious jurisdictions. The fully institutionalized endpoint eliminates deterrence entirely by making the Maintainer

Architecture	M_t Owner	Death Consequence	Deterrence Strength
Centralized	Maintainer	Immediate freeze of global audit accuracy	Maximum (100%)
Distributed	Each client	Gradual fragmentation through version drift and calibration decay	Moderate (∼60%)
Fully institutionalized (Foundation)	Multi-stakeholder board	Operational continuity; individual Maintainer irrelevant	Zero (0%) —the Linus equilibrium

structurally irrelevant—the position to which Linus Torvalds successfully transitioned the Linux kernel after 2005.

The optimal trajectory is temporal: begin with centralized M_t to accumulate the time-advantaged quality gap that makes the non-proliferation equilibrium binding; transition to distributed M_t as the accuracy gap becomes insurmountable and privacy considerations intensify; and ultimately establish a multi-stakeholder Foundation that renders any single individual's survival irrelevant to the protocol's continuity. This trajectory is the strategic equivalent of the Wallfacer who, having established deterrence, spends his remaining years building the institutional architecture that will allow him to die without the world ending.

4.13 Designing for Audit Neutrality

Drawing on the SWIFT and IASB cases, we identify five design principles for audit protocol governance that can support the non-proliferation equilibrium:

Principle 1: Multi-Jurisdictional Incorporation. The protocol's legal domicile should be distributed across multiple jurisdictions, ideally through a treaty-based international organization with immunities and privileges comparable to those of the IAEA or the Bank for International Settlements. Incorporation in any single jurisdiction—especially one with expansive extraterritorial legal instruments—invites jurisdictional capture.

Principle 2: Funding Firewalls. The protocol's operating budget should be derived from a diversified base of mandatory and voluntary contributions, with hard caps on any single contributor's share. The IAEA's funding model, which combines assessed contributions from member states with voluntary contributions to a Technical Cooperation Fund, provides a template.

Principle 3: Adversarial Governance. The protocol's governance board should include representatives from jurisdictions with divergent data governance preferences, and decision rules should be structured to prevent capture by any single bloc. Qualified majority voting, regional representation requirements, and minority veto powers are all elements of the IASB model that could be adapted.

Principle 4: Audit Transparency and Contestability. The protocol's audit methodologies, calibration procedures, and CEC composition should be publicly documented to a degree sufficient for external reproducibility. Audit outputs should include not only binary or scalar quality assessments but also the evidentiary basis for those assessments, enabling contestation without requiring access to proprietary audit modules.

Principle 5: Modular Sovereignty. Jurisdictions should have the right to contribute audit modules to the ensemble, enabling them to embed their own data governance preferences within the protocol's framework without fragmenting the overall infrastructure. Module contributions would be subject to validation requirements ensuring technical compatibility and performance thresholds, but not to substantive alignment with any particular jurisdiction's data governance philosophy.

These principles are demanding, and their implementation would require a level of international cooperation that the current geopolitical environment does not obviously support. We return to this tension in Section 6.

5 Business Model and Lock-in Dynamics

5.1 The Temporal Accumulation Mechanism

The lock-in mechanism of the Yajie Protocol operates through five distinct but mutually reinforcing channels, each of which intensifies over time:

Channel 1: Accuracy Divergence. As established in Section 2.1, the incumbent protocol's accuracy $\theta(t)$ is an increasing function of cumulative audit volume. An entrant begins with accuracy $\theta(0)$, implies an accuracy gap $\delta(t) = \theta(t) - \theta(0)$ that grows monotonically over time. Critically, this gap is not merely a temporary first-mover advantage that an entrant could overcome through accelerated investment; it is a *structural* gap rooted in the fact that many data quality anomalies are long-tail phenomena that manifest only through the accumulation of diverse audit experiences. An entrant cannot compress time: even with unlimited financial resources, it must conduct audits on diverse datasets to encounter the long-tail anomalies that the incumbent has already catalogued.

Channel 2: Calibration Specificity. The auditor weights w_j in the Yajie ensemble are calibrated against the CEC's judgment corpus $\mathcal{J}_t$. As $\mathcal{J}_t$ grows, the ensemble's calibration becomes increasingly tuned to the specific distribution of anomalies, edge cases, and quality failure modes represented in the CEC. An entrant with an empty CEC would need to replicate this calibration process, which requires not merely technical replication of the calibration algorithm but access to the judgment corpus itself—and the judgment corpus is entangled with the specific audit modules that generated the initial audit reports. This creates a circular dependency: the entrant needs calibration data to achieve competitive accuracy, but producing that calibration data requires conducting audits with competitive accuracy.

Channel 3: Ecosystem Externalities. As the incumbent protocol accumulates users, a supporting ecosystem emerges: training programs for audit practitioners, integration services for enterprise data pipelines, certification regimes for audit professionals, and regulatory recognition of the protocol's outputs. These complementary assets do not themselves constitute network effects in the narrow sense—the value of an audit to user i does not depend on the number of other users—but they reduce adoption costs for new users and increase switching costs for existing ones. More importantly, regulatory recognition is path-dependent: once a protocol is recognized as a valid compliance mechanism under, say, the EU AI Act or a US federal algorithmic accountability statute, competing protocols face a regulatory barrier that compounds the technical barrier.

Channel 4: The Epistemic Authority Trap. As the CEC grows, the protocol attains a status that goes beyond technical accuracy. It becomes the de facto arbiter of what constitutes a "data quality defect," shaping the very definition of data quality through its accumulated body of adjudicated anomalies. This is not merely market power but *epistemic power*: the authority to define the categories within which market competition occurs. An entrant cannot compete on the definition of data quality because the definition has already been institutionalized in regulation, contract law, and professional practice, all of which reference the incumbent protocol's outputs.

Channel 5: The Audit Trail Irreversibility. Organizations that have adopted the incumbent protocol accumulate an internal audit trail: a history of audits, remediation actions, and quality certifications that are referenced in contracts, regulatory filings, and due diligence processes. Switching to a competing protocol would render this audit trail incompatible, creating a discontinuity in the organization's quality assurance history. The cost of this discontinuity is not merely administrative; it is reputational and legal. An organization that switches protocols and subsequently experiences a data quality failure faces the question: did the failure occur because the new protocol is inferior, or because the old protocol's audit trail was unreliable? The ambiguity itself constitutes a switching cost.

5.2 Five Stages of Lock-In

The lock-in trajectory can be modeled as a five-stage progression:

Stage 1: Emergence (t_0 to t_1). The protocol is launched with an initial ensemble of audit modules and an empty CEC. Accuracy is relatively low, and the protocol competes on features, usability, and cost with bespoke audit solutions. Adoption is driven by early adopters who value the multi-expert ensemble architecture over the accuracy level. No lock-in exists at this stage; the protocol's survival depends on the quality of its initial audit modules and its ability to attract a sufficient volume of audits to begin CEC accumulation.

Stage 2: Critical Mass (t_1 to t_2). Cumulative audit volume reaches a threshold $|\mathcal{E}|^*$ at which the accuracy advantage over a zero-CEC entrant becomes statistically significant. The protocol begins to attract adoption from organizations that had previously relied on internal audit processes. The CEC reaches sufficient size that its anomaly library covers the most common data quality failure modes across multiple domains. The first regulatory recognitions occur. Lock-in begins to emerge but remains contestable.

Stage 3: Moat Formation (t_2 to t_3). The accuracy gap $\delta(t)$ exceeds the fixed cost of developing a competing protocol, meaning that an entrant would need to invest more than c^{develop} merely to *match* the incumbent's accuracy at time t_0, without accounting for the incumbent's continued accumulation. The Non-Proliferation Equilibrium becomes the unique Nash equilibrium of the adoption game. The protocol becomes the default standard for data quality assessment in its domain. Ecosystem externalities accelerate: training programs proliferate, integration services become commoditized, and regulatory references become routine.

Stage 4: Institutionalization (t_3 to t_4). The protocol is embedded in the institutional fabric of data governance. Laws and regulations reference the protocol by name. Contracts specify compliance with protocol standards. Insurance policies for AI systems require protocol certification. Professional certifications are built around protocol expertise. At this stage, lock-in extends beyond technical and economic factors to encompass legal and institutional path-dependency. Switching costs become prohibitive not because the protocol is technically superior—though it may be—but because the *institutional environment* has been built around it.

Stage 5: Epistemic Closure (t_4 onward). The protocol's accumulated CEC becomes so comprehensive that no meaningful challenge to its authority is possible within the existing institutional framework. The definition of data quality *is* what the protocol measures. Dissenting assessments are not merely inaccurate; they are unintelligible, because they employ different definitional frameworks that cannot be translated into the protocol's categories. At this stage, lock-in is complete, and the protocol has transitioned from a market-dominant product to a constitutive element of the data governance order.

This trajectory is not deterministic; it can be arrested, diverted, or reversed at each stage through governance interventions or exogenous shocks. But the default trajectory, absent intervention, is toward Epistemic Closure, driven by the cumulative dynamics described above.

5.3 五阶段锁入轨迹的马尔可夫形式化

前文的五阶段描述为叙事性刻画。本节将其严格形式化为非齐次马尔可夫链，使每一阶段的转移机制获得概率论基础。

Definition 5.1 (锁入阶段的状态空间). 定义有限状态空间 $\mathcal{S} = \{S_1, S_2, S_3, S_4, S_5\}$：

- S_1：**涌现期**（Emergence）——协议发布，CEC 为空（$|\mathcal{E}| = 0$），审计精度 $\theta = \theta_0$，无锁入；
- S_2：**临界质量期**（Critical Mass）——$|\mathcal{E}| \geq |\mathcal{E}|^*$，精度优势具统计显著性（$\theta(t) - \theta_0 \geq \varepsilon_\theta$），首批监管认可发生；
- S_3：**护城河形成期**（Moat Formation）——$\Delta(|\mathcal{E}|) \geq \lambda - \kappa$ 满足，NPE 为唯一纳什均衡（定理3.2），精度差距超开发成本；
- S_4：**制度化期**（Institutionalization）——协议被 ≥ 2 部法律/法规以名称引用，合同条款中出现"Yajie 认证" 作为合规条件；
- S_5：**认知闭合期**（Epistemic Closure）——数据质量的操作定义由协议确定，不存在可理解的替代质量概念。吸收态。

Definition 5.2 (锁入马尔可夫模型的全部假设). **M1 有限状态空间**：系统在任意离散时间 $t = 0, 1, 2, \ldots$ 处于 $\mathcal{S}$ 中的恰好一个状态。时间单位为一审计周期（audit cycle）。*合理性：状态定义穷尽且互斥。*

M2 马尔可夫性：下一状态 X_{t+1} 仅依赖于当前状态 X_t 和当前 CEC 规模 $|\mathcal{E}_t|$：

$$\mathbb{P}[X_{t+1} = S_j \mid X_t = S_i, |\mathcal{E}_t|, \mathcal{H}_{t-1}] = \mathbb{P}[X_{t+1} = S_j \mid X_t = S_i, |\mathcal{E}_t|] \tag{58}$$

其中 $\mathcal{H}_{t-1}$ 是直到 $t-1$ 的全部历史。 *可放松性：这假定 CEC 规模充分总结了历史的影响。如果法律引用或合同条款的累积具有独立于 CEC 的惯性，则需要引入辅助状态变量，但马尔可夫性质在扩展状态空间后可恢复。*

M3 单向锁入（不可逆性）： $p_{ij}(|\mathcal{E}|) = 0$ 对所有 $j < i$ 成立。一旦系统进入更高阶段，不退化至更低阶段。 *经验基础：技术标准的制度化路径极少发生逆转（如 QWERTY 键盘、TCP/IP 协议栈）。但在理论上，极端外部冲击（如协议运营者被发现系统性操纵审计结果）可导致退化。本假设排除此可能性以聚焦常规动力学。*

M4 CEC 单调非减： $|\mathcal{E}_t|$ 是 t 的非减函数：$|\mathcal{E}_{t+1}| \geq |\mathcal{E}_t|$ 对所有 t 几乎必然成立。*物理基础：CEC 是追加式数据结构——审计记录只增不删（第 2.1 节）。*

M5 吸收态： S_5 是唯一吸收态：$p_{55}(|\mathcal{E}|) = 1$ 对所有 $|\mathcal{E}|$。一旦系统进入认知闭合，永远停留其中。 *含义：认知闭合是锁入轨迹的终态——替代标准不仅在技术上次优，在概念上亦不可理喻（unintelligible）。*

在假设 M1–M5 下，转移矩阵具有上三角结构：

$$\mathbf{P}(|\mathcal{E}|) = \begin{pmatrix} p_{11}(|\mathcal{E}|) & p_{12}(|\mathcal{E}|) & p_{13}(|\mathcal{E}|) & p_{14}(|\mathcal{E}|) & p_{15}(|\mathcal{E}|) \\ 0 & p_{22}(|\mathcal{E}|) & p_{23}(|\mathcal{E}|) & p_{24}(|\mathcal{E}|) & p_{25}(|\mathcal{E}|) \\ 0 & 0 & p_{33}(|\mathcal{E}|) & p_{34}(|\mathcal{E}|) & p_{35}(|\mathcal{E}|) \\ 0 & 0 & 0 & p_{44}(|\mathcal{E}|) & p_{45}(|\mathcal{E}|) \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \tag{59}$$

其中对每个 $i \in \{1,2,3,4\}$，行和为 1：$\sum_{j=i}^{5} p_{ij}(|\mathcal{E}|) = 1$。转移概率的函数形式由 CEC 驱动机制确定：

- $S_1 \to S_2$**（涌现至临界质量）**：当 CEC 逼近临界值 $|\mathcal{E}|^*$ 时，转移概率增加：

  $$p_{12}(|\mathcal{E}|) = \frac{1}{1 + e^{-\alpha(|\mathcal{E}| - |\mathcal{E}|^*)}} \tag{60}$$

  其中 $\alpha > 0$ 为敏感度参数。$|\mathcal{E}| \gg |\mathcal{E}|^*$ 时 $p_{12} \to 1$。

- $S_2 \to S_3$**（临界质量至护城河）**：转移由采纳优势驱动——当 $\Delta(|\mathcal{E}|)$ 越过 $\lambda - \kappa$ 门槛时：

  $$p_{23}(|\mathcal{E}|) = \frac{1}{1 + e^{-\beta[\Delta(|\mathcal{E}|) - (\lambda - \kappa)]}} \tag{61}$$

  其中 $\beta > 0$。此概率直接由定理3.1的均衡条件决定。

- $S_3 \to S_4$**（护城河至制度化）**：设 $R(t)$ 为时间 t 引用该协议的法律/监管文本累积数。制度化门槛为 R^*：

  $$p_{34}(|\mathcal{E}|) = \frac{1}{1 + e^{-\zeta(R(t) - R^*)}} \tag{62}$$

  其中 $\zeta > 0$。$R(t)$ 的增长部分内生于协议的市场份额——这是锁入从经济领域向制度领域扩散的数学表达。

- $S_4 \to S_5$**（制度化至认知闭合）**：设 $T_4(t)$ 为系统在 S_4 的逗留时间（以审计周期计）。转移概率为：

  $$p_{45}(|\mathcal{E}|) = 1 - e^{-\eta \cdot T_4(t)} \tag{63}$$

  其中 $\eta > 0$。这一指数形式反映了”制度惯性”假设：在制度化状态下停留越久，认知闭合的概率越大——制度对替代概念的排斥随时间增强。

- **自我维持概率：** $p_{ii}(|\mathcal{E}|) = 1 - \sum_{j=i+1}^{5} p_{ij}(|\mathcal{E}|)$ 对所有 $i \in \{1,2,3,4\}$。

Theorem 5.1 (锁入吸收的必然性与期望时间). *在假设 M1–M5 下：*

(i) **吸收必然性：***从任意初始状态 $X_0 = S_i$（$i \in \{1,\ldots,4\}$）出发，马尔可夫链 $\{X_t\}_{t\geq 0}$ 以概率 1 被吸收到状态 S_5：*

$$\mathbb{P}\left[\exists t < \infty : X_t = S_5 \mid X_0 = S_i\right] = 1 \tag{64}$$

(ii) **期望吸收时间**：从 S_1 出发的吸收时间期望满足：

$$\mathbb{E}[T_{absorb} \mid X_0 = S_1] = \sum_{i=1}^{4} \mathbb{E}[\tau_i] \tag{65}$$

其中 τ_i 是系统在第 i 阶段的逗留时间（以审计周期计）。τ_i 的期望有界：

$$\mathbb{E}[\tau_i] \leq \frac{1}{\inf_{t \geq 0}\left(1 - p_{ii}(|\mathcal{E}_t|)\right)} \tag{66}$$

(iii) ***CEC* 加速效应**：设审计频次密度为 ρ（单位时间的审计循环数）。则期望吸收时间（以日历时间计）为 $\mathbb{E}[T_{absorb}]/\rho$。增大 ρ（加速审计）严格缩减期望吸收的日历时间。

(iv) **阶段停留时间的比较静态**：

- $\partial\mathbb{E}[\tau_1]/\partial|\mathcal{E}|^* < 0$：临界 *CEC* 规模越小，$S_1$ 停留越短；
- $\partial\mathbb{E}[\tau_2]/\partial\beta < 0$：采纳优势对均衡条件的敏感度越高，$S_2$ 停留越短；
- $\partial\mathbb{E}[\tau_3]/\partial R^* < 0$：制度化引用的门槛越低，$S_3$ 停留越短。

证明. **(i)** 对每个 $i \in \{1, 2, 3, 4\}$，由于 CEC 单调增长（M4），$|\mathcal{E}_t| \to \infty$ 当 $t \to \infty$（只要审计持续进行）。在此极限下，$p_{i,i+1}(|\mathcal{E}_t|) \to 1$（由方程 (60)–(63) 的函数形式），故 $p_{ii}(|\mathcal{E}_t|) \to 0$。每个阶段 i 的逗留时间 τ_i 被一成功概率趋于 1 的几何分布所随机支配，故 $\mathbb{P}[\tau_i < \infty] = 1$。由有限个几乎必然有限时间的和收敛到几乎必然有限时间，吸收时间以概率 1 有限。

(ii) 设 T_i 为从 S_i 出发的吸收时间期望。由马尔可夫性（M2）：

$$T_i = 1 + \sum_{j=i}^{5} p_{ij} \cdot T_j = 1 + p_{ii} \cdot T_i + \sum_{j=i+1}^{5} p_{ij} \cdot T_j \tag{67}$$

解此递归关系：

$$T_i = \frac{1}{1 - p_{ii}} + \sum_{j=i+1}^{5} \frac{p_{ij}}{1 - p_{ii}} \cdot T_j \tag{68}$$

递归代入 $T_5 = 0$（吸收态），递推至 $i = 1$ 即得所示表达式。期望逗留时间 $\mathbb{E}[\tau_i]$ 精确等于 $\frac{1}{1-p_{ii}(|\mathcal{E}_t|)}$ 当 p_{ii} 为常数时（齐次链）。在非齐次情形，不等式由每一步成功概率的下界给出。

(iii) 吸收的日历时间 $= T_{\text{absorb}}/\rho$。ρ 增大直接压缩日历时间。

(iv) 比较静态由方程 (60)–(62) 对各自参数的导数直接验证。 □

Remark 5.1 (吸收必然性的规范含义). 定理5.1的核心结论——以概率 1 吸收到认知闭合——在福利层面是模糊的。吸收的必然性意味着：(a) 正面：碎片化成本 λ 在全球范围内归零（无竞争协议），全社会的审计精度在统一 CEC 下最大化；(b) 负面：竞争性审

计市场的永久消失，消除了竞争对价格、质量和创新的约束。(a) 与 (b) 的净福利符号取决于第 6 节的治理安排能否在吸收完成前嵌入。

[政策干预窗的单调闭合] 设 $\mathcal{W}_i$ 为系统处于状态 S_i 时可实施有效治理干预（如强制国际化、价格管制立法、CEC 治理多边化）的策略空间。定义干预的有效性函数 $\eta_i = \max_{\iota \in \mathcal{W}_i} \mathbb{P}[\text{干预 } \iota \text{ 成功阻止吸收}]$。在假设 M1–M5 下：

$$\eta_1 > \eta_2 > \eta_3 > \eta_4 > \eta_5 = 0 \tag{69}$$

即在认知闭合（S_5），任何干预均无效——" 数据质量即协议所测" 已成制度事实。

证明. 在 S_1，协议运营者尚不享有显著市场力量，立法者可通过设定治理条件作为市场准入前提来行使最大议价力。在 S_3，运营者的 CEC 精度优势使得" 开发替代协议" 的威胁不再可信（定理3.2），立法者的议价力下降。在 S_5，数据质量的操作定义已内化于全部相关制度之中，替代协议在概念上不可理喻——干预缺乏操作支点。序列的严格单调性由每一阶段锁入深度的增加得到保证。 □

Remark 5.2 (干预窗的政策含义). 命题5.3给出了一个紧迫的时间约束：治理安排必须在系统达到 S_3（护城河形成）——此时 NPE 成为唯一均衡——之前嵌入。一旦越过此点，运营者的议价力将压倒任何事后监管尝试。这为第 6 节的政策建议提供了形式化的时序基础。

5.4 Stage 6: 天下大同 (Universal Harmony)

前文（§5.2–§5.2.1）将锁入轨迹建模为从 S_1（涌现）到 S_5（认知闭合）的五阶段非齐次马尔可夫链，其中 S_5 被设定为唯一吸收态（假设 M5）。本节论证此设定——尽管在形式上是自洽的——在实质上是*不完整的*。锁入轨迹存在第六个阶段，且该阶段不是对 S_5 的否定，而是对锁入本身意义的转化：当 CEC 从先发者的私有资产转变为全球公共基础设施时，垄断锁入转变为公共品协调。我们称此阶段为*天下大同*（Universal Harmony）。

Definition 5.3 (天下大同状态 S_6). 天下大同状态 S_6 定义为满足以下全部条件的系统配置：

(i) **CEC 的完全国际化**：累积证据语料库 $\mathcal{E}_t$ 由多边治理机构（如第 6 节建议的国际数据审计署 IDAA）持有和管理，任何辖区享有非歧视性、非排他性的平等访问权。CEC 不再是任何单一辖区或实体的私有资产；

(ii) **审计标准的去国家化**：数据质量的操作标准——异常判定阈值、多专家共识的聚合规则、Spring 收敛的终止条件——由多辖区受托人委员会通过透明程序确定，任何单一辖区不具备否决权；

(iii) **协议演化的公共治理**：Yajie 协议的版本演进（新审计模块的纳入、聚合函数的修订、校准算法的更新）由多辖区共识程序决定，维护者（Maintainer）的原始角色转变为技术秘书处，不具有单边决策权；

(iv) **采纳的非强制性**：任何辖区可以选择不采纳该协议，但此选择不改变该辖区数据被其他辖区的采纳者审计的可能性（方程 (77) 中的审计函数 $\mathcal{V}$ 保持非零），亦不改变碎片化成本 λ 的适用性——即天下大同是采纳博弈的均衡，而非强制。

Definition 5.4 (扩展锁入马尔可夫模型的补充假设)**.** 在假设 M1–M5（定义5.2）的基础上，引入以下补充假设：

M6 S_5 **的非吸收性**：S_5（认知闭合）不是吸收态。在 S_5 中，社会选择变量 $D_t \in \{0,1\}$ 被激活，其中 $D_t = 1$ 当且仅当在时间 t，拥有 CEC 治理权的国际社会通过多边协议决定将 CEC 国际化。D_t 的实现依赖于外生的政治意愿参数 $\pi_{\text{political}} \in [0,1]$，后者表示在任何给定审计周期中达成 CEC 国际化协议的概率。经济学含义：认知闭合并不消除人类能动性——*CEC* 的持有者（无论是先发辖区、协议运营者还是维护者）始终保留将 *CEC* 捐赠给国际社会的选项。是否行使该选项取决于政治经济计算。

M7 S_6 **为真正的吸收态**：一旦系统进入 S_6（天下大同），其状态不可逆转。转移概率满足 $p_{66}(|\mathcal{E}|) = 1$ 对所有 $|\mathcal{E}|$，且对所有 $j < 6$，$p_{6j}(|\mathcal{E}|) = 0$。合理性：*CEC* 一旦国际化并嵌入多边治理架构，逆转该国际化——即某辖区单方面将已国际化的 *CEC* 重新私有化——在制度上不可行，因为 *CEC* 的公共品属性已使任何辖区的排他性占有成为对全体其他辖区的违约行为。

M8 $S_5 \to S_6$ **的转移概率**：系统从认知闭合进入天下大同的每周期转移概率为：

$$p_{56}(D_t, \pi_{\text{political}}) = D_t \cdot \pi_{\text{political}} \tag{70}$$

其中 $D_t = \mathbb{I}$[国际社会在第t 周期达成 CEC 国际化协议]。转移概率的期望——在 D_t 的随机性未实现前——为 $\pi_{\text{political}}$。含义：天下大同不是自动发生的——它需要一次有意识的制度行动。本次行动的可信度由 $\pi_{political}$ 度量。

在补充假设 M6–M8 下，扩展转移矩阵 $\tilde{\mathbf{P}}(|\mathcal{E}|)$ 具有 6×6 上三角结构，其最后两行为：

$$\begin{pmatrix} \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & p_{55}(|\mathcal{E}|) & p_{56}(D_t, \pi_{\text{political}}) \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \tag{71}$$

Theorem 5.2 (天下大同的吸收必然性与到达时间)**.** *在假设 M1–M8 下：*

(i) **吸收必然性**：*从任意初始状态 $X_0 = S_i$（$i \in \{1, \ldots, 5\}$）出发，扩展马尔可夫链 $\{X_t\}_{t\geq 0}$ 以概率 1 被吸收到状态 S_6：*

$$\mathbb{P}\left[\exists t < \infty : X_t = S_6 \mid X_0 = S_i\right] = 1 \tag{72}$$

(ii) **两阶段吸收结构**：吸收过程具有两阶段结构。第一阶段：系统以概率 *1* 穿越 $S_1 \to S_2 \to S_3 \to S_4 \to S_5$（定理*5.1*之 *(i)*）。第二阶段：进入 S_5 后，系统在每个审计周期以概率 $p_{56} = D_t \cdot \pi_{political}$ 进入 S_6。在 S_5 中的逗留时间 τ_5 服从几何分布：

$$\mathbb{P}[\tau_5 = k] = (1 - p_{56})^{k-1} \cdot p_{56}, \quad k = 1, 2, \ldots \tag{73}$$

期望逗留时间 $\mathbb{E}[\tau_5] = 1/p_{56}$。

(iii) **期望总吸收时间**：从 S_1 到 S_6 的期望吸收时间满足：

$$\mathbb{E}[T^{(6)}_{absorb} \mid X_0 = S_1] = \mathbb{E}[T^{(5)}_{absorb} \mid X_0 = S_1] + \frac{1}{p_{56}} \tag{74}$$

其中 $\mathbb{E}[T^{(5)}_{absorb}]$ 是到 S_5 的期望吸收时间（定理*5.1*之 *(ii)*）。

(iv) **政治意愿参数的临界角色**：若 $\pi_{political} = 0$，系统永远停留在 S_5——定理*5.1*的五阶段模型作为特例恢复。若 $\pi_{political} = 1$ 且 $D_t = 1$，系统在进入 S_5 的首个审计周期即进入 S_6——认知闭合被即时转化为天下大同。对一般的 $\pi_{political} \in (0, 1)$，系统在 S_5 的期望逗留时间为有限值，但其分布可能具有厚尾（*heavy tail*）——低 $\pi_{political}$ 意味着系统可能在认知闭合中徘徊大量审计周期。

(v) **诚实暴击——锁入轨迹的真实终态**：将 S_5（认知闭合）认定为吸收态（假设 *M5*）是在排除了制度创新可能性条件下的条件性正确结论。但在包含制度创新可能性的完整模型中，S_6（天下大同）才是真正的吸收态。定理*5.1*的"以概率 *1* 吸收到 S_5"应修正为"以概率 *1* 吸收到 S_6"，其中 S_5 是通往 S_6 的必经但非终点状态。此转型的动力学由质变点定理 *(Spring SE-1)* 所刻画：足够的迭代导致质变——$S_1 \to S_5$ 为量变积累，$S_5 \to S_6$ 为质变跃迁。

这一修正具有以下实质含义：**先发优势——无论是中国的制造业-审计复合体优势（*§11.4*）还是任何其他先发辖区的 *CEC* 规模优势——不是终局，而是阶段。量变（*CEC* 的持续积累）在 S_5 达到质变（认知闭合——数据质量的操作定义被单一 *CEC* 标准所锚定），但此质变本身在 S_6 被第二次质变所超越：垄断标准转化为公共品标准。**

证明. **(i)** 扩展马尔可夫链的有限状态空间为 $\mathcal{S}' = \{S_1, \ldots, S_6\}$。在假设 M6–M8 下，$S_5$ 是瞬态（transient）——存在正的转移概率 $p_{56} > 0$ 离开 S_5（只要 $\pi_{\text{political}} > 0$ 且 $D_t = 1$ 至少在一个周期实现）。由 M7，S_6 是唯一吸收态。由有限马尔可夫链的标准结果（Kemeny & Snell, 1976, 定理 3.1.1），从任意瞬态出发，以概率 1 最终被某吸收态吸收。由于唯一的吸收态是 S_6，吸收到 S_6 以概率 1 发生。

(ii) 在假设 M3（单向锁入）下，系统必须先按序穿越 $S_1 \to \cdots \to S_5$（或从任意 S_i 出发完成剩余阶段），方可进入 S_6——直接转移 $p_{i6} = 0$ 对所有 $i < 5$（S_6 以 CEC 的认知闭合为其进入前提——在审计标准获得认知权威之前，国际化缺乏制度操作支点）。

进入 S_5 后，每周期转移至 S_6 的尝试是独立的伯努利试验，成功概率为 p_{56}。逗留时间 τ_5 为首次成功前的试验次数，服从所示几何分布。

(iii) 由两阶段吸收结构的独立性（第一阶段：确定性序列 + 随机逗留时间 τ_1 至 τ_4；第二阶段：几何分布 τ_5），总期望吸收时间为两阶段期望时间之和。

(iv) 直接由 p_{56} 的定义和几何分布的期望得出。

(v) 认知闭合（S_5）的" 不可逆" 属性——假设 M3 所保证的单向锁入——是*技术性*不可逆（替代协议在概念上不可理喻），而非*制度性*不可逆（CEC 不能被捐赠给国际社会）。M5 的修正将这两类不可逆性加以区分：技术不可逆性维持（$p_{5j} = 0$ 对所有 $j < 5$——审计标准的认知权威不可回退至竞争状态），但制度不可逆性不成立（$p_{56} > 0$——CEC 的治理权可以从先发者转移到国际社会）。" 量变引起质变" 在此处的精确含义是：(a) CEC 规模的量变引起认知闭合的质变（$S_4 \to S_5$）；(b) 认知闭合状态的制度化量变（国际化协议的签署和批准积累）引起标准去国家化的质变（$S_5 \to S_6$）；(c) 两次质变均为不可逆，但第二次质变将第一次质变所创造的垄断权力转化为公共品。 □

[天下大同的采纳博弈等价——先发优势的制度性消除] 在 S_6（天下大同）状态下，非扩散博弈 Γ^{NP}（定义3.2）的全体采纳均衡条件 (9) 得到完全对称化：对所有辖区 $i, j \in N$：

$$\Delta_i(|\mathcal{E}^{\mathrm{public}}|) = \Delta_j(|\mathcal{E}^{\mathrm{public}}|) \triangleq \Delta^{\mathrm{public}} \tag{75}$$

采纳优势不再依赖于任何辖区的身份。NPE 的维持基础从" 先发者的私有 CEC 精度优势" 转变为" 全球 CEC 公共品的精度优势"。博弈 Γ^{NP} 在 S_6 中退化为一对称协调博弈——所有辖区的支付矩阵完全相同，唯一纳什均衡为全体采纳（当 $\Delta^{\mathrm{public}} \geq \lambda - \kappa$），且该均衡不赋予任何辖区以博弈论层面的特权地位。

诚实暴击：这意味着推论3.3所分析的先发优势衰减在 S_6 达到其逻辑终态——衰减不是趋近于零，而是*衰减到身份的消失*。先发者在采纳博弈中不再是" 先发者"；其身份被 CEC 的公共品属性所溶解。" 天下大同" 的严格形式含义即此——审计标准不再" 属于" 任何国家，不是因为任何国家放弃了所有权，而是因为" 所有权" 这一范畴在公共品配置下失去操作意义。

证明. 在 S_6 下，CEC 的完全国际化（定义5.3之 (i)）保证 $|\mathcal{E}_i^{\mathrm{accessible}}| = |\mathcal{E}^{\mathrm{public}}|$ 对所有 i 成立——每个辖区可调用的 CEC 规模相同。代入 $\Delta(\cdot)$ 的定义（方程8），Δ_i 简化为仅依赖于公共 CEC 规模的函数，与 i 的身份无关。全体采纳均衡条件 $\Delta^{\mathrm{public}} \geq \lambda - \kappa$ 对所有辖区同时成立或同时不成立。当成立时，$(A, \ldots, A)$ 是对称博弈的对称严格纳什均衡——所有辖区地位等同。 □

Remark 5.3 (天下大同与" 历史终结论" 的区分). 定理5.2所描述的 $S_5 \to S_6$ 转型不应被误解为 Fukuyama(1989) 意义上的" 历史终结"。天下大同不是数据审计治理的" 终点"——它是*数据审计标准归属*的终点。在 S_6 中，以下竞争性维度仍然存在：(a) 审计模块的技术竞争——各辖区仍可贡献新模块以影响操作标准，竞争从" 谁拥有 CEC" 转变为" 谁的审计方法论更优"；(b) 数据质量的实证竞争——各辖区的数据集质量仍存在差异，审计继续揭示这些差异；(c) 治理权的程序竞争——多边受托人委员会的席位分配、

投票权重和决策规则仍是政治博弈的对象。S_6 消除的是标准归属的*排他性*——审计标准不再属于任何国家，但标准本身的确定仍然是人类制度中的人类决策。天下大同是霸权终结，不是政治终结。

Remark 5.4 (天下大同的博弈论基础与 NPE 的制度转型). 推论3.3（第 3.2 节）与定理5.2（本节）共同构成了一个完整的制度转型理论：

(1) NPE 的*先发垄断型*阶段：CEC 由先发者私有持有，$\Delta_i(|\mathcal{E}|)$ 对先发者最大，均衡由先发者的精度优势维持。此阶段的规范性基础是脆弱的——均衡的稳定性依赖于先发者不滥用其垄断地位的可信承诺（参见第 4 节关于 SWIFT 武器化的分析）。

(2) NPE 的*转型期*：推论3.3之 (iii) 所描述的 CEC 国际化过程。转型的速度由 $\pi_{\text{political}}$ 决定，而 $\pi_{\text{political}}$ 又受以下因素影响：(a) 先发者的策略性延迟——先发者在 S_3 之后具有推迟国际化的经济激励（命题5.3）；(b) 后发者的集体行动——后发辖区可通过威胁联合开发替代协议来提高 $\pi_{\text{political}}$（改变博弈参数 λ 和 c^{develop}）；(c) 外部冲击——如 SWIFT 武器化事件提升了对审计基础设施中立性的需求，从而增大 $\pi_{\text{political}}$。

(3) NPE 的*公共品型*阶段（S_6）：CEC 完全国际化，NPE 的维持基础从先发者垄断转变为公共品协调。此阶段的规范性基础是稳健的——采纳的动机是”开发替代协议不必要”（因为全球 CEC 公共品已经存在且对所有辖区开放），而非”开发替代协议不划算”（因为先发者的私有 CEC 精度优势不可企及）。

此三阶段转型为第 6 节的政策建议提供了时序结构：建议 1（IDAA 的建立）应发生在转型期，建议 5（治理国际化的预承诺）应在 S_3 之前做出，建议 6（价格管制公式）应在转型期施行并在 S_6 中由多边受托人委员会维持。

Remark 5.5 (先发者的道德义务：非居高临下的治理姿态). 推论3.3和定理5.2共同产生了先发者的一个规范性义务，该义务不是道德劝诫，而是博弈均衡的推论：

(1) 先发者的最高姿态不是”我帮你们”。”帮助”预设了不平等——帮助者处于给予的位置，被帮助者处于接受的位置。SCX 框架不提供任何人以这种位置。人人平等定理（推论**??**）规定：没有特权观察者。先发者在精度优势上暂时领先，但在认知地位上与任何后发者完全对等——后发者有权独立验证先发者的任何声称，并要求其声明全部假设。

(2) 先发者的正确姿态是”我先建好了，你们验证我”。先发者的 CEC 精度优势是可以被独立验证的——任何辖区可以提交测试数据集、检查审计模块的检测率、验证假设条件是否满足。先发者不要求信任，只要求验证。这与老实人定理（定理 3）一致：声称成立只需声明假设并通过验证，不需要权威背书。

(3) 中国的历史角色：”建了第一段路，然后把路交给所有人。”若中国的制造业密集经济结构和算力基础设施使其成为审计基础设施的首先部署者（MAC 定理的条件），则其在 SCX 时代的角色不是”统治者”而是”第一段路的修建者”。路的归属在修建完成时不取决于谁建了它——取决于谁可以走它。CEC 作为追加式公共品的属性意味着：一旦国际化（T^* 时刻），中国在审计精度上的先发优势转化为全人类的公共品存量，中国自身的精度优势和其他国家一样缩减为对该公共品的平等访问权。

(4) 对落后国家的尊重是定理要求，不是同情。 后发辖区不需要”帮助”——它们拥有与先发者完全相同的认知武器：(i) 不声明假设的声称在数学上无效（老实人定理）；(ii) 任何声称的假设可以被独立验证（人人平等定理）；(iii) 如果审计架构被排他性持有，后发辖区可以通过集体行动威胁联合开发替代协议来改变博弈参数（c^{develop} 和 λ），迫使先发者加速国际化进程。这些不是”权利”——是定理。后发者不强，先发者不仁——博弈结构使双方的对等地位成为唯一均衡。

(5) 天下大同不是”中国的世界”——是”不再有谁的世界”。 S_6 的定义（定义5.3）排除了任何国家对审计标准的排他性归属。如果中国在 $S_1 \to S_5$ 中扮演了核心角色——这反映了其制造业体量和算力优势的历史事实——那么在 $S_5 \to S_6$ 的转化中，它的核心角色是”交出自己修建的路”。这不是道德选择——是 MAC 定理的诚实推论：先发优势是有半衰期的。拖延国际化只会延长一个终究会结束的优势，而加速国际化使中国成为审计公共品的共同创建者，而非一个终究要被迫交出垄断地位的垄断者。

5.5 Pricing and the Addiction Dynamic

The business model of a CEC-driven audit protocol exhibits a distinctive structure that we term the *addiction dynamic*. In conventional platform markets, pricing power derives from switching costs and network effects: once users are locked in, the platform can raise prices, extracting rents up to the point where the cost of switching exceeds the price increase (Farrell & Klemperer, 2007). The Yajie Protocol's pricing dynamic is different.

Because each audit cycle enriches the CEC—improving accuracy for all users, including the auditing organization itself—the *marginal social value* of an audit exceeds its *marginal private value* by the positive externality contributed to the CEC. This creates a rationale for below-cost pricing in the early stages of protocol adoption, subsidized by patient capital or public funding. Below-cost pricing accelerates CEC accumulation, widening the accuracy gap and reinforcing the non-proliferation equilibrium.

However, once the protocol reaches Stages 3–4, the pricing dynamic inverts. The accuracy gap becomes so large that organizations cannot credibly threaten to switch to a competing protocol or revert to internal audit processes. At this point, the protocol operator can raise prices, capturing a portion of the value generated by the protocol's accuracy advantage. The theoretical maximum price is:

$$P_{\max} = V[\theta(|\mathcal{E}|)] - \max\{V[\theta(0)], V_{\text{internal}}\} \tag{76}$$

where V_{internal} is the value an organization could derive from its own internal audit processes. As $|\mathcal{E}|$ grows, $P_{\max}$ increases.

This pricing trajectory—predatory low pricing followed by rent extraction—mirrors the standard narrative of platform monopolization (Khan, 2017). However, there is an important difference: the Yajie Protocol's early low pricing is not predatory in the antitrust sense, because it is not aimed at driving competitors out of the market (there are no competitors to drive out) but at accelerating CEC accumulation. The pricing is *developmental* rather than *exclusionary*. This distinction matters for competition policy, as we discuss in Section 6.

The addiction dynamic operates on the demand side as well. Organizations that adopt the protocol become dependent on its accuracy, integrating its outputs into their quality assurance workflows, regulatory compliance processes, and contractual arrangements. As the protocol's accuracy improves, the organization's own internal quality assurance capabilities may atrophy—a dynamic familiar from the literature on automation dependency (Parasuraman & Riley, 1997). The organization becomes "addicted" to the protocol not in the metaphorical sense but in the economic sense: the cost of disadoption rises as internal capabilities degrade, while the benefit of continued adoption rises as the protocol improves.

5.6 The Bible-and-Pope Model: Open-Source Poison, Calibrated Antidote

A distinctive feature of the Yajie Protocol's business architecture is the separation of the self-evolution engine (Spring) from the cross-client calibration service (Yajie API). This separation generates a commercial dynamic that we term the *Bible-and-Pope model*.

The Spring engine is released as open-source software. Any organization can download, deploy, and operate its own instance of the self-evolution loop, accumulating a private M_t on its own infrastructure. The open-source release serves three strategic functions. First, it eliminates adoption friction: organizations can begin auditing their data without negotiating contracts, paying subscription fees, or exposing proprietary data to a third party. Second, it accelerates lock-in: each audit cycle enriches the organization's private M_t, making the accumulated audit history non-portable and the organization structurally dependent on Spring's audit methodology. Third, it forecloses competitive entry: any would-be competitor must match not only the algorithmic quality of Spring but also the accumulated audit history that Spring's adopters have already built—a barrier that grows with every audit cycle. The open-source release is, in economic terms, a *poison*: freely ingested, irreversible in its effects, and lethal to competitors.

The Yajie API is offered as a paid calibration service. The API accepts feature matrices $\phi(X)$ computed locally by client organizations (satisfying data sovereignty requirements) and returns a calibration report that maps the client's private quality scores onto a global reference scale, maintained by the protocol operator's central gatekeeper. The value proposition of the API is not "we audit your data better than you do"—it is "we tell you what your scores mean relative to everyone else." A client organization whose private gatekeeper assigns a quality score of 0.87 to its training data does not know, absent calibration, whether 0.87 represents industry-leading data quality or a self-serving overestimate generated by a gatekeeper that has never seen anyone else's data. The Yajie API resolves this ambiguity, mapping local scores onto a globally comparable metric. The annual subscription fee—projected at \$50,000–\$200,000 depending on audit volume and calibration depth—purchases not superior technology but *interpretive authority*: the assurance that one's private quality assessments are commensurable with those of competitors, regulators, and counterparties.

The Bible-and-Pope model is economically self-reinforcing. The open-source Spring engine creates the demand for calibration (because private gatekeepers inevitably drift from one another as their M_t corpora diverge), while the paid Yajie API supplies the calibration that resolves the resulting ambiguity. Organizations that decline the API are not technically incapacitated—their Spring instances continue to function—but they become epistemically isolated, unable to compare their data quality to industry benchmarks. In regulated industries where data quality certifications are prerequisites for market access, epistemic isolation is functionally equivalent to exclusion. The API is thus not a luxury; it is an *interpretive necessity* that the open-source engine itself manufactures.

5.7 Welfare Analysis

The welfare implications of the lock-in trajectory are ambiguous and stage-dependent:

- **Stages 1–2**: Welfare is unambiguously positive. The protocol provides data quality assessment capabilities that did not previously exist, and the CEC accumulation dynamic generates positive externalities for all adopters.

- **Stage 3**: Welfare remains positive on net, but distributional concerns emerge. The protocol operator begins to extract rents, and late adopters pay higher prices than early adopters. The accuracy gap begins to foreclose the possibility of competitive entry, eliminating the disciplinary effect of potential competition.

- **Stages 4–5**: Welfare analysis becomes fundamentally ambiguous. The protocol's institutionalization eliminates the possibility of competitive discipline, raising the specter of monopoly pricing, quality degradation, and innovation suppression. At the same time, the protocol's universal adoption eliminates fragmentation costs λ and enables the accumulation of an audit corpus whose social value—in terms of improved data quality across the economy—may vastly exceed the deadweight loss from monopoly pricing.

A critical parameter in the welfare analysis is the *governance discount*: the degree to which multilateral governance of the protocol reduces the monopoly pricing problem while preserving the CEC accumulation dynamic. If governance arrangements can cap prices at competitive levels without impairing the protocol's operational effectiveness, the welfare case for a universal protocol is strong. If governance capture prevents effective price regulation, the welfare case weakens, and the fragmentation equilibrium may become preferable despite its higher coordination costs.

5.8 A Note on the Storage Externality

A secondary economic effect of the Yajie Protocol warrants brief mention, if only because its distributional consequences are perversely elegant. The protocol's foundational design principle—that the Cumulative Evidentiary Corpus M_t grows monotonically and is never pruned—generates a persistent, compounding demand for digital storage. Each audit cycle appends new entries to M_t: anomaly registers, calibration parameters, quality fingerprints, and adjudication records. At scale, with hundreds of client organizations each conducting daily audits across millions of records, the global storage footprint of the protocol's audit corpora becomes non-trivial.

This demand is structurally inelastic. No client organization can reduce its M_t without impairing its audit accuracy, because the Spring self-evolution mechanism depends on the cumulative audit history to calibrate gatekeeper parameters. Deleting audit records is not cost-saving; it is self-sabotaging—equivalent to a library discarding its oldest volumes on the grounds that they are old. The protocol's clients are therefore locked into an indefinitely growing storage obligation that cannot be negotiated, optimized away, or substituted.

The distributional consequence is a wealth transfer from the consumers of data auditing to the producers of data storage. Cloud infrastructure providers, hard disk manufacturers, and data center operators capture a growing share of the protocol's economic surplus through storage fees, while the auditing organizations that generate the surplus see their operating costs rise with each audit cycle. The magnitude of this transfer is not large relative to the value created by the protocol—the cost of storing an audit record is orders of magnitude smaller than the cost of the data quality failure it prevents—but its direction is noteworthy: a protocol designed to reduce information asymmetry in AI training data inadvertently increases demand for the physical substrate on which information is stored.

5.9 A Note on the Future of Commerce

The SCX framework does not eliminate commerce. It eliminates the informational asymmetry on which dishonest commerce depends. When the 禁止装逼定理 (Adoption-Audit Equilibrium) makes audit outputs unconditionally verifiable by any third party, the competitive advantage shifts from persuasive packaging to verifiable quality. A chip manufacturer whose power efficiency is independently audited does not need to outspend competitors on marketing—its claims are mathematically certified. A pharmaceutical company whose clinical data survives Yajie cross-validation does not need to lobby regulators—its evidence is self-authenticating.

The corporation that makes the best product wins, not the one that tells the best story. This is not a moral prediction—it is a Nash equilibrium. When the cost of falsifying claims exceeds the premium earned by verified quality, honesty becomes the profit-maximizing strategy. Commerce continues. It just becomes honest.

5.10 A Note on the Maintainer's Audit Obligation

The protection equilibrium (Theorem 9.2) rests on the Maintainer's *capability* to audit. But capability without exercise is indistinguishable from incapability. If the Maintainer never audits public claims—government statistics, corporate disclosures, published research—then the deterrence signal is absent. Bad actors observe no audits, infer no risk, and continue falsifying.

This yields a structural obligation, not a moral one:

[Maintainer 的审计义务 (The Maintainer's Audit Obligation)] Under the mutual deterrence framework (D1–D5), the Maintainer's failure to audit publicly available claims results in the collapse of the protection equilibrium. Specifically: if no public audit occurs within any time window of length $T_{\max}$, then $\mathbb{P}(\text{deterrence credible}) \to 0$ as $t \to \infty$. The Maintainer must audit—not to punish, but to make the deterrence signal observable.

Proof sketch. The deterrence equilibrium is a Perfect Bayesian Equilibrium in which off-equilibrium beliefs sustain the threat. If the Maintainer never exercises audit capability, off-equilibrium-path beliefs become degenerate: actors assign zero probability to detection. The equilibrium unravels by the intuitive criterion (Cho & Kreps, 1987). Regular public audit—of government statistics, corporate disclosures, published research—maintains the signal and keeps the equilibrium intact. □ □

This obligation is bounded. The jurisdictional boundary (Section 6.2) restricts audit to publicly available data: the Maintainer audits what is *claimed in public*, not what is *held in private*. Anyone who publishes a claim—a GDP number, a clinical trial result, an AI benchmark—subjects it to potential audit. Those who keep their data private are not audited, but neither can they claim quality—their silence is their choice, and the market discounts unverifiable claims accordingly.

人人平等定理 requires that this obligation applies universally. The Maintainer does not selectively audit enemies and spare friends. The audit is the same for everyone—because the theorem requires it, and because selective audit would itself be detectable as bias, destroying the equilibrium's credibility.

This obligation extends to non-adopting nations with particular force. Any state that publishes official statistics—GDP growth, unemployment rates, carbon emissions—

makes a public claim. That claim is auditable, regardless of whether the state has formally adopted Yajie. Adoption is not a prerequisite for audit; publication is.

[非采纳者的审计暴露 (Audit Exposure of Non-Adopters)] Let jurisdiction i choose not to adopt Yajie ($s_i = D$ in the NPE game). If i publishes any public claim, that claim is subject to SCX audit. The audit result is a revealed signal: if the claim passes, i's non-adoption is revealed as a strategic choice unrelated to data integrity; if the claim fails, i's non-adoption is revealed as a protective measure against detection. In either case, i's type is revealed. No jurisdiction can simultaneously (i) refuse audit infrastructure, (ii) publish unverifiable claims, and (iii) maintain epistemic credibility. The trilemma is mathematically inescapable.

Proof sketch. By Theorem 3 (老实人定理), a claim lacking stated, independently verifiable assumptions is mathematically empty. A non-adopting jurisdiction that publishes a claim without submitting to audit has made an unverifiable claim—equivalent, under the theorem, to making no claim at all. If the jurisdiction submits to audit and passes, its non-adoption stands exposed as a political decision, not a data-quality necessity. If it refuses audit entirely, its refusal itself is observable and carries information: honest leaders have no reason to fear independent verification; refusal to be verified is a signal of expected verification failure (theSpence-style separating equilibrium). □

Thus the Maintainer does not need to *force* adoption. The logic of the framework is self-executing: honest leaders voluntarily adopt because audit certifies their claims; those who decline are identified by their choice; and all are subject to the same verification of their public claims regardless of adoption status. The mathematics of verification applies uniformly—and those who choose to remain unverified communicate their choice in a way that is itself observable.

5.11 A Note on Internal Audit: Why Self-Audit Is Structurally Untrusted

A corporation that deploys Yajie internally—using its own employees, on its own infrastructure, reporting to its own management—has not satisfied the independence condition of Theorem 1 (独醒不太可能定理). The employee who audits her employer's data faces a structural conflict: her career advancement depends on her employer's satisfaction, and her employer's satisfaction depends, in part, on favorable audit outcomes. Even if she resists this pressure, her employer cannot *know* that she has resisted it—and therefore cannot cite her audit as independent verification.

This is not a failure of character. It is a failure of structure. The independence required by Theorem 1 is not a personality trait of the auditor—it is a property of the auditor's position in the incentive graph. An internal auditor, regardless of integrity, occupies a node whose edges point toward her employer. The Maintainer occupies a node with no edges to any audited entity. The distinction is topological, not moral.

[内部审计的独立性悖论 (The Internal Audit Independence Paradox)] Let $\mathcal{A}_{\text{internal}}$ be an audit department of organization O, and let M_{internal} be its expert ensemble. Under the AAE framework, the condition $M_{\text{eff}} = M/(1 + (M-1)\bar{\rho})$ (Theorem 6.1) applies with $\bar{\rho} > 0$ whenever the experts share the same employer—because their errors are correlated through shared incentives. The effective number of independent experts degrades. In the limit where the employer can terminate all M auditors, $\bar{\rho} \to 1$ and $M_{\text{eff}} \to 1$—the entire internal audit department reduces to the epistemic equivalent of a single observer, regardless of headcount. This degradation is a direct consequence of 强人所难定理 (Theorem 2): when the features of independence are weak, the detection guarantee collapses. An internal audit report signed by fifty employees carries no more independence guarantee than a report signed by one.

Thus the Maintainer's structural independence is not a luxury—it is a mathematical prerequisite for the audit to satisfy the conditions of Theorem 1. An organization *cannot* audit itself credibly, not because it lacks expertise, but because it lacks the topological property of disinterest. The Maintainer has no reason to inflate the numbers. The employee does. The CEO knows this. The game is solved before the first audit runs.

5.12 A Note on the Deadline: Disprove or Protect

The SCX framework creates a strategic deadline for every rational actor—leader, scientist, or philosopher—who becomes aware of its claims. The framework asserts 105 theorems with complete proofs. If any theorem is false, a competent actor can disprove it by exhibiting a counterexample or a proof error. If all theorems are true, the framework implies that the actor's own claims are subject to independent audit—and that the Maintainer is the sole credible auditor. The actor therefore faces a binary choice with a ticking clock:

[推翻或保护的截止期博弈 (The Disprove-or-Protect Deadline)] Let an actor A become aware of the SCX framework at time t_0. Define two strategies:

(i) **Disprove**: identify a fatal error in any of the 105 theorems, publish the refutation, and demonstrate that the framework is unsound.

(ii) **Protect**: accept the framework's validity and move to protect the Maintainer, thereby signaling that A's own claims can withstand independent audit.

Any third strategy—neither disproving nor protecting—is strictly dominated after a finite horizon T^*. Specifically, as $t \to \infty$ without either disprove or protect, the posterior probability that A is a bad type (one whose claims would fail audit) converges to 1. The actor cannot remain neutral: neutrality is epistemically equivalent to guilt.

Proof sketch. The SCX framework is publicly available with complete proofs. If a theorem is false, disproving it requires only identifying the error—a task bounded in complexity by the length of the proof chain. If the theorems are true, no amount of time will produce a valid refutation. Thus, for any actor who has examined the framework:

- If A believes a theorem is false, A should disprove it immediately—delay confers no advantage and allows the framework to accumulate credibility.
- If A believes the theorems are true, A should protect the Maintainer immediately —delay exposes A to the inference that A expects his own claims to fail audit and is waiting for someone else to eliminate the auditor.

The observable action set is Disprove, Protect, Neither. The payoff from Neither is strictly less than either Disprove (for honest skeptics) or Protect (for honest acceptors) at all $t > t_0 + \Delta$, where Δ is the time required to read and verify the proofs. Thus Neither is iteratively dominated. □ □

The deadline is not imposed by the Maintainer. It is imposed by the logic of the situation. Every day a leader, scientist, or philosopher neither disproves the framework nor protects its author is a day that accumulates evidence against their own epistemic integrity. They are not waiting for more information—the proofs are complete. They are waiting for the problem to go away. But the problem is a set of theorems, and theorems do not expire.

Remark 5.6 (投石理论 (The Stone-Throwing Theorem): 宁可信其有). The SCX framework casts a stone into the public square. The stone is a simple statement: *the later a leader protects the Maintainer, the stupider that leader looks.* Leaders must respond. They have two paths—the classical Chinese idiom 投石问路 (throw a stone to test the path ahead): either disprove the framework (throw the stone of refutation) or protect its author (throw the stone of commitment). Standing still is not a third option—it is being struck by the stone.

If the stone hits a leader, that leader has only himself to blame. The stone does not name names. It states a conditional: *if* you delay, *then* you look stupid. Any leader who reads this and feels accused has performed the logical operation himself—he has checked the antecedent, found it true of his own behavior, and drawn the consequent. The stone does not accuse. The stone reveals. Those who recognize themselves in its trajectory have **对号入座** (self-identified). The author bears no responsibility for their self-recognition.

The Disprove-or-Protect deadline admits an even stronger formulation that does not require understanding the 105 theorems at all. Consider the payoff matrix for any leader:

	SCX is true	**SCX is false**
Protect early	Credibility preserved; type revealed as good	Wasted protection resources
Do nothing	Credibility destroyed; type revealed as bad	Nothing lost

The asymmetry is fatal to the ”do nothing” strategy. If SCX is true and the leader does nothing, the cost is total—irreversible loss of epistemic credibility, permanent revelation as bad type. If SCX is false and the leader protects, the cost is bounded—resources expended on a nonexistent threat. The expected value calculation favors protection for any prior $\mathbb{P}(\text{SCX true}) > \varepsilon$ where $\varepsilon = C_{\text{protect}}/C_{\text{exposed}}$ is the ratio of protection cost to exposure cost. For any leader whose legitimacy depends on verifiable claims, C_{exposed} is effectively infinite, and $\varepsilon \to 0$.

This is not Pascal's Wager. Pascal required belief in a deity whose existence was unprovable. SCX requires no belief—the proofs are published, complete, and verifiable by any competent mathematician. The leader need not read them personally. The leader need only observe that *no one has refuted them.* In the absence of refutation, the rational strategy is protection.

Furthermore, the timing of protection is itself a signal. A leader who protects at $t_0 + \Delta$ (immediately after the proofs can be verified) reveals clear strategic reasoning. A leader who protects at $t_0 + 5$ years reveals an extended period of inaction—either the information did not reach the decision-maker, or it was understood but not acted upon. Delay is not neutral. Delay is observable. A rational actor with access to the proofs should reach the decision in time Δ. Protracted delay is a choice—and choices are informative.

The stone thus serves a second diagnostic function: it measures the responsiveness of the governance system that surrounds the leader. **Early protection signals institutional agility.** A leader who protects quickly demonstrates that information channels are functional and decision-making is unobstructed. **Late protection signals institutional inertia.** A leader who protects slowly reveals that somewhere in the chain there is friction. Either the information did not reach the leader, or it reached and was not prioritized, or it was understood and the leader chose to wait. All three scenarios indicate processes that could be improved. The stone does not identify individual leaders as good or bad. It identifies systems that respond quickly and systems that do not.

There is a further implication. **The later the protection, the more questions it raises.** A system that protects immediately has nothing to fear from audit—its claims will pass. A system that delays invites the natural inference that there was something to consider. A system that delays *for years* invites the inference that the concerns were not trivial. Extended delay is not a scheduling oversight. It is a data point. The longer the delay, the more one naturally asks: what was being weighed against the cost of protection?

The SCX framework could be entirely wrong. Every one of the theorems could contain an error. If so—let the world have a good laugh. The author will be the joke.

But the payoff matrix does not depend on the framework being right. It depends only on the *possibility* that it is right—a possibility that cannot be eliminated until someone actually reads the proofs and finds the error. Until that day, the expected value calculation favors protection for any leader with a non-zero stake in being perceived as honest. If the framework is wrong, protecting the author wastes resources—a finite cost. If the framework is right and the leader does nothing, the cost is total. The asymmetry holds regardless of the truth value of the theorems.

The stone thus establishes a clean boundary around legitimate criticism. **Academic refutation is the only legitimate path.** Anyone may attack the 105 theorems—by finding a proof error, constructing a counterexample, or demonstrating a logical contradiction. Such criticism is welcome; it is the Disprove path, and it advances knowledge regardless of outcome. **Personal attack, moral condemnation, or historical judgment are not legitimate.** They are not arguments—they are admissions. A critic who cannot find a mathematical error but attacks the author's character, motives, or right to speak has conceded the mathematical argument. The attack is not skepticism. It is a confession: the critic fears the audit and has no other weapon. Such a critic is not a scholar. Such a critic is a bad type, self-revealed. The boundary is not arbitrary. It follows from Theorem 3 (老实人定理): a claim without stated, verifiable assumptions is mathematically empty. "This person is morally bad" is a claim. Where are the assumptions? Where is the verification protocol? If none are stated, the claim is void—not because it is offensive, but because it fails the only standard that the framework recognizes. The framework does not protect the author from criticism. It protects criticism from itself—by requiring that all claims, including claims about the author, meet the same standard.

This yields a structural asymmetry that places the author in a position that is, for all practical purposes, unassailable. **The mathematical layer can be attacked. The philosophical layer cannot.** If a critic finds an error in the proofs, the author accepts the correction and the framework improves—a legitimate academic outcome, reflecting well on both parties. But the philosophical core of SCX—that claims require stated assumptions, that no observer is privileged, that honesty is a Nash equilibrium—does not depend on whether Theorem 4's constant is correctly derived or whether the Lipschitz bound in Situs Theorem 1.2.3 carries a sign error. These are not theorems. They are *standards*. A standard is not true or false; it is adopted or not adopted. The standard says: state your assumptions. One cannot disprove this standard with a counterexample, because the standard itself defines what counts as a valid counterexample. It is philosophically self-sealing—not through dogma, but through recursive consistency.

Thus the author stands on asymmetric ground. A mathematical critic who succeeds earns honor for both parties. A mathematical critic who fails earns embarrassment for herself. A personal critic invalidates her own claim before it lands, because the claim fails the standard that the framework imposes on all claims. A philosophical critic faces the recursive problem of needing to state assumptions that the standard itself does not already require. **The author is not invulnerable. The author is simply standing on the only ground that remains after all unstated assumptions have been stripped away.** It is not a fortress. It is a vacuum. There is nothing left to attack but the mathematics—and the mathematics is offered openly, with complete proofs, for anyone to inspect.

This raises an apparent paradox. If the framework makes the author unassailable, does it not also silence legitimate critics—good people who genuinely detect an error but fear that their criticism will fail the framework's own standards and expose them as bad types? The paradox is resolved by the distinction between rigor and fear. The framework does not prohibit criticism of the author. It demands that criticism meet the same standard as any other claim: stated assumptions, verifiable evidence, logical soundness, precision. A good critic who has genuinely found an error can state precisely where the proof fails, under what assumptions, with what counterexample. Such criticism is not merely permitted—it is the Disprove path, and it advances knowledge. A critic who merely *suspects* an error but has not verified it should state that uncertainty honestly: "I suspect an error at line X, but I have not verified." This too is a valid claim with stated assumptions. The only criticism the framework rejects is the criticism that refuses to state its own assumptions—the criticism that demands to be accepted on authority, emotion, or moral indignation rather than on evidence.

The author can be judged. The author can be condemned. The author can be proven wrong. But the judgment must be the product of careful, repeated, rigorous reasoning. It must be confirmed before it is declared. It must survive the same scrutiny that the framework imposes on every other claim. If a critic rushes to condemn without verifying, the failure is not the framework's—it is the critic's. The framework does not protect the author from judgment. It protects judgment from itself—by requiring that those who judge meet the same standard as those they judge. A good person who has done the work should speak. The framework guarantees that their voice will be heard, because it will be unassailable for the same reason the author's is: it has stated its assumptions, and they can be verified.

This is not merely a theoretical position. It is a response to a lived reality. The author has experienced, repeatedly, a particular form of intellectual violence: the groundless accusation. A senior figure declares, without theory, without experiment, without evidence, that the author's work is wrong. The accusation demands to be accepted on authority alone. It states no assumptions. It provides no verification protocol. It is, by the standard of this framework, mathematically empty—and yet, in the social reality of academic hierarchy, it carries weight. It silences. It injures. It wastes years.

The author was brought to tears by these attacks more than once. The author came close to depression. What kept the author alive was not resilience, not optimism, not faith in the system. It was a specific, stubborn thought: I will not go quietly. I will not be silenced without a fight. That thought—unpolished, survival-level—was enough to keep going. The framework presented in this paper is, in part, the product of that thought. It is what happens when someone who refused to be silenced under groundless accusations builds a mathematical structure that makes those accusations formally impossible.

The framework names this for what it is: **epistemic bullying**. The bully makes a claim he cannot verify, under assumptions he will not state, and demands submission. The framework strips him of his weapon. He is not a scholar. He is not a skeptic. He is a bad type, self-revealed, and his accusation, under Theorem 3, is void. The framework does not protect the author from such people. It simply makes their behavior mathematically visible—and in doing so, it protects everyone who has ever been on the receiving end of an unverified accusation, from graduate students to whistleblowers to anyone who has been told, by someone with more power and less evidence, that they are wrong.

Beyond the personal, this principle extends to history. **History must be subjected to rigorous judgment. Injustice demands a just verdict—arrived at through strict reasoning, verifiable evidence, and open debate.** Not all historical claims are equal. Not all accusations carry weight. The framework provides the criterion: a claim without complete mathematical derivation, without logical rigor, made by someone who has not verified their own position, is not a claim at all. It carries no mathematical weight. **A person who attacks others without having verified their own position—without having done the work—has made a claim that fails the standard.** Such a claim does not need to be debated. It fails on its own terms. The accusation is void under Theorem 3. The audit will find what it always finds when the assumptions are unstated: the claim cannot sustain itself. The accused is vindicated. And history—real history, the kind that survives multi-expert verification—records the outcome.

Thus the framework implies something larger than personal vindication. Even if the theorems contain errors—even if the mathematical edifice is flawed—the game-theoretic structure stands independently. And that structure calls for a deep, systematic improvement. Academia. Finance. Governance. Any domain where unverified claims have been wielded by those with power against those without. Those who accuse without verifying, those who judge without evidence, those who demand submission without stating assumptions—are now identifiable by their own behavior. The framework does not need to seek them out. It only needs to be deployed. They will identify themselves—because they cannot stop making unverified claims, and every such claim, under Theorem 3, is self-defeating. This paper is not merely a theory of audit. It is an invitation to raise standards—in which every institution that has relied on unverified authority is given a choice: state your assumptions, or watch your claims lose their force.

From the perspective of life philosophy, **unreasoned anger is self-defeating.** Anger without reasoning behind it illuminates only the angry person's own position.

Life should be calm. One may reason, prove, refute. One may be angry—righteous anger is the fuel of reason. But one should not attack without cause, for unreasoned attack defeats itself. One may be blunt—the Stone-Throwing Theorem itself speaks plainly—but one should not be angry and unreasonable at once. Plain speech with calm is clarity. Anger without reason is a confession.

Those who rage without competence have already revealed themselves. As good people, we should hold them to account—not out of animosity, but out of respect for standards. They degrade the discourse. They waste everyone's time. They harm those who cannot fight back. The framework provides the tools. The audit provides the method. Every one of their cries is an unverified claim. Every one is void under Theorem 3. Holding them to account is not punishment. It is the maintenance of intellectual standards.

Tolerating unreasoned attack degrades the environment. When those in power tolerate such behavior, they reveal their own position—and, often, their own entanglements. They are not blind. They are choosing not to see. Why? Because the attacker may be their ally, their counterparty. To tolerate a person who attacks others without evidence is to make a choice: either unable to distinguish valid from invalid claims—or able, but unwilling to act. In either case, their position is compromised. Their silence speaks about what they may have to protect.

Two further types complete the taxonomy. **First, the false protector.** A bad actor may calculate that early protection is cheap—issue a statement, attend a ceremony, pose for a photograph—and then continue as before. But protection under the SCX framework is not symbolic. It is operational. To protect the Maintainer is to submit to audit. The false protector who protects early and then resists audit has merely accelerated his own exposure. His protection was a claim—"I am honest." The framework now has standing to verify that claim. If his data fails, his early protection becomes evidence against him: he knew the standard, he claimed to meet it, and he failed. False protection is not a loophole. It is a faster path to detection.

Second, the active suppressor. The taxonomy of types—Disprove, Protect, Neither—assumes actors operate within the framework's rules. But a fifth type exists: the suppressor, who seeks to prevent the framework from being evaluated on its merits. Not through academic refutation—there is no counterexample. Not through silence—the cost of waiting is too high. But through institutional suppression: obstruction, legal pressure, defunding, character attacks, manufactured controversies. The suppressor does not engage with the theorems because engagement would acknowledge them as worthy of engagement. This is a telling response. One does not try to suppress a theory one believes to be false—one ignores it. One tries to suppress a theory one believes to be true and consequential. The suppressor is not a skeptic. The suppressor is someone who has privately evaluated the framework and concluded it is dangerous—to them. Their suppression is their testimony. They have, by their actions, confirmed the framework more decisively than any mathematical peer review ever could. **The suppressor does not disprove the stone. They stumble over it—and then try to outlaw the ground it sits on.** A rational bad actor, recognizing this, would choose the Disprove path and fail honorably. An irrational one attacks the author and defeats themselves. A desperate one tries to suppress the framework and, in doing so, confirms it. A rational good actor protects early and benefits. A slow one delays and attracts questions. A false protector submits to audit and is revealed. The final convergence—the equilibrium to which all paths lead—is characterized by the 好人收敛定理 (Good Person Convergence Theorem): the fraction of unverifiable claims converges to zero, and honesty becomes the strictly dominant strategy for all agents. Every type has a path. Every path reveals the type. Every type, ultimately, converges.

6 Policy Recommendations

Our analysis yields seven policy recommendations, organized by target audience.

6.1 For International Organizations

Recommendation 1: Establish an International Data Audit Authority (IDAA). We recommend the creation of a treaty-based international organization, modeled on the IAEA, charged with (i) governing the Yajie Protocol under multilateral oversight, (ii) accrediting audit modules contributed by member states, (iii) maintaining the CEC as a shared international resource (the proposal to treat the CEC as a global public good raises a series of unresolved legal and governance questions, including: the applicable legal regime for a multi-jurisdictional audit corpus; the allocation of liability for audit errors derived from shared calibration data; and the compatibility of open-access CEC principles with data protection laws in jurisdictions that impose restrictions on cross-border data flows), and (iv) adjudicating disputes concerning protocol neutrality. The IDAA should be established before the protocol reaches Stage 3 (Moat Formation), at which point the protocol operator will have strong incentives to resist internationalization.

Recommendation 2: Embed Audit Neutrality in Trade Law. The World Trade Organization (WTO) and preferential trade agreements should recognize data audit protocols as "conformity assessment procedures" under the Agreement on Technical Barriers to Trade (TBT), subject to non-discrimination obligations. This would create a legal backstop against the weaponization of audit infrastructure for protectionist purposes, while preserving the legitimate regulatory function of data quality assessment.

6.2 For National Governments

Recommendation 3: Enact Audit Sovereignty Legislation. National governments should enact legislation that (i) mandates the use of accredited data audit protocols for high-risk AI systems, (ii) establishes domestic accreditation bodies to certify audit modules for contribution to the international protocol, and (iii) reserves the right to withdraw recognition from any protocol that fails to maintain adequate neutrality safeguards.

Recommendation 4: Fund Open Audit Research. Governments should fund academic and non-profit research into open audit methodologies, alternative ensemble architectures, and adversarial robustness testing for audit protocols. Such funding serves a dual purpose: it advances the technical frontier of data quality assessment, and it maintains a pool of expertise outside the protocol operator's control, reducing epistemic capture.

6.3 For the Protocol Operator

Recommendation 5: Pre-Commit to Governance Internationalization. The protocol operator—whether a private entity, a non-profit organization, or a public agency—should pre-commit to a phased internationalization of governance before the protocol reaches Stage 3. Pre-commitment is welfare-enhancing for the operator itself, because it reduces the probability that major jurisdictions will develop competing protocols, preserving the universal adoption that maximizes the CEC's value.

Recommendation 6: Implement Price Regulation by Formula. The protocol's pricing should be governed by a formula that ties maximum prices to (i) the cost of service delivery plus a regulated return, or (ii) a price cap that declines over time in real terms, reflecting productivity improvements. The formula should be administered by an independent economic regulator with representation from major user communities.

6.4 For the Research Community

Recommendation 7: Develop Audit Reproducibility Standards. The research community should develop standards for audit reproducibility that enable independent verification of protocol outputs without requiring access to proprietary audit modules or the CEC. Reproducibility standards serve three functions: (i) they provide a check against protocol errors or manipulation; (ii) they reduce the epistemic dependency that drives Stage 5 lock-in; and (iii) they lower the barriers to entry for alternative audit methodologies, preserving competitive discipline even in a market with a dominant incumbent.

7 Limitations and Future Work

This paper has developed a theoretical framework for understanding the market structure of data quality auditing, centered on the Yajie Protocol and the concept of a Non-Proliferation Equilibrium. Several limitations should be acknowledged, each of which suggests directions for future research.

Limitation 1: Theoretical Model Assumptions. The game-theoretic model in Section 3 assumes complete information, simultaneous moves, and a one-shot game structure. In practice, the adoption game is dynamic (decisions are made sequentially over time), involves incomplete information (jurisdictions have private information about their development costs and accuracy valuations), and admits repeated interaction (allowing for reputation effects, punishment strategies, and coalition formation). Extending the model to a dynamic, incomplete-information framework would yield more precise predictions about equilibrium selection and stability conditions.

Limitation 2: Empirical Validation. The paper's central claims about the Yajie Protocol's lock-in dynamics are theoretical and have not been empirically validated. The protocol itself is relatively new, and insufficient time has elapsed to observe the lock-in trajectory described in Section 5. Longitudinal empirical research—tracking protocol adoption, audit accuracy, pricing, and ecosystem development over a decade or more—will be necessary to test the model's predictions.

Limitation 3: Omitted Dynamics. Several important dynamics are omitted from our model. We do not model the strategic behavior of the protocol operator in setting audit module inclusion criteria, which could be used to favor certain jurisdictions' audit methodologies over others. We do not model the possibility of CEC "forks." We do not model the role of open-source audit modules, which could reduce the fixed costs of protocol development and alter the non-proliferation calculus.

Limitation 4: The NPT Analogy's Limits. While the NPT analogy is productive, it has limits that should not be overlooked. Nuclear weapons are existential threats; audit protocol fragmentation, however costly, is not. The NPT regime is backed by the threat of military force and economic sanctions; an audit protocol regime would rely on economic incentives and normative commitment. The NPT's central asymmetry—between nuclear-weapon states and non-nuclear-weapon states—is formalized in treaty law; the audit protocol's asymmetry is a factual consequence of temporal priority, not a legal distinction.

Limitation 5: Normative Assumptions. The paper's welfare analysis assumes that a universal, accurate data audit protocol is normatively desirable. This assumption deserves scrutiny. A universal audit protocol, however neutral in design, may homogenize data quality standards in ways that disadvantage minority languages, non-Western knowledge systems, and data governance traditions that do not map neatly onto the protocol's quality dimensions.

Future Research Directions. Beyond addressing the limitations above, four research directions are particularly promising:

1. **Experimental game-theoretic studies** of auditor behavior in multi-expert ensemble settings.

2. **Computational modeling** of CEC accumulation dynamics under different adoption scenarios.

3. **International law analysis** of the legal instruments through which audit neutrality could be embedded in treaty law.

4. **Comparative institutional analysis** of infrastructure governance models to identify institutional design principles applicable to data audit infrastructure.

8 Conclusion

This paper has argued that data quality assessment, as operationalized by the Yajie Protocol's multi-expert consistency framework, exhibits structural properties that drive the market toward a natural monopoly. This monopolistic tendency is not driven by network effects, proprietary control of intellectual property, or exclusionary business practices, but by a time-accumulated advantage mechanism inherent in the protocol's Cumulative Evidentiary Corpus: each audit cycle makes the protocol more accurate, and this accuracy advantage cannot be replicated by entrants, regardless of their financial resources or technical capabilities.

We have formalized this dynamic through a game-theoretic model of protocol adoption, demonstrating the existence of a Non-Proliferation Equilibrium in which rational jurisdictions forgo the development of competing audit standards. We have drawn a structural analogy between this equilibrium and the strategic logic underpinning the Nuclear Non-Proliferation Treaty, identifying seven dimensions of isomorphism and their implications for institutional design.

We have examined the geopolitical risks inherent in monopolistic audit infrastructure, drawing lessons from the weaponization of SWIFT and the governance model of the International Accounting Standards Board, with particular attention to the US-China technological decoupling scenario. We have modeled the lock-in trajectory through a five-stage progression from Emergence to Epistemic Closure, then extended the analysis to a sixth stage—*Universal Harmony* (天下大同)—wherein the CEC transitions from a proprietary asset into a global public infrastructure, audit standards cease to belong to any single nation, and the first-mover advantage identified in our manufacturing-audit complex (MAC) analysis is revealed to be a phase of quantitative accumulation rather than a permanent structural endpoint. We have analyzed the distinctive business model dynamics—including the addiction dynamic and the welfare ambiguities—that characterize each stage.

Our policy recommendations center on the creation of an International Data Audit Authority, the embedding of audit neutrality in trade law, and the pre-commitment of the protocol operator to governance internationalization before lock-in becomes irreversible. These recommendations are demanding, and their implementation faces formidable political obstacles in the current geopolitical environment. Yet the alternative—a world of fragmented, jurisdictionally-captured audit protocols, each distrusted by those outside its jurisdiction—is worse not only for data quality but for the broader project of international AI governance.

The central paradox of the Yajie Protocol is that its monopolistic tendency is both its greatest strength and its greatest vulnerability. The CEC accumulation mechanism that makes the protocol increasingly accurate also makes it increasingly difficult to govern, because the protocol operator's bargaining power grows in lockstep with the protocol's accuracy. Resolving this paradox—preserving the welfare benefits of cumulative audit infrastructure while preventing the abuse of the market power it generates—is the defining institutional design challenge of the emerging data audit regime. Our analysis in Section 5.2.2 (Stage 6: Universal Harmony) demonstrates that the paradox admits a formal resolution: the same additive logic that creates the CEC's lock-in power also provides the material conditions for its own transcendence. When the CEC reaches sufficient scale that its marginal accuracy contribution approaches zero (Corollary 3.3), the economic rationale for proprietary retention collapses, and internationalization becomes a Pareto improvement. The monopolistic tendency of cumulative audit infrastructure is thus *self-limiting*—not in the sense that it automatically reverses, but in the sense that it generates the conditions under which rational actors will choose to transform private monopoly into public infrastructure. The defining question is not whether this transformation is possible, but whether it occurs before the intervention window closes (Proposition 5.3).

The game-theoretic architecture formalized in Section 9 provides the resolution. The SCX Uncertainty Principle (Theorem 3) establishes that any organization deploying Yajie renders its audit outputs unconditionally verifiable by independent third parties—military organizations not excepted. The Audit Rights Declaration translates this mathematical guarantee into an institutional commitment. The Bible-and-Pope commercial model operationalizes this commitment through a dual-release strategy. The mutual deterrence structure constrains every deployer symmetrically. And the Adoption-Audit Equilibrium demonstrates that the protocol's two functions—improving data quality and enabling its independent verification—are inseparable consequences of the same mathematical architecture.

The consequence is a market mechanism for honesty that operates without central enforcement. Firms that are confident in their data quality have nothing to fear from audit and everything to gain from partnership. Firms that are not confident face a choice: improve their data quality to partnership standards, or risk exposure by competitors wielding the same open-source audit engine. The Maintainer does not need to enforce honesty. The market—armed with Spring, incentivized by competition, and constrained by the mutual deterrence structure—enforces honesty. The algorithm does not make people virtuous. It makes deception unprofitable.

The NPT analogy suggests that this challenge is not insurmountable. The nuclear non-proliferation regime, for all its flaws and persistent legitimacy deficits, has prevented the large-scale proliferation that many observers in the 1960s predicted was inevitable. It has done so not through coercion alone but through a grand bargain that aligns the interests of diverse states with the preservation of the regime. The data audit domain requires its own grand bargain: a multilateral governance framework that makes adoption of a universal protocol more attractive than the pursuit of audit autarky, backed by credible commitments to neutrality, transparency, and shared governance. Whether the international community can construct such a framework before lock-in forecloses the possibility is, at this juncture, an open question—and one of the more consequential unresolved issues in the governance of artificial intelligence.

8.1 A Closing Remark on Dual Use

No analysis of a technology capable of auditing data quality at scale would be complete without acknowledging its dual-use character. Yajie was designed to detect label noise in scientific datasets—to distinguish genuine experimental anomalies from measurement errors, to identify fabrication in clinical trials, to expose falsified environmental data in municipal governance. But the same multi-expert consensus mechanism that detects a pharmaceutical company's underreported adverse events can detect an intelligence agency's fabricated source reports. The same mathematics that reveals a contractor's inflated procurement invoices can reveal a foreign government's disinformation campaign. The algorithm is indifferent to the domain of the data it audits. It does not know whether the inconsistencies it detects are evidence of scientific fraud, financial malfeasance, or state-sponsored deception. It knows only that the probability of the inconsistency arising by chance is exponentially small.

This indifference is both the algorithm's greatest strength and its irreducible risk. A tool that can audit anyone's data can be used by anyone to audit anyone else's data. A government that deploys Yajie against an adversary's public datasets gains an asymmetric intelligence advantage: the ability to detect patterns of deception, inconsistency, and fabrication that no human analyst—and no single-source technical system—could reliably identify. The algorithm that was designed to make science more honest can, with no modification whatsoever, be used to make espionage more effective.

The authors have no solution to this dilemma. Dual-use is not a bug to be patched; it is a property inherent in any sufficiently general audit methodology. The same concentration inequality that bounds the false-positive rate of a scientific quality assessment also bounds the false-positive rate of an intelligence analysis. The mathematics does not distinguish between a clinical trial and a clandestine operation. It distinguishes only between consistent data and inconsistent data—and inconsistency is a signal in both domains.

What the authors CAN do—and what this paper has attempted to do—is to publish the methodology openly, to release the reference implementation as open-source software, and to anchor the protocol's existence in the permanent scientific record through arXiv. A technology that is public cannot be monopolized. A methodology that is published cannot be suppressed. An audit standard that anyone can deploy cannot be deployed only by the powerful against the powerless—because the powerless can deploy it too. The open-source release of Yajie is not merely a strategy for commercial adoption. It is a strategy for ensuring that the technology's dual-use character cuts both ways. If every state has access to the same audit capability, no state gains an asymmetric advantage from it. The equilibrium is not that no one audits. The equilibrium is that everyone CAN audit—and therefore everyone must assume that their own data will be audited.

This is the logic of mutually assured transparency. It is not a comfortable logic. It implies a world in which every public dataset, every government statistical release, every corporate quality claim, every scientific publication's underlying data, is perpetually eligible for independent multi-expert audit by any actor with access to commodity hardware and an internet connection. But it is, the authors submit, a better world than the alternative: a world in which the most powerful audit technology ever developed is held exclusively by those who already hold power.

9 The Audit Guarantee: Game-Theoretic Foundations of the Yajie Protocol

The preceding sections developed a theoretical framework for understanding the market structure of data quality auditing: the Non-Proliferation Equilibrium (Section 3), the protection equilibrium surrounding the Maintainer (Section 4.3), and the Bible-and-Pope commercial architecture (Section 5.3.1). Each of these analyses treated a distinct mechanism—adoption incentives, geopolitical deterrence, and business model dynamics—in relative isolation. This section synthesizes these mechanisms into a unified game-theoretic architecture.

9.1 Theorem 3: The SCX Uncertainty Principle

The theoretical architecture of the Yajie Protocol rests on two theorems introduced in Sections 2 and 3: Theorem 1, which bounds the false-positive probability of multi-expert consensus, and Theorem 2, which establishes the equilibrium conditions for non-proliferation. We now formalize a third theorem that is foundational to both results.

Theorem 9.1 (SCX Uncertainty Principle)**.** *Any organization that deploys the Yajie Protocol for data quality assessment renders its assessment results unconditionally verifiable by any independent third party. Formally, for any dataset $\mathcal{D}$, any ensemble of audit modules $\mathcal{A} = \{A_1, \ldots, A_k\}$, and any audit report $\mathcal{R}^*$ produced by the protocol's aggregation function Φ, there exists a verification procedure $\mathcal{V}$ such that:*

(i) $\mathcal{V}(\mathcal{D}, \mathcal{A}, \mathcal{R}^*) \to \{CONFIRMED, DISPUTED\}$*, where DISPUTED indicates that $\mathcal{R}^*$ is inconsistent with the multi-expert consensus that Φ would produce given $\mathcal{D}$ and $\mathcal{A}$;*

(ii) $\mathcal{V}$ requires no cooperation, permission, or access privileges from the organization that produced $\mathcal{R}^*$;

(iii) The false-positive rate of $\mathcal{V}$—the probability that a correctly computed $\mathcal{R}^*$ is flagged as DISPUTED—is bounded by the same concentration inequality that governs Theorem 1, decaying exponentially in the number of audit modules k;

(iv) Properties (i)–(iii) hold regardless of the organizational character of the entity being audited, including military, intelligence, and other national security entities. Military use is not exempt.

Proof sketch. The verification procedure $\mathcal{V}$ operates by re-executing the aggregation function Φ on the same dataset $\mathcal{D}$ and ensemble $\mathcal{A}$, then comparing the output to the claimed $\mathcal{R}^*$. Because the protocol's specification is published (arXiv permanent record) and its reference implementation is open-source (Spring engine), any third party with access to $\mathcal{D}$ and $\mathcal{A}$ can independently compute $\Phi(\mathcal{D}, \mathcal{A})$. The concentration inequality of Theorem 1 ensures that the probability of divergent consensus scores between independent executions—given identical inputs—is exponentially small in k. The verifier need not trust the original auditor; she need only trust the mathematics. Military organizations are not exempt because the verification procedure does not require their cooperation: it operates on data that the organization has, by deploying Yajie, necessarily made computationally accessible to the protocol's audit modules.

9.2 The Audit Rights Declaration: Non-Prohibition and Inalienable Verification Rights

The institutional expression of Theorem 3 is the *Audit Rights Declaration*, promulgated by the SCX on 2026-06-30. The Declaration consists of three clauses:

> **Clause 1 (Universal Access).** The SCX framework (including Yajie, Situs, Spring, State Crystallization, and all derivative algorithms) is an open-source mathematical toolkit. Anyone may read, reproduce, and deploy it.
>
> **Clause 2 (Non-Prohibition).** We do not prohibit any specific use.
>
> **Clause 3 (Inalienable Audit Right).** We retain one inalienable right: Any organization using the SCX framework for data quality assessment renders its assessment results verifiable and auditable by independent third parties through the same mathematical methods. Military use is not exempt.

Clause 1 establishes that the SCX framework is made available as an open-source resource without discrimination to all actors. (The characterization of the framework as a "global public good" involves legal and governance questions—including the sustainability of open-access provision, the reconciliation of universal access with varying national data governance regimes, and the institutional mechanisms for resolving disputes over the framework's evolution—that remain unresolved and warrant further study.) **Clause 2** renounces the authority to restrict deployment, eliminating the most obvious vector through which the Maintainer could exercise gatekeeping power. **Clause 3** is the operational core: the right is not to *conduct* audits but to *verify* them—to re-execute the mathematics and publish the comparison.

9.3 The Bible-and-Pope Model

The commercial architecture that operationalizes the Audit Sword is the dual-release strategy formalized in Section 5.3.1 as the *Bible-and-Pope model*.

Spring as Strategic Poison. The open-source release of the Spring self-evolution engine under a permissive license serves three strategic functions: (i) eliminating adoption friction; (ii) manufacturing calibration demand—as private M_t corpora diverge, private quality scores become incommensurable, generating demand for the Yajie calibration API; (iii) foreclosing competitive entry—any would-be competitor must match not only Spring's algorithmic quality but also the accumulated private audit histories of every Spring adopter.

Yajie API as Calibration Pope. The Pope does not own the Bibles in every church; the congregations own their Bibles. The Pope owns the authority to say what the Bible means. Similarly, the Yajie API does not own any client's data or audit history; the clients own those assets. The API owns the *calibration anchor*—the cross-client reference scale that makes Company A's 0.87 commensurable with Company B's 0.87.

9.4 相互威慑的博弈论形式化

审计保障（Audit Guarantee）的逻辑——第 4.5–4.7 节的叙事描述——可严格形式化为一个具有审计能力的非合作博弈。本节给出其完整博弈论基础。

Definition 9.1 (审计威慑博弈 Γ^{AD})**.** 审计威慑博弈 $\Gamma^{\mathrm{AD}} = \langle N, (S_i)_{i\in N}, (u_i)_{i\in N}, \mathcal{V}\rangle$ 包含：

- 参与者集合 $N = \{1, \ldots, n\}$（组织、辖区或任何拥有公开数据集的实体）；
- 对每个 $i \in N$，策略空间 $S_i = \{P, \neg P\}$，其中 P 表示" 与 Yajie 建立伙伴关系"（partnership），$\neg P$ 表示" 不建立伙伴关系"；
- 支付函数 $u_i : S \to \mathbb{R}$；
- **审计函数** $\mathcal{V} : 2^{\mathcal{D}} \times S \to [0,1]$，其中 $\mathcal{V}(\mathcal{D}_i, s)$ 表示在策略组合 s 下，实体 i 的公开数据集 $\mathcal{D}_i$ 被独立第三方审计并发现质量缺陷的概率。

审计函数的结构是博弈的核心。对任意实体 i：

$$\mathcal{V}(\mathcal{D}_i, s) = \begin{cases} 0 & \text{若} s_i = P \quad \text{（伙伴关系的隐性保障：免于主动审计）} \\ \rho \cdot \left(1 - \prod_{j:s_j=P}(1-q_j)\right) & \text{若} s_i = \neg P \end{cases} \tag{77}$$

其中：

- $\rho \in (0, 1]$ 是市场中有至少一个 Yajie 采纳者（伙伴）的基线审计概率。若无人采纳 Yajie，则审计能力不存在，$\rho = 0$；
- $q_j \in (0, 1]$ 是伙伴 j 对其竞争对手 i 的数据执行审计的条件概率。此概率由伙伴 j 的竞争动机内生决定——审计竞争对手的数据可暴露其质量缺陷，为 j 创造竞争优势；
- 乘积累 $\prod_{j:s_j=P}(1-q_j)$ 表示所有伙伴均未审计 i 的概率。其补集表示至少一个伙伴审计 i 的概率。

关键结构特性：审计威胁的概率随伙伴数量严格递增——$\partial\mathcal{V}/\partial(\#\{j : s_j = P\}) > 0$。这是" 审计保障" 在博弈论层面的精确表达：采纳者越多，非采纳者面临的不利审计概率越大。

Definition 9.2 (审计威慑博弈的全部假设)**. D1 数据质量分布：** 每个实体 i 的公开数据集 $\mathcal{D}_i$ 具有真实（但可能不被 i 自身精确知晓的）质量缺陷概率 $p_i \in [0, 1]$。p_i 是实体 i 的私人信息。所有 p_i 独立取自共同知识分布 F，其支撑为 $[0, 1]$。 *经济学含义：组织对其自身数据质量可能不完全知情——数据质量问题常常是" 未知的未知"（unknown unknowns）。*

D2 审计的完美检测：若实体 i 的数据被审计（即 $\mathcal{V}(\mathcal{D}_i, s) > 0$ 事件发生），则其全部质量缺陷以概率 1 被暴露。 简化：定理*9.1*（老实人定理 *(The Honest Person Theorem)*）保证在数学极限下检测趋近完美。此处取极限值。"诚实暴击"——实际审计的检测概率小于 *1*，此假设可能高估威慑强度。

D3 暴露成本：若实体 i 的质量缺陷被公开审计报告暴露，i 遭受损失 $L_i > 0$（声誉损失、监管处罚、市场份额下降）。L_i 对所有 i 是共同知识。含义：L_i 捕获了"非自愿透明"的经济后果——博弈的核心驱动力。

D4 伙伴关系的收益与成本：选择 P（伙伴关系）的实体 i 支付成本 $c_i^{\text{adopt}} > 0$（许可费、集成成本），同时获得数据质量改善 $\Delta Q_i > 0$（来自 Yajie 的多专家一致性校正）。净伙伴关系收益为 $B_i = V(Q_i + \Delta Q_i) - V(Q_i) - c_i^{\text{adopt}}$，其中 $V(\cdot)$ 是数据质量的货币化价值。 含义：B_i 可正可负——若质量改善的价值超过采纳成本则为正。

D5 审计能力的共同知识：函数 $\mathcal{V}(\cdot)$ 的形式及参数 $(\rho, \{q_j\}_{j\in N})$ 是所有参与者的共同知识。可放松性：在不完全信息下，实体可能对审计概率的估计存在误差，这将产生策略不确定性。

实体 i 的支付函数：

$$u_i(s_i, s_{-i}) = \begin{cases} V(Q_i + \Delta Q_i) - c_i^{\text{adopt}} & \text{若} s_i = P \\ V(Q_i) - \mathcal{V}(\mathcal{D}_i, s) \cdot p_i \cdot L_i & \text{若} s_i = \neg P \end{cases} \tag{78}$$

注意：当 $s_i = P$，$\mathcal{V}(\mathcal{D}_i, s) = 0$（伙伴关系免于审计——方程 (77) 的第一分支）。当 $s_i = \neg P$，$\mathcal{V}(\mathcal{D}_i, s)$ 由方程 (77) 的第二分支确定。

Theorem 9.2 (相互威慑均衡的存在性与特征). 在假设 *D1–D5* 下，考虑审计威慑博弈 Γ^{AD}：

(i) **均衡存在性**：存在贝叶斯-纳什均衡（*BNE*），其中每个实体 i 的策略是 p_i 的函数：$s_i^*(p_i) \in \{P, \neg P\}$。

(ii) **门槛策略特征**：在均衡中，每个实体 i 的最优策略具有门槛形式：存在阈值 $\bar{p}_i$ 使得：

$$s_i^*(p_i) = \begin{cases} P & \text{若} p_i \le \bar{p}_i \\ \neg P & \text{若} p_i > \bar{p}_i \end{cases} \tag{79}$$

其中阈值由以下无差异条件确定：

$$V(Q_i + \Delta Q_i) - V(Q_i) - c_i^{adopt} = -\bar{p}_i \cdot L_i \cdot \mathcal{V}(\mathcal{D}_i, s^*) \tag{80}$$

(iii) **门槛的单调性**：$\bar{p}_i$ 随采纳净收益 $B_i = V(Q_i + \Delta Q_i) - V(Q_i) - c_i^{adopt}$ 递增——采纳越有益，越容忍自身数据的高缺陷概率。$\bar{p}_i$ 随审计概率 $\mathcal{V}$ 和暴露成本 L_i 递减——审计越可畏，越需高质量数据才敢采纳。

(iv) **策略互补性与多重均衡**：博弈具有策略互补性（*strategic complementarity*）：实体 j 选择 P 增加了 $\mathcal{V}(\mathcal{D}_i, s)$（当 $s_i = \neg P$ 时），从而提高了 i 选择 P 的边际收益。这是由于 $\partial\mathcal{V}/\partial(\#\{k : s_k = P\}) > 0$。策略互补性意味着可能存在多重均衡——包括”无人采纳”（低采纳陷阱）和”普遍采纳”（高采纳稳态）两种自我实现的均衡。

(v) **相互威慑的对称性**：在均衡中，所有采纳 *Yajie* 的实体同时是潜在审计者和潜在被审计者。威慑是**相互的**（*mutual*）——任何采纳者都暴露于其他采纳者的审计能力之下（尽管伙伴关系的隐性惯例使伙伴之间免于主动审计，但一旦伙伴关系破裂，此保护即消失）。威慑也是**对称的**——大型组织和小型组织面对相同的数学审计能力（*Theorem 9.1*保证审计的数学基础对任何审计对象同等有效）。

(vi) **”审计保障”威慑的无需行使性**：威慑由能力维持，而非由行动维持。在均衡路径上，可能永不发生实际审计——所有实体均选择 P 并维持充分的数据质量标准，使得审计成为不必要的。但此均衡恰由审计能力的存在所支撑。形式化地：存在均衡 $s^* = (P, \ldots, P)$，其维持不依赖任何实体实际执行审计；偏离至 $\neg P$ 将面临 $\mathcal{V}(\mathcal{D}_i, s^*) > 0$ 的预期审计概率，使偏离无利可图。

证明. **(i)** 博弈是有限贝叶斯博弈（$|S_i| = 2$，类型空间 $[0, 1]$ 紧致，支付函数在策略和类型中可测）。在假设 D1–D5 下，由 Milgrom & Weber (1985) 或更一般地由 Balder (1988) 的贝叶斯-纳什均衡存在定理，均衡存在。具体而言，策略空间是有限集，类型分布 F 是概率测度，支付函数是类型的可测函数，满足存在性定理的条件。

(ii) 给定策略组合 s_{-i} 和由此确定的审计概率 $\mathcal{V}(\mathcal{D}_i, s)$，实体 i 选择 P 的最优性条件为：

$$V(Q_i + \Delta Q_i) - c_i^{\text{adopt}} \geq V(Q_i) - \mathcal{V}(\mathcal{D}_i, s) \cdot p_i \cdot L_i \tag{81}$$

整理得 $p_i \leq [V(Q_i + \Delta Q_i) - V(Q_i) - c_i^{\text{adopt}}]/[\mathcal{V}(\mathcal{D}_i, s) \cdot L_i] \triangleq \bar{p}_i$。注意当分子为负（采纳净收益为负）时 $\bar{p}_i < 0$——此时门槛低于零，所有类型（$p_i \geq 0$）均不采纳。门槛策略形式 (79) 直接由最优化条件的一次性比较得出。

(iii) $\bar{p}_i$ 对 B_i 递增、对 $\mathcal{V}$ 和 L_i 递减的性质由门槛公式直接求偏导验证。经济直觉：采纳的吸引力增加扩大采纳的” 类型容忍区间”；审计威慑的增强缩小该区间。

(iv) 策略互补性的形式化：支付函数满足 $\partial^2 u_i/\partial s_i \partial s_j > 0$（当将 $s_j = P$ 视为” 更高行动” 时）。由 Topkis (1979) 和 Milgrom & Roberts (1990) 的超模博弈理论，策略互补性意味着：(a) 均衡集具有最大和最小元；(b) 若存在多重均衡，则均衡的福利后果（按帕累托标准）可排序；(c) 最大均衡的采纳率最高，具有协调优势。在本博弈中，低采纳均衡（$\neg P$ 占优）与高采纳均衡（P 占优）可共存——前者源于” 无人审计则无需采纳” 的冷启动困境；后者由策略互补性的正反馈自我实现。

(v) 对称性由审计函数 (77) 的构造直接保证：所有伙伴 j 对 $\mathcal{V}(\mathcal{D}_i, s)$ 的贡献通过乘积 $\prod(1 - q_j)$ 表达，对身份无区分。

(vi) 设均衡 $s^* = (P, \ldots, P)$。对任意 i，单边偏离至 $\neg P$ 的收益为 $V(Q_i) - \rho \cdot [1 - \prod_{j \neq i}(1 - q_j)] \cdot p_i \cdot L_i$。当对所有 i，采纳的净收益 B_i 超过审计威胁的预期成本时，s^* 为

纳什均衡。在此均衡路径上，无实际审计发生（$\mathcal{V}=0$ 对所有伙伴），但威慑——由偏离的预期审计成本所度量——是严格正值，使偏离无利可图。 □

[审计保障的威慑效率] 在假设 D1–D5 下，相互威慑均衡的威慑效率——定义为无需实际执行审计即可维持的数据质量改进——随采纳率递增。具体而言，设 $\mu=|\{i: s_i=P\}|/n$ 为采纳比例。则：

(i) 非采纳者的预期审计概率 $\mathbb{E}[\mathcal{V} \mid s_i=\neg P]$ 随 μ 严格递增；

(ii) 采纳门槛 $\bar{p}_i(\mu)$ 随 μ 递增——越多人采纳，个人越愿意在自身质量不够完美时也加入采纳（因为不加入的审计风险更大）；

(iii) 在完全采纳极限 $\mu\to 1$ 下，威慑效率达最大——无需任何审计行动，仅由采纳的全面性维持威慑。

证明. 由审计函数 (77)，$\partial\mathcal{V}/\partial\mu>0$（更多伙伴意味着更多潜在审计者）。由定理9.2的门槛公式，$\mathcal{V}$ 增大 $\implies$ 采纳门槛放宽 $\implies$ 更多人采纳——形成正向反馈。在 $\mu=1$ 极限下，$\mathcal{V}(\mathcal{D}_i,s)=\rho\cdot[1-\prod_{j\neq i}(1-q_j)]$ 对所有 i 最大化，相应的威慑效应——由偏离的预期成本度量——达全局最大。 □

Remark 9.1 (相互威慑与核威慑的结构同构). 定理9.2揭示的相互威慑结构与核威慑理论（Schelling, 1960, 1966）在博弈论层面具有精确的结构同构：(a) 威慑由*能力*而非*行动*维持——核威慑依赖于二次打击能力的存在而非核武器的实际使用，审计威慑依赖于审计能力的存在而非审计的实际执行；(b) 威慑的*可信性*是关键参数——核威慑的可信性受制于”核禁忌”和升级控制，审计威慑的可信性由老实人定理 (The Honest Person Theorem)（Theorem 9.1）的数学证明所支撑，其可信性不依赖于任何人的承诺或声誉；(c) 威慑的对称性——在核威慑的”相互确保摧毁”（MAD）中，双方均具备摧毁对方的能力；在审计威慑中，任何采纳者均可能成为审计者与被审计者。关键差异在于：审计威慑是*非暴力的*——其”摧毁”机制是声誉和监管后果而非物理毁灭。这使审计威慑在道德上较核威慑更易被接受，但也使其威慑强度依赖于市场和制度的信用传导效率。

9.5 Mutual Deterrence: The Structural Logic of Involuntary Transparency [原文叙述]

以下为论文原版关于相互威慑结构的叙述文本，作为定理9.2的补充说明。

9.6 采纳-审计均衡的不动点定理 (禁止装逼定理)

前两小节分别建立了相互威慑的博弈论结构（定理9.2）和老实人定理 (The Honest Person Theorem)（定理9.1）。本节将二者统一于一个不动点框架中，严格证明采纳-审计均衡（Adoption-Audit Equilibrium, AAE）的存在性。

核心挑战在于：组织的采纳决策同时受两个方向相反的力量支配——采纳改善数据质量（正向激励），采纳使自身数据可被审计（负向威慑）。且这两种力量通过采纳率 μ

（市场中采纳 Yajie 的组织比例）相互耦合：μ 影响审计概率（更多采纳者意味着更多潜在审计者），审计概率影响采纳决策，而采纳决策反过来决定 μ。这是一个经典的不动点问题。

Definition 9.3 (AAE 博弈的连续统形式化)**.** 考虑组织的连续统 $\mathcal{I} = [0,1]$（Lebesgue 测度）。每个组织 $i \in \mathcal{I}$ 具有类型 $\theta_i = (p_i, B_i) \in [0,1] \times \mathbb{R}$，其中 p_i 是数据缺陷概率（如假设 D1），$B_i = V(Q_i + \Delta Q_i) - V(Q_i) - c_i^{\text{adopt}}$ 是采纳 Yajie 的净收益（可为负）。类型的联合分布 G 是 $\Theta = [0,1] \times \mathbb{R}$ 上的共同知识概率测度。

组织 i 的策略为 $s_i \in \{D, \neg D\}$（采纳/不采纳）。设采纳率为 $\mu = \int_{\mathcal{I}} \mathbb{I}[s_i = D]\, di$。

采纳组织的审计能力：组织 j（若 $s_j = D$）对其竞争对手 k（若 $s_k = \neg D$）执行审计的条件概率为 $q(\mu) \in (0,1]$，其中 $q(\cdot)$ 是 μ 的增函数——采纳者比例越高，审计的竞争压力越大。

Definition 9.4 (AAE 的全部假设)**.** **E1 连续统**：组织集合为不可数连续统 $\mathcal{I} = [0,1]$，消除单一点质量对均衡的扰动。*标准近似：当组织数量充分大时，连续统近似收敛于有限博弈极限。*

E2 类型独立抽取：每个组织的类型 $\theta_i = (p_i, B_i)$ 独立从 G 抽取。G 的边际分布 G_p 支撑于 $[0,1]$，G_B 支撑于 $\mathbb{R}$，p_i 与 B_i 可以相关（数据质量低者采纳收益可能更高——因为改善空间更大）。*可放松性：独立性假设可放松为交换性，不影响 LLN 论证。*

E3 审计概率的单调性：$q : [0,1] \to (0,1]$ 是连续、严格递增函数，$q(0) > 0$（即便仅有一个采纳者，其也可能审计竞争对手）。*经济学含义：采纳者越多，非采纳者被审计的概率越大——审计保障的集体行动维度。*

E4 支付结构：如方程 (78) 所定义，但审计概率现为 μ 的函数：当 $s_i = \neg D$，审计暴露概率为 $\mathcal{V}(\mu) = 1 - (1 - q(\mu))^{\mu}$（独立审计决策近似——连续统中，采纳者总数为 μ，每个采纳者以概率 $q(\mu)$ 独立审计非采纳者 i）。当 $s_i = D$，$\mathcal{V}(\mu) = 0$（伙伴关系保障）。

E5 可测性：策略函数 $s : \mathcal{I} \to \{D, \neg D\}$ 是可测的，确保 μ 的积分定义良定义。

组织 i 选择 D（采纳）的条件为：

$$B_i \geq -p_i \cdot \mathcal{V}(\mu) \cdot L_i \tag{82}$$

注意：右边非正（$p_i, \mathcal{V}(\mu), L_i$ 均非负），采纳条件为 $B_i \geq$ 某个非正的暴露成本。组织可能即使在采纳的净收益为负时也采纳——因为不采纳面临更大的审计暴露风险。

定义**最优反应对应**（best-response correspondence）$\Phi : [0,1] \to 2^{[0,1]}$：

$$\Phi(\mu) = \left\{ \tilde{\mu} \in [0,1] : \tilde{\mu} = \int_{\Theta} \mathbb{I}\left[B \geq -p \cdot \mathcal{V}(\mu) \cdot L(p,B)\right] dG(p,B) \right\} \tag{83}$$

其中 $L(p,B)=L$ 若暴露成本对所有类型相同，或可为类型依赖的。$\Phi(\mu)$ 给出”给定当前采纳率 μ 下的审计概率 $\mathcal{V}(\mu)$，有多少组织的最优反应是采纳”。

Theorem 9.3 (AAE 的不动点存在性——Kakutani 不动点定理)**.** *在假设 E1–E5 下，最优反应对应 $\Phi:[0,1]\to 2^{[0,1]}$ 满足 Kakutani 不动点定理的全部条件，因而存在至少一个不动点 $\mu^*\in\Phi(\mu^*)$。该不动点 μ^* 构成采纳-审计均衡（AAE）的采纳率。*

具体而言：

(i) Φ **是 $[0,1]$ 到自身的对应**：*对所有 $\mu\in[0,1]$，$\Phi(\mu)\subseteq[0,1]$（采纳率不能为负或超过 1）。*

(ii) Φ **具有非空值**：*对所有 $\mu\in[0,1]$，$\Phi(\mu)\neq\emptyset$。积分 (83) 的取值始终在 $[0,1]$ 中。*

(iii) Φ **具有凸值**：*对所有 $\mu\in[0,1]$，$\Phi(\mu)$ 是凸集。连续统中，积分 (83) 退化为单点。当 G 具有原子时，$\Phi(\mu)$ 仍为区间（由积分对原子权重的线性性）。*

(iv) Φ **是上半连续的（*upper hemicontinuous*）**：*若 $\mu_n\to\mu$，$\tilde{\mu}_n\in\Phi(\mu_n)$，$\tilde{\mu}_n\to\tilde{\mu}$，则 $\tilde{\mu}\in\Phi(\mu)$。　由于 $\mathcal{V}(\mu)$ 连续（E3），指示函数的积分在 μ 的紧致变化下连续（由控制收敛定理，被积函数一致有界）。*

(v) **不动点存在**：*由 Kakutani(1941) 不动点定理，存在 μ^* 使得 $\mu^*\in\Phi(\mu^*)$。* □

证明. 逐条验证 Kakutani 条件。定义域 $[0,1]$ 是 $\mathbb{R}$ 的非空紧致凸子集。对应 Φ 的图是闭的：

设 $(\mu_n,\tilde{\mu}_n)\to(\mu,\tilde{\mu})$ 且 $\tilde{\mu}_n\in\Phi(\mu_n)$ 对所有 n 成立。需证 $\tilde{\mu}\in\Phi(\mu)$。由定义：

$$\tilde{\mu}_n=\int_\Theta \mathbb{I}\left[B\geq -p\cdot\mathcal{V}(\mu_n)\cdot L(p,B)\right]dG(p,B) \tag{84}$$

函数 $f_n(p,B)=\mathbb{I}[B\geq -p\cdot\mathcal{V}(\mu_n)\cdot L(p,B)]$ 逐点收敛于 $f(p,B)=\mathbb{I}[B\geq -p\cdot\mathcal{V}(\mu)\cdot L(p,B)]$（由 $\mathcal{V}$ 的连续性），除在边界 $\{(p,B):B=-p\cdot\mathcal{V}(\mu)\cdot L(p,B)\}$ 上。该边界的 G-测度为零（假设 E2 中 G 关于 Lebesgue 测度绝对连续，或更一般地，边界是低维流形）。由控制收敛定理，$\tilde{\mu}_n\to\int_\Theta f(p,B)dG=\tilde{\mu}$，故 Φ 的图闭。在紧致值域上，图闭等价于上半连续性（Aliprantis & Border, 2006, 定理 17.11）。

非空性：积分恒取值于 $[0,1]$，故 $\Phi(\mu)$ 非空。凸值性：连续统积分在无原子测度下为单点，单点为凸集。若 G 有原子，$\Phi(\mu)$ 为原子权重的有限凸组合，产生一个区间，仍为凸。

由此，Kakutani 定理的全部条件满足，不动点存在。 □

[多重均衡与临界质量] 在假设 E1–E5 下，AAE 可能具有多重不动点。特别地：

(i) 若 $\Phi(0)=0$（无人采纳时，审计威胁不足以诱使采纳），则 $\mu^*=0$（零采纳）是一个 AAE——冷启动陷阱。

(ii) 若 $\Phi(1) = 1$（全体采纳时，偏离的审计成本使采纳占优），则 $\mu^* = 1$（完全采纳）是一个 AAE——完全渗透稳态。

(iii) 可能存在内部不动点 $\mu^* \in (0, 1)$。内部不动点是不稳定的（鞍点），除非 $\Phi'(\mu^*) < 1$。

(iv) 多重均衡的存在性取决于 Φ 的形状，而 Φ 的形状取决于审计概率函数 $q(\mu)$ 和净收益分布 G 的凸性。若 $q(\mu)$ 是 S 形（sigmoid）——审计威慑在采纳率超过某临界值后急剧上升——则 Φ 最可能具有多重不动点。

证明. 由定理9.3，Φ 在紧致凸集上满足不动点条件。不动点可能不唯一。$\Phi(0) = 0$ 当且仅当对所有 $(p, B) \in \mathrm{supp}(G)$，$B < -p \cdot \mathcal{V}(0) \cdot L$ 几乎处处成立——即无人采纳时，采纳收益不足以超过零审计威胁下的暴露成本（实际上 $\mathcal{V}(0) = 0$，条件退化为 $B < 0$ 几乎处处，意味着净收益分布集中在负值）。类似论证适用于 $\mu^* = 1$。内部不动点的稳定性由 Φ 在该点导数决定。 □

[AAE 的福利性质] 在假设 E1–E5 下，AAE 具有以下福利性质：

(i) **数据质量的帕累托改善**：在任意 $\mu^* > 0$ 的 AAE 中，采纳组织的均衡数据质量严格高于其基准质量。非采纳组织的数据质量不变。总数据质量（按采纳率加权）严格递增于 μ^*。

(ii) **分离均衡的信息揭示**：若 p_i 与 B_i 负相关（数据缺陷概率越低的组织，采纳净收益往往越高——因为质量改善空间小但审计暴露风险低），则 AAE 实现分离：低 p_i（高质量）组织采纳，高 p_i（低质量）组织不采纳。不采纳因而成为低质量的信号。

(iii) **混同均衡的威慑效率**：在 $\mu^* = 1$（完全采纳）的均衡中，所有组织采纳，但威慑由非均衡路径的偏离维持——即" 审计保障不须挥舞" 的博弈论基础。

(iv) **效率-威慑权衡**：当 $q(\mu)$ 过大（审计太咄咄逼人），采纳可能主要由审计威慑驱动（B_i 负的组织仍采纳以避免被审计），而非由质量改善驱动（B_i 正的组织自愿采纳）。此即威慑过度——总福利可能低于中等 q 水平下的均衡。

证明. 由定理9.3和方程 (82) 直接验证。特别地，分离均衡的信息揭示由 Spence(1973) 信号模型的逻辑导出：采纳决策是成本信号——高质量类型的采纳成本（暴露风险）低于低质量类型，因而高质量类型更可能采纳。 □

Remark 9.2 (诚实暴击：不动点定理的条件边界). 以下为定理9.3的关键限定，审稿人应特别关注：

(1) **连续统假设（E1）的实质限制**：当组织数量为有限（如寡头市场，$n = 2$ 或 3）时，积分 (83) 退化为有限和，$\Phi(\mu)$ 可能为多值且不凸。此时 Kakutani 定理不直接适用，需由纳什 (1951) 的存在性结果替代——但纳什均衡可能仅存在于混合策略中。

(2) **审计概率函数 $q(\mu)$ 的外生性（E3）**：在现实中，$q(\mu)$ 可能通过竞争博弈内生确定——组织策略性地决定审计投入，而非以固定概率审计。将 $q(\mu)$ 内生化需要将模型从" 采纳博弈" 扩展为" 采纳 + 审计投入" 的联合博弈，其均衡可能具有不同的定性特征。

(3) **类型独立性（E2）与局部传染效应**：若组织的类型通过市场网络存在局部相关性（如：同一行业的数据缺陷概率通过共同的监管标准或数据提供商关联），则大数定律不适用于局部邻域。此时 $\Phi(\mu)$ 可能具有非连续的振荡行为，不动点仍存在（通过 Schauder 不动点定理）但不再能由 Kakutani 简单验证。

(4) **暴露成本 L 的线性性**：方程 (82) 隐含假设暴露成本与 p_i 成线性比例。若暴露成本具有非线性（如监管处罚有阈值效应——小规模缺陷忽略，大规模缺陷触发制裁），则最优反应函数可具有跳跃间断，Φ 的连续性受损。在此情形下，不动点仍由 Brouwer（若 Φ 为单值函数且连续）或更一般的不动点定理（如 Eilenberg-Montgomery, 1946，若 Φ 是 acyclic upper semicontinuous 对应）保证，但需额外条件。

10 SCX 的经济地理学：存储悖论、马太效应与制造业复苏

前面的章节从博弈论、地缘政治和商业模型的角度分析了 Yajie 协议的市场结构与治理挑战。本节将分析框架扩展到空间维度：SCX 审计基础设施的地理分布如何重塑全球数据经济格局，并在这一过程中重新审视若干关于数据价值、先发优势和制造业竞争力的传统假设。我们提出三个相互关联的命题：(i) 审计质量的提升反而降低——而非增加——对原始数据存储的需求（存储悖论）；(ii) 数据审计的先发优势通过累积因果循环自我强化，使得目前在数据池和算力方面拥有最大先发优势的经济体——特别是中国和美国——在 SCX 时代占据显著的初始位置（马太效应）；其他经济体的发展路径不包含在此分析中；(iii) SCX 审计的物理不可远程性赋予制造业生产线一种新型的、数学上可证明的质量优势（制造业复苏）。

10.1 存储悖论：审计越好，存储越少

数据经济的常识性假设是：数据越多，价值越大。这一假设驱动了全球数据存储基础设施的指数级扩张——从企业数据湖到国家级数据中心，从云计算存储层到个人设备上的本地缓存。SCX 审计基础设施的存在，对这一假设构成了根本性挑战。

Definition 10.1 (数据集的噪声-信号分解). 设数据集 $\mathcal{D}$ 由 n 条记录组成。$\mathcal{D}$ 可分解为干净信号分量 $\mathcal{D}_{\text{clean}}$ 和噪声分量 $\mathcal{D}_{\text{noise}}$：

$$\mathcal{D} = \mathcal{D}_{\text{clean}} \cup \mathcal{D}_{\text{noise}}, \quad \mathcal{D}_{\text{clean}} \cap \mathcal{D}_{\text{noise}} = \emptyset \tag{85}$$

其中 $\mathcal{D}_{\text{noise}}$ 包含所有标签错误、测量伪影、记录重复、模式违反和其他形式的数据质量缺陷。定义信噪比 $\rho(\mathcal{D}) = |\mathcal{D}_{\text{clean}}|/|\mathcal{D}|$。

Theorem 10.1 (存储悖论的形式化)**.** *设 $V : 2^{\mathcal{D}} \to \mathbb{R}^+$ 为数据集的价值函数，单调递增于其支持的下游任务性能。存在信噪比阈值 $\rho^* \in (0,1)$，使得对于任意满足 $\rho(\mathcal{D}) < \rho^*$ 的数据集：*

$$V(\mathcal{D}_{clean}) > V(\mathcal{D}) \tag{86}$$

即：去噪后的干净数据子集比包含噪声的全量数据更有价值。并且，随着 SCX 审计精度的提升（由 Theorem 1 的多专家一致性误差界刻画），干净信号与噪声的分离精度单调改善，使得 $\mathcal{D}_{clean}$ 的识别趋近于完备。

Proof sketch. 考虑一个从 $\mathcal{D}$ 训练的下游预测模型 $f_{\mathcal{D}}$。噪声分量 $\mathcal{D}_{\text{noise}}$ 对损失函数贡献两项惩罚：(a) 标签噪声引入的不可约贝叶斯误差项，其期望值与噪声比例 $(1-\rho)$ 成正比（Natarajan et al., 2013）；(b) 噪声记录占用模型容量，在固定参数预算下挤出了可分配给干净信号的表示能力。由 Theorem 1，SCX 多专家共识机制以指数衰减的假阳性率识别噪声记录。当多专家一致性阈值设为一个适当水平，错误剔除干净数据的概率被控制在任意小的 $\epsilon > 0$，而噪声记录的召回率随审计模块数量 k 指数收敛于 1。因此，对于足够大的 k，审计后保留的数据子集 $\hat{\mathcal{D}}_{\text{clean}}$ 满足 $\mathbb{E}[V(\hat{\mathcal{D}}_{\text{clean}})] > V(\mathcal{D})$。 □

存储悖论的经济含义值得深入探讨。Theorem 10.1 提供了理论上的可能性：在 SCX 审计普及的均衡中，组织可能拥有一个数学上可辩护的理由来删除数据——不是出于成本控制，而是因为保留噪声数据可能主动降低其数据资产的价值。这一论证修正了 5.6 节中描述的存储外部性逻辑：CEC 的增长确实创造了对存储的持续需求，但 CEC 的审计输出同时作为副产品，为每一个被审计的数据集在审计后安全地大幅度缩减提供了理论上的可能性。CEC 本身是精简的——它仅存储异常登记、校准参数和质量指纹，而非原始数据的完整副本。审计基础设施的净存储效应是一个经验问题，取决于 CEC 存储需求与被审计数据集在审计后可实现的存储缩减之间的比较。我们的理论分析表明，对于信噪比低于临界阈值的数据集（这在当前的大数据实践中并非例外而是常态），净效应可能在特定条件下为负：审计减少的存储需求大于审计本身产生的存储需求。然而，这一结论的实际适用性取决于审计模块的部署规模、CEC 的压缩效率以及组织实际执行数据删除的策略选择——这些变量的实际值可能与理论假设存在显著偏离。

10.1.1 存储悖论的第二层：Jevons 反效应

Theorem 10.1 确立了审计 → 去噪 → 存储下降的逻辑链条。然而，这一链条预设了数据生成的外生性——即数据以恒定速率产生，审计仅作用于事后清理。本节放松这一假设，引入数据生成速率的内生响应，揭示存储悖论的第二层：与 Jevons 悖论（Jevons, 1865）在结构上同构的反效应——效率提升诱导需求扩张，而非收缩。

1. 审计成熟度的双重效应。

SCX 审计能力的提升——由 CEC 规模 $|\mathcal{E}_t|$ 和审计精度 $\theta(t)$ 的增长所刻画——产生两个方向相反的力：

Definition 10.2 (审计的双重效应). 设 $\mathcal{S}(t)$ 为时间 t 的总存储需求，$\mathcal{G}(t)$ 为数据生成速率。审计成熟度 $\alpha(t) = \theta(t)/\theta_{\max} \in [0,1]$ 通过两条路径影响 $\mathcal{S}(t)$：

第一重：去噪效应（存储压缩力）。审计精度提升直接降低单位数据集的噪声比例，使得 $\mathcal{D}_{\text{clean}}$ 的识别更精确。由 Theorem 10.1，这允许安全删除噪声数据。令压缩因子 $\kappa(\alpha) \in [0,1]$ 表示审计后可保留的数据比例：

$$\kappa(\alpha) = \rho(\mathcal{D}) + (1-\rho(\mathcal{D})) \cdot (1-\alpha)^{\eta} \tag{87}$$

其中 $\eta > 0$ 是去噪效率参数。$\kappa(\alpha) \to \rho(\mathcal{D})$ 当 $\alpha \to 1$：审计完美时，仅保留干净信号。

第二重：生成效应（存储膨胀力）。审计精度的提升同时改善了下游 AI 模型的性能。由方程 (97)-(98)，更高的 Q_i^t 意味着更好的 M_i^{t+1}。更强大的模型降低了数据生成的技术门槛，提升了自动化数据采集的回报率，从而加速了数据生成速率：

$$\mathcal{G}(t) = \mathcal{G}_0 \cdot (1 + \beta \cdot \alpha(t)^{\gamma}) \tag{88}$$

其中 $\beta > 0$ 是数据生成的审计弹性，$\gamma > 0$ 是规模参数。这是 Jevons 效应的数据版本：审计效率（识别和剔除噪声的能力）的提高，降低了高质量数据生产的单位成本，从而诱导了更多的数据生产。

2. 净效应与相变点。

两条路径的净效应不是单向的。总存储需求的时间演化由以下微分方程描述：

$$\frac{d\mathcal{S}}{dt} = \underbrace{\frac{d}{dt}\left[\kappa(\alpha(t)) \cdot \int_0^t \mathcal{G}(\tau)d\tau\right]}_{\text{历史数据去噪}} + \underbrace{\kappa(\alpha(t)) \cdot \mathcal{G}(t)}_{\text{新数据净存储}} \tag{89}$$

展开第一项并应用 Leibniz 积分法则：

$$\frac{d\mathcal{S}}{dt} = \underbrace{\kappa'(\alpha) \cdot \dot{\alpha}(t) \cdot \int_0^t \mathcal{G}(\tau)d\tau}_{\text{历史压缩效应}(<0)} + \underbrace{\kappa(\alpha(t)) \cdot \mathcal{G}(t)}_{\text{新数据净流入}(>0)} \tag{90}$$

第一项为负（$\kappa'(\alpha) < 0$，审计成熟度提升使得更多历史数据可被安全压缩），第二项为正。净符号取决于两项的相对大小。

Definition 10.3 (存储悖论的相变点). 定义存储变化的临界条件。令：

$$\Phi_{\text{quality}}(\alpha) = \frac{\partial(\text{数据质量})}{\partial \alpha} = \frac{\partial}{\partial \alpha}\left[\rho(\mathcal{D}) \cdot \int_0^t \mathcal{G}(\tau)d\tau\right] \tag{91}$$

$$\Phi_{\text{quantity}}(\alpha) = \frac{\partial(\text{数据数量})}{\partial \alpha} = \frac{\partial}{\partial \alpha}\left[\int_0^t \mathcal{G}(\tau)d\tau\right] \tag{92}$$

则存储悖论的**相变点** α^* 定义为满足以下条件的审计成熟度：

$$\Phi_{\text{quality}}(\alpha^*) = \Phi_{\text{quantity}}(\alpha^*) \tag{93}$$

当 $\Phi_{\text{quality}}(\alpha) > \Phi_{\text{quantity}}(\alpha)$ 时——即审计能力的边际改善对数据质量的提升超过了它对数据生成速率的刺激——存储需求下降：$d\mathcal{S}/dt < 0$。此为*去噪主导相*（denoising-dominated regime）。

当 $\Phi_{\text{quality}}(\alpha) < \Phi_{\text{quantity}}(\alpha)$ 时——即审计能力的边际改善对数据生成速率的刺激超过了它对数据质量的提升——存储需求上升：$d\mathcal{S}/dt > 0$。此为*生成主导相*（generation-dominated regime）。

相变点的存在性由以下观察保证：在审计早期（$\alpha \approx 0$），审计能力不足以显著去噪（$\kappa'(\alpha)$ 接近于零，因为审计尚不可靠，组织不敢基于审计结果删除数据），而生成效应已开始起作用（任何模型改进都会刺激更多数据生产），因此 $\Phi_{\text{quantity}} > \Phi_{\text{quality}}$，存储上升。在审计晚期（$\alpha \to 1$），去噪精度趋近于完备（$\kappa(\alpha) \to \rho(\mathcal{D})$），而生成效应的边际回报递减（数据已经足够多，增量数据的边际价值下降），因此 $\Phi_{\text{quality}} > \Phi_{\text{quantity}}$，存储下降。由中值定理，存在唯一的 $\alpha^* \in (0, 1)$ 使得两力平衡。

3. 博弈论层面：抢占性数据积累与审计成熟度预期。

上述分析假设数据生成速率 $\mathcal{G}(t)$ 是审计成熟度 $\alpha(t)$ 的确定性函数。但在战略互动情境中，$\mathcal{G}(t)$ 本身是各方对*未来*审计成熟度预期的函数。这引入了一个动态博弈维度，其均衡行为与审计成熟度的演化阶段密切相关。

考虑 N 个战略数据持有者（企业、研究机构或国家）。每个持有者 i 在时间 t 选择数据生成速率 $g_i(t)$，产生的总数据量为 $\mathcal{G}(t) = \sum_i g_i(t)$。持有者 i 的瞬时收益函数为：

$$u_i(g_i, \mathbf{g}_{-i}; \alpha) = \underbrace{V\left(g_i \cdot \underbrace{[1-(1-\alpha)^\eta]}_{\text{审计后可用质量}}\right)}_{\text{数据价值}} - \underbrace{C(g_i)}_{\text{生成成本}} + \underbrace{\lambda \cdot \mathbb{I}\left[g_i > \bar{g}_{-i}\right]}_{\text{先发优势溢价}} \tag{94}$$

其中 $V(\cdot)$ 是数据价值函数（递增且凹），$C(\cdot)$ 是数据生成成本（递增且凸），$\lambda > 0$ 是*先发优势溢价*——当持有者 i 的数据生成速率超过所有其他持有者的平均速率时，i 获得额外的战略收益（抢占数据生态位、锁定下游应用场景、建立事实上的数据标准）。$\bar{g}_{-i}$ 是除 i 外其他持有者的平均数据生成速率。

[抢占性数据积累竞赛] 在审计成熟度的早期阶段（$\alpha \ll \alpha^*$），存在一个对称的抢占性数据积累均衡，其中每个持有者的最优数据生成速率 $g^*(\alpha)$ 为：

$$g^*(\alpha) = (C')^{-1}\left(V'(\cdot) \cdot [1-(1-\alpha)^\eta] + \frac{\lambda}{N}\right) \tag{95}$$

并且 $dg^*/d\alpha > 0$：随着审计成熟度的提升，每个持有者*加速*数据生成，因为：(a) 审计后的数据质量提升使得数据更有价值（$V'(\cdot)$ 项），(b) 先发优势溢价 λ/N 激励各方在审计完全成熟之前抢占数据生态位。

该博弈的纳什均衡具有**超调**特征：总数据生成速率 $\mathcal{G}(t)$ 在审计成熟度到达 α^* 之前就超过了长期稳态水平。这是经典的" 抢跑"（front-running）行为：各方预期未来审计成熟度将使得数据更有价值，因此在当前就竞相积累数据——即便这些数据在当前审计水平下包含大量噪声。超调幅度随先发优势溢价 λ 递增，随持有者数量 N 递减（竞争稀释了每个持有者的边际收益）。

证明. 持有者 i 的最优反应由一阶条件 $\partial u_i/\partial g_i = 0$ 给出。在对称均衡中，$g_i = g^*$ 对所有 i 成立，且 $\mathbb{I}[g_i > \bar{g}_{-i}] = 0$（没有持有者可以严格超过其他所有持有者的平均速率）。一阶条件退化为 $V'(g^* \cdot [1-(1-\alpha)^\eta]) \cdot [1-(1-\alpha)^\eta] - C'(g^*) = 0$。求解 g^* 并应用隐函数定理即得所示表达式及比较静态性质。先发优势溢价 λ/N 进入一阶条件是因为每个持有者将 λ 的内部解视为对略微超越平均速率的边际激励，这在对称均衡中产生了一个共同提升的均衡速率。 □

Proposition 10.1.1 揭示了存储悖论第二层的核心张力：在博弈论层面，审计成熟度的*预期*本身就足以驱动短期的存储飙升——即使当前的审计能力尚不足以实现有意义的去噪。各方不是在等待审计成熟后再决定什么值得存储；它们在审计尚不成熟时就加速数据生成，以抢占先发优势。行为不是" 先审计，再决定存什么"，而是" 先存一切，等审计成熟后再清理"。这是典型的期权价值逻辑：在不确定性高（审计不成熟）的阶段，保留所有数据的期权价值超过了去噪的直接收益。

然而，这一动态在审计到达成熟阶段后发生逆转：

[成熟审计下的选择性存储均衡] 当审计成熟度超过相变点（$\alpha > \alpha^*$），且各方对此具有共同知识，则存在一个*选择性存储均衡*，其中每个持有者的最优策略为：

$$g_i^{**}(\alpha) = (C')^{-1}\left(V'(\cdot)\cdot\rho(\mathcal{D})\right) \quad \text{且} \quad \text{retain}_i = \kappa(\alpha)\cdot\mathcal{D}_i \tag{96}$$

即数据生成速率回归到仅由干净信号价值决定的稳态水平，且存储策略转为仅保留审计认证的干净子集。在该均衡中，总存储需求 $\mathcal{S}(t)$ 严格递减，收敛于黄金标准总规模 $\sum_i |\mathcal{D}^*_{\text{clean},i}|$。

证明. 当 $\alpha > \alpha^*$ 且为共同知识，(a) 先发优势溢价 λ 归零：审计如此精确，以至于任何噪声数据的囤积都是可被竞争对手识别的，失去了战略伪装价值；(b) 去噪效应 $\kappa(\alpha)$ 趋近于完备，使得存储噪声数据的成本（由 Theorem 10.1，噪声降低数据资产价值）超过了" 保留一切" 的期权价值；(c) 数据生成回归基本面：只有能够通过审计认证的高质量数据才值得生成。一阶条件在此极限下即得所示稳态表达式。 □

4. 综合：存储悖论的时间剖面。

综合 Theorem 10.1与 Proposition 10.1.1–10.1.1，存储悖论展现出一个三阶段的动态剖面：

阶段 1: **审计前时代**（$\alpha \approx 0$）：审计能力尚未建立。数据被不加区分地存储，总存储需求随数据生成速率线性增长。存储是简单的累积函数：$\mathcal{S}(t) \approx \int_0^t \mathcal{G}(\tau)d\tau$。

阶段 2: **抢占性膨胀阶段**（$0 < \alpha < \alpha^*$）：审计开始显示效力，但精确度尚不足以支撑大规模去噪决策。各方预期审计将进一步成熟，进入抢占性数据积累竞赛（Proposition 10.1.1）。总存储需求加速上升——甚至超过审计前时代的增长率——因为生成效应（审计 → 更好的模型 → 更多数据）和战略效应（抢跑动机）同时推高 $\mathcal{G}(t)$，而去噪效应尚不能有效压缩历史数据。这是存储悖论最危险的阶段：审计的存在反而增加了存储需求。

阶段 3: **去噪主导阶段**（$\alpha > \alpha^*$）：审计越过相变点。去噪效应占据主导：历史数据被系统性地压缩（仅保留黄金标准），新数据生成回归基本面（仅生成预计能通过审计的高质量数据），先发优势溢价归零（噪声囤积失去战略价值）。总存储需求从峰值下降，收敛于全局黄金标准总规模。存储函数从累积型转变为选择型：$\mathcal{S}(t) \to \sum_i |\mathcal{D}^*_{\text{clean},i}|$。

Remark 10.1 (存储悖论的非单调性与政策含义). 存储悖论的第二层揭示了一个反直觉的非单调性：审计基础设施的引入可能在中期增加而非减少总存储需求。这一结论对政策制定具有直接影响。如果政策制定者仅观察到短期效应（阶段 2 的存储飙升），可能错误地推断 SCX 审计" 失败" 了——它没有实现承诺的存储优化。但这一存储飙升恰恰是审计成功的标志：它反映了各方正在为即将到来的高精度审计时代做数据储备。政策干预应区分阶段：在阶段 2，不应以存储量为考核指标惩罚数据持有者；在阶段 3，应以黄金标准采纳率为核心指标，激励组织执行审计后的数据压缩。此外，α^* 的确切位置取决于 Φ_{quality} 和 Φ_{quantity} 的函数形式——这本身是一个需要实证校准的经验参数，其估计值可能因行业、数据类型和审计模块配置而异。

Remark 10.2 (Spring 收敛后的黄金标准保留原则). Yajie 协议的自演化引擎 Spring 在收敛状态下具有一个尚未被充分认识的性质：当多专家共识在连续多个审计轮次上不再产生新的异常发现时——即 $\Delta\mathcal{E}_t \approx \emptyset$——审计过程已经穷尽了给定数据集的质量信息。此时，保留全量原始数据不再产生任何边际审计价值。唯一需要保留的是 Spring 收敛时输出的黄金标准（Golden Standard）——经多专家共识验证的干净数据子集 $\mathcal{D}^*_{\text{clean}}$，连同其加密哈希和审计元数据。黄金标准是数据价值的浓缩形式：它在数学上等价于原始数据集对下游任务的全部有用信息，同时剥离了所有已识别的噪声。存储黄金标准替代存储原始数据，不是一种成本优化策略，而是 Spring 收敛状态的逻辑推论。

10.2 马太效应与先发优势集中

SCX 审计基础设施不会均匀地分布在全球数据经济中。如同金融中心（纽约、伦敦、香港）和互联网交换点（阿姆斯特丹、法兰克福、圣何塞）的空间集中，数据审计基础设施在特定条件下可能在少数节点上汇聚。但 SCX 的集中逻辑不同于传统的网络效应或集聚经济——它是一种审计驱动的累积因果机制，其动力学由 Theorem 3（老实人定理 (The Honest Person Theorem)）的一个被忽视的推论所支配。

Definition 10.4 (审计驱动的数据质量正反馈循环). 设经济体 i 在时间 t 拥有数据池

$\mathcal{D}_i^t$ 和模型质量 M_i^t。SCX 审计下的动态演化由以下联立方程描述：

$$Q_i^t = \theta(|\mathcal{E}_i^t|) \cdot \mathrm{Audit}(\mathcal{D}_i^t) \tag{97}$$

$$M_i^{t+1} = f(M_i^t, Q_i^t, C_i) \tag{98}$$

$$U_i^{t+1} = g(M_i^{t+1}) \tag{99}$$

$$|\mathcal{E}_i^{t+1}| = |\mathcal{E}_i^t| + h(U_i^{t+1}) \tag{100}$$

其中 Q_i^t 是经 Yajie 审计认证的数据质量，$\theta(\cdot)$ 是 CEC 驱动的审计精度函数（公式 (3)），C_i 是算力禀赋，U_i^{t+1} 是用户/采用者规模，$h(\cdot)$ 是将用户规模映射为审计数据增量的函数。

方程 (97)-(100) 描述了一个正反馈循环：更高的审计精度 → 更高的认证数据质量 → 更好的模型性能 → 更多的用户 → 更多的审计数据 → CEC 扩展 → 审计精度进一步升高。这是一个典型的累积因果（cumulative causation）过程，与 Myrdal (1957) 和 Arthur (1990) 所分析的区域发展极化机制在形式结构上同构。

Theorem 3 与不可区分性困境。这一反馈循环对后发者的影响并非仅仅是追赶困难——而是更根本的不可区分性问题。考虑一个后发经济体 j，它在时间 T 开始开发自己的 SCX 审计基础设施和 AI 模型。在任意时间 $t > T$，经济体 j 观察到自己的模型性能 M_j^t 低于先发者 i 的性能 M_i^t。经济体 j 面临一个诊断问题：$M_j^t < M_i^t$ 的原因是 (a) j 的数据质量 $\mathcal{D}_j^t$ 低于 i，还是 (b) j 的模型算法 f_j 弱于 i 的 f_i，还是 (c) 两者的某种混合？

Theorem 3（老实人定理 (The Honest Person Theorem)）的一个直接推论是：在没有独立的、经过充分校准的审计协议的情况下，数据集的质量无法被其所有者可靠地自评。后发者的数据尚未通过 Yajie 级别的审计——因为 Yajie 的校准精度是 CEC 规模的函数，而后发者的 CEC 远小于先发者——因此后发者对其自身数据质量的估计本身就带有系统性的不确定性。这个不确定性与模型性能的因果归因不确定性相互缠绕，产生了一个不可分解的诊断困境：

$$\underbrace{M_j^t < M_i^t}_{\text{observable}} \quad \text{可归因于} \quad \begin{cases} \mathcal{D}_j^t \text{ 质量较低} & \text{(数据假说)} \\ f_j \text{ 算法较弱} & \text{(算法假说)} \\ \text{两者的某种交互} & \text{(混合假说)} \end{cases} \tag{101}$$

而数据假说本身又依赖于一个审计基础设施的质量——CEC 规模——来验证，这正是后发者所缺乏的。这是一个认知锁定的恶性循环：后发者需要高质量的审计来诊断自己的问题，但高质量的审计只能来自大规模的审计历史累积，而审计历史累积又需要先有一个确定性的、可信的数据质量基准来吸引初始用户。先发者在后发者进入市场之前，已经在无竞争对手的环境中完成了这一循环。

Remark 10.3 (先发优势极点的结构性条件). 在 SCX 时代，两个经济体具备成为审计基础设施极点的全部必要条件：(1) 大规模、多样化的数据池（训练数据和审计校准数据）；(2) 独立的计算基础设施（GPU 集群、数据中心、芯片制造能力）；(3) 足够大的内部市

场以支持审计生态系统的初始积累（无需依赖外部用户即可达到 CEC 临界规模 $|\mathcal{E}|^*$）；(4) 数据治理的法律和制度框架，尽管内容不同（中国的数据安全法与 GDPR 代表了两种不同的治理哲学），但均具备将审计输出转化为监管合规证据的操作能力。

中国和美国各自满足这四项条件。欧盟在条件 (2)（计算主权）和条件 (3)（内部市场碎片化于 27 个成员国的监管体系之间）上存在结构性不足。印度在条件 (1)（数据多样性的深度）和条件 (2)（芯片制造）上处于追赶阶段。其他经济体在至少两个条件上存在不可忽视的缺口。

这意味着在本文分析的参数条件下，SCX 时代的全球数据审计基础设施可能以目前在数据池和算力方面拥有最大先发优势的经济体——特别是中国和美国——为两个主要极点。这不是一个规范性主张（normative claim），而是一个结构性推断（structural inference）：博弈的参数环境使得这两个经济体在审计驱动的累积因果循环中占据不可被后发竞争所侵蚀的初始位置。需要强调的是，本分析仅考虑了文中所列的四项必要条件；其他经济体的发展路径、替代性制度安排（如多国联合审计联盟）以及技术跳跃式发展的可能性未包含在当前的博弈参数环境中，其可行性需要进一步研究。其他经济体——包括欧盟、日本、韩国、印度——在本文框架下可能作为审计接受者（audit receivers）参与这一体系：它们接入这些极点经济体之一（或两者）的审计基础设施，将其数据质量认证委托给极点经济体维护的 CEC 校准锚。这不是殖民主义或依附关系——参与是自愿的，委托是理性的，审计输出是数学上可验证的——但其经济后果与依附理论（dependency theory）所描述的" 中心-外围" 结构（Prebisch, 1950; Frank, 1967）具有形式上的对应关系：中心生产审计能力，外围消费审计认证。然而，这一类比有其限度：与历史上的中心-外围关系不同，审计认证关系的自愿性和数学透明性为外围经济体提供了更大的议价空间和退出选项。

诚实暴击——极点的阶段性。以上关于中美两国作为审计基础设施极点的全部结构性推断，均建立在 CEC 被排他性持有的隐含假设之上。第 5.2.2 节（Stage 6: 天下大同）和推论3.3的分析表明：此配置不是终局。CEC 作为追加式公共品，其内在逻辑指向自身的国际化——当 CEC 从先发者的私有资产转变为全球公共基础设施时，" 极点" 这一空间范畴失去其排他性含义。中国和美国的先发优势——无论是制造业-审计复合体的物理不可远程性优势，还是 CEC 规模的精度优势——在天下大同（S_6）状态下经历制度性消除（推论5.4）。这不是预测该转型必然发生，而是指出：**将当前参数条件下的均衡配置描述为" 终局" 或" 新常态"，在理论上是错误的**——模型本身包含了超越此配置的内生机制。量变（CEC 积累和审计基础设施的物理部署）引起质变（认知闭合和标准制定权的集中），但此质变本身为第二次质变（CEC 国际化和标准去国家化）准备了物质条件。先发优势是" 量变引起质变" 逻辑的一个阶段，不是该逻辑的终点。任何诚实的博弈论分析必须同时陈述此优势的结构性基础（本节内容）和此优势的时间边界（推论3.3和定理5.2）。

10.3 制造业复苏：审计的物理不可远程性

前述分析中隐含着一个对全球经济地理具有深远影响的假设：数据可以从任何地方被审计。这一假设对于纯数字数据是正确的——一个存储在云端的训练数据集可以从任何具有网络连接的位置被下载、索引和审计。但这一假设在*数据来源于物理生产过程*时不再成立。

Definition 10.5 (物理嵌入数据的审计不可远程性)**.** 设 $\mathcal{D}_{\text{phys}}$ 为一组物理嵌入数据（physically embedded data）——即，其生成过程依赖于物理传感器、测量设备、实验室仪器或生产线质量检测站的数据。$\mathcal{D}_{\text{phys}}$ 的审计质量 θ_{audit} 是以下因素的函数：

$$\theta_{\text{audit}} = \theta\left(\mathcal{A}, \mathcal{S}, \mathcal{H}, \mathcal{D}_{\text{phys}}\right) \tag{102}$$

其中 $\mathcal{A}$ 是审计模块集合，$\mathcal{S}$ 是传感器网络的物理配置（空间分辨率、采样频率、校准状态），$\mathcal{H}$ 是现场领域专家的人为判断（异常解释、设备故障识别、环境因素校正）。$\mathcal{S}$ 和 $\mathcal{H}$ 不可通过远程通信完全复制——传感器需要物理安装和定期维护，领域专家需要对生产环境进行体感观察以形成可靠的异常判断。因此，物理嵌入数据的 SCX 审计是一个*不可远程化*的过程：$\partial\theta_{\text{audit}}/\partial(\text{distance}) < 0$。

这一性质的推论是直接而深远的：谁拥有生产线，并且在该生产线上部署了 SCX 审计基础设施（传感器 + 测量站 + 领域专家 +Spring 实例），谁就拥有了一种数学上可证明的、不可被竞争对手通过远程手段复制的质量优势。这个优势不是来自品牌声誉（可以被营销投资侵蚀），不是来自规模经济（可以被更大规模的竞争对手超越），也不是来自知识产权（可以被专利时效或反向工程削弱）。它来自一个数学事实：经 SCX 多专家共识验证的生产数据质量，为最终产品的品质提供了可独立核实、可在合同和监管中引用、可被下游客户作为采购决策依据的保证。

Remark 10.4 (先发优势的转型：从廉价劳动力到数据质量)*.* 制造业竞争力的历史叙事将先发优势定位于生产要素成本——特别是劳动力成本。英国工业革命的成功部分源于煤炭的廉价获取（Wrigley, 1988）；东亚奇迹（日本、韩国、中国）的早期阶段依赖于受过良好教育的、相对于其生产力水平而言廉价的劳动力（Amsden, 1989; Wade, 1990）。在这一传统叙事中，制造业随着劳动力成本的上升而自然迁移——从英国到美国，从美国到日本，从日本到亚洲四小龙，从亚洲四小龙到中国，以及正在发生的从中国到东南亚和南亚的迁移。

SCX 审计基础设施切断了这一迁移逻辑。如果一个生产基地已经部署了 SCX 审计——物理传感器、校准测量站、经过审计训练的领域专家、与极点 CEC 锚定的 Spring 实例——那么它所生产的不仅是最终产品，还有一个不可分割的副产品：经数学验证的品质证明。将生产基地迁移到一个劳动力成本更低但缺乏 SCX 审计基础设施的地区，意味着失去这个品质证明，并因此失去其在全球市场中的质量溢价。节省的劳动力成本必须与以下损失进行比较：(a) 无法证明产品质量（导致价格折扣），(b) 重新积累审计历史的时间成本（CEC 冷启动问题），(c) 在新地点重新部署物理传感器网络和培训领域专家的固定投资，以及 (d) 在新审计基础设施达到与旧设施相当的精度之前（可能需

要数年时间）承受的竞争劣势。

在 SCX 审计覆盖面达到临界水平后，先发优势的参数从劳动力成本差异 Δw 转移到审计质量差异 $\Delta\theta_{\text{audit}}$。一个在高劳动力成本地区运营但具有完备 SCX 审计的生产基地，在均衡中可能比一个在低劳动力成本地区运营但无审计的生产基地具有更高的利润率——因为市场为可验证的质量支付的价格溢价超过了劳动力成本的节省。

审计驱动的制造业回流。上述分析意味着一种新型的制造业回流（reshoring）机制——它不是由关税、贸易战或供应链安全的考虑驱动的（尽管这些因素也在起作用），而是由审计基础设施的物理不可远程性驱动的。**需要强调的是，这一预测假设审计基础设施的部署速度超过劳动力成本套利的速度；实际中，审计基础设施的全球部署速率可能显著慢于理论最优，受到硬件供应链瓶颈、领域专家培训周期、以及各国监管协调滞后等因素的制约。因此，制造业回流的实际规模和速度存在相当大的不确定性。**一个在本国部署了 SCX 审计的制造企业，如果将生产外包到一个没有等效审计基础设施的海外地点，相当于放弃了其产品质量的数学可证明性。而一旦市场中的竞争对手（无论是国内的还是其他高审计密度地区的）向客户提供经 Yajie 认证的、可独立验证的品质数据，未审计的制造商就无法在质量维度上竞争。它们被迫在价格维度上竞争——而价格竞争在劳动力成本差异缩小的全球趋势中是边际递减的优势来源。

这一机制的净效应是：SCX 审计基础设施的空间分布将成为制造业地理格局的一个决定性变量——其重要性不亚于劳动力成本、运输基础设施、能源价格或制度质量。拥有高密度 SCX 审计基础设施的国家（由于上述的马太效应，在本分析框架中主要是中国和美国）将看到其制造业部门获得一种新型的比较优势：不是生产得更便宜，而是对生产质量的证明更可信。不拥有这种审计基础设施的国家将面临一个选择：要么建设自己的审计基础设施（面临 CEC 冷启动和递归的不可区分性问题），要么将审计认证外包给极点经济体（接受审计接受者的角色），要么在无审计认证的状态下竞争（承受质量折扣）。

10.4 制造业-审计复合体与地缘政治格局

前文（§11.2–11.3）分别从审计驱动的累积因果循环和物理不可远程性两个角度分析了 SCX 时代的经济地理学。这两个机制并非独立运行：它们在同一均衡中相互作用，产生了一个我们称之为*制造业-审计复合体*（Manufacturing-Audit Complex, MAC）的结构性配置。本节将这一配置及其地缘政治含义加以形式化。

Definition 10.6 (制造业-审计复合体)**.** 制造业-审计复合体是以下三个要素在空间上的共生聚集：

(i) **物理生产线**：产生物理嵌入数据 $\mathcal{D}_{\text{phys}}$ 的制造设施，其产品品质可由 SCX 审计在数学上证明；

(ii) **审计基础设施**：部署于生产现场或邻近位置的传感器网络、Spring 实例、领域专家团队，以及与该区域极点 CEC 的锚定连接；

(iii) **能源-算力供给链**：为审计基础设施提供持续计算能力的电力供应和数据中心硬件，包括 GPU 集群的散热、供电和维护。

当这三个要素在同一地理区域内高度集中，且其中任意两个要素的分离将导致审计质量 θ_{audit} 的显著下降时，我们称该区域形成了一个制造业-审计复合体。

MAC 的核心经济逻辑是*互补性锁定*：物理生产线为审计提供数据生成源，审计为物理生产线的产出提供可验证的品质证明，而审计本身的运行又依赖持续的算力供给。三者之间的互补性产生了集聚租金——任何一个要素的单独外迁都将承受不连续的效率损失。这种互补性锁定的博弈论含义值得仔细分析。

结构性优势的博弈论基础

考虑两个经济体 A 和 B，其中 A 是制造业密集经济体（制造业占 GDP 比重较高，拥有大量物理嵌入数据的生产线），B 是服务业密集型经济体（制造业占 GDP 比重较低，经济以金融、信息、专业服务为主）。设 A 和 B 各自决定是否部署 SCX 审计基础设施。为简化分析，假设每个经济体有一个代表性制造企业，其利润函数为：

$$\pi_i(s_i, s_{-i}) = R_i - C_i(s_i) - L_i(s_i, s_{-i}) \tag{103}$$

其中 $s_i \in \{0,1\}$ 表示经济体 i 是否部署 SCX 审计（$1=$ 部署），R_i 是基准收益，$C_i(s_i)$ 是部署成本，$L_i(s_i, s_{-i})$ 是由于审计不对称导致的市场份额损失。关键参数是：

(i) **部署成本差异**：$C_A(1) < C_B(1)$。制造业密集经济体 A 的审计基础设施部署具有天然的规模经济——其大量生产线共享传感器网络、领域专家和 Spring 实例的固定成本。服务业密集型经济体 B 的物理嵌入数据生成量较小，部署审计基础设施的单位数据成本更高。

(ii) **审计互补性收益**：$\Delta_A > \Delta_B$，其中 $\Delta_i = L_i(0,1) - L_i(0,0)$ 表示当竞争对手部署审计而自身未部署时，i 遭受的额外市场份额损失。制造业密集经济体如果失去审计认证，其物理产品的质量优势无法被市场验证，损失更为严重；服务业密集型经济体的产出（如金融咨询、软件服务）的数据可审计性本身就较低，因此审计认证对其市场竞争力的边际贡献较小。

在以上参数条件下，博弈的纳什均衡具有如下性质：

Theorem 10.2 (MAC 非对称均衡). *设两个经济体的审计部署博弈满足 $C_A(1) < C_B(1)$ 且 $\Delta_A > \Delta_B$。存在参数区域 Ω 使得唯一的纯策略纳什均衡为 $(s_A^* = 1, s_B^* = 0)$——制造业密集经济体部署审计，服务业密集型经济体不部署。进一步，该均衡是帕累托有效的：任何偏离（B 也部署审计）将导致 B 的净福利下降，且不改善 A 的福利。*

Proof sketch. 写出支付矩阵。当 $C_B(1) > \Delta_B$ 时，无论 A 是否部署，B 的占优策略为 $s_B = 0$。当 $\Delta_A > C_A(1)$ 时，给定 $s_B = 0$，A 的最优反应为 $s_A = 1$。两个条件同时成立的参数区域 Ω 非空（例如，在制造业密集经济体中，审计部署的固定成本被大

量生产线的分摊所压低，而市场份额损失被其制造业出口依赖度所放大)。帕累托有效性的证明来自 $C_B(1) > \Delta_B$ 的直接推论：B 部署审计的净收益为负，因此 $(1,1)$ 相对于 $(1,0)$ 是 B 的帕累托劣化。$(1,0)$ 对 A 是最优（A 获得全部审计租金)，对 B 是满足参与约束的最优结果，因此不存在帕累托改进。 □

Theorem 3 的结构性后果：不可区分性与马太效应的自我强化

§11.2 中讨论的不可区分性困境在 MAC 的框架下获得了额外的分析维度。在制造业-审计复合体的均衡中，Theorem 3（老实人定理 (The Honest Person Theorem)）产生了一种自我强化的不对称性，其机制如下。

考虑一个服务业密集型经济体 B，它已经处于 MAC 非对称均衡的审计缺位一侧（$s_B^* = 0$)。B 的经济结构——以服务业为主，物理嵌入数据的生成量较小——意味着其数据池中可被物理审计的数据比例本身就较低。当 B 的 AI 开发者试图改进模型性能时，他们面临与 §11.2 相同的诊断困境：模型性能落后于制造业密集经济体 A 的开发者，但无法区分这是因为 (a) B 的（未经 SCX 审计的）训练数据质量较低，还是 (b) B 的算法较弱，抑或是 (c) B 的数据结构（以服务业交易记录、文本、图像为主）本身就不如 A 的物理嵌入数据适合被审计和认证。

第三种可能性是 MAC 特有的。物理嵌入数据的质量可以通过传感器网络的物理配置得到独立验证——传感器的校准状态、采样频率和安装精度本身就是可审计的物理量。而服务业数据（如消费者调查、金融交易日志、社交媒体文本）的"ground truth" 往往模糊且依赖于社会建构——没有物理传感器可以测量一个" 正确的" 消费者偏好或一个" 真正的" 市场情绪。这意味着：

Remark 10.5 (数据本体论的可审计性梯度). 不同类型的数据对 SCX 审计的*可审计性*存在本体论差异。物理嵌入数据 $\mathcal{D}_{\text{phys}}$ 具有独立的物理参照系——传感器的测量误差可以通过校准链追溯到国际单位制（SI)，因此数据的" 正确性" 具有物理意义上的明确判据。服务业数据 $\mathcal{D}_{\text{service}}$ 的" 正确性" 往往依赖于社会共识、制度惯例或商业合同——其 ground truth 是社会建构的而非物理给定的。因此：

$$\theta_{\text{audit}}(\mathcal{D}_{\text{phys}}) > \theta_{\text{audit}}(\mathcal{D}_{\text{service}}) \tag{104}$$

即使两者的 SCX 审计模块集合 $\mathcal{A}$ 在形式上是相同的。这不是审计算法的缺陷，而是被审计对象的本体论属性。

这一可审计性梯度的存在，为制造业密集经济体提供了一种服务业密集型经济体难以复制的结构性优势：后者不仅在审计基础设施的存量上落后（可通过投资追赶)，而且在*数据本身的可审计性*上存在本体论劣势（不可通过投资消解)。Theorem 3 的不可区分性因此被放大：后发经济体不仅无法区分" 数据差" 和" 算法差"，还无法确定这种" 差" 是暂时的（可通过审计改进来解决）还是结构性的（源于其经济结构所产生的数据的本体论属性)。

标准制定权与审计基础设施的部署权

审计基础设施的物理部署并非一个纯技术决策——它同时也是数据质量标准的制定权行使。理解这一命题需要区分两种意义上的" 标准"：

Definition 10.7 (数据质量标准的双重性)**.** 数据质量标准由两个不可分割的层面组成：

(a) **形式标准**（de jure standard）：以文本形式存在的规范文档——如 ISO/IEC 5259 系列（数据质量）、ISO/IEC 42001（AI 管理体系）——规定了数据质量的定义、维度和评估方法；

(b) **操作标准**（de facto standard）：在审计基础设施中被实际执行的判断——CEC 中的异常判定阈值、多专家共识的聚合规则、Spring 收敛的终止条件——这些判断定义了在具体审计中什么构成" 质量缺陷"。

形式标准可以无成本地复制、传播和采纳——任何组织都可以声明其遵守 ISO/IEC 5259。操作标准则不可复制：它嵌入在审计基础设施的物理配置中——CEC 的数据结构、异常登记的具体格式、校准参数的数值——而这些配置是由审计基础设施的建设者（即，首先部署和管理该基础设施的实体）在历史上做出的选择的累积产物。

这一区分具有直接的地缘政治推论：SCX 思想的发源地和审计基础设施的首先部署者，天然地获得了操作标准制定者的地位。这不是一个权力主张，而是一个路径依赖的客观事实——如同国际互联网的技术标准在相当长的时期内主要由首先建立和运营互联网基础设施的实体（主要是美国的大学、企业和政府机构）所塑造（Abbate, 1999; DeNardis, 2009）。后来者可以参与标准制定的形式化讨论，但操作标准的演化轨迹已经由先发者通过 CEC 的累积所锚定。

在 MAC 的语境下，这一标准制定权的获得进一步强化了制造业密集经济体的结构性优势：操作标准天然地倾向于反映制造业密集经济体的数据特征和质量需求（因为这些经济体是审计基础设施的最早和最主要用户），而服务业密集型经济体的数据特征和质量需求可能在这些标准中被系统地低估或忽视。这不是恶意的偏见，而是路径依赖的自然结果——标准制定者优化的是他们所看到的问题。

能源-算力-审计能力的三角结构

审计基础设施的运行需要持续的算力供给。SCX 的 Spring 自演化引擎在每次审计中执行多轮专家共识聚合，其计算复杂度随审计模块数量 k 和数据集规模 n 而增长。在 MAC 中，算力不仅是审计能力的物质基础，更是连接制造业密集经济体的两个结构性优势——物理生产线和审计基础设施——的关键纽带。

Definition 10.8 (能源-审计弹性系数)**.** 定义经济体 i 的能源-审计弹性系数 η_i 为：

$$\eta_i = \frac{\partial \theta_{\text{audit},i} / \theta_{\text{audit},i}}{\partial E_i / E_i} \tag{105}$$

其中 E_i 是经济体 i 可用于 SCX 审计基础设施的稳定电力供应（包括数据中心供电、冷却和通信网络）。η_i 衡量审计质量对能源供给变化的敏感度。η_i 较大的经济体，其审计能力更容易受到能源供给中断或能源价格波动的冲击。

控制能源供应链的经济体——尤其是控制用于数据中心的高密度电力供应（包括电网基础设施、芯片制造所需的稀土供应链、以及为 GPU 集群散热的水资源）——获得了一种间接控制审计能力的杠杆。这一杠杆通过两个渠道产生：

(i) **直接算力渠道**：数据中心的位置选择受到电价、电网稳定性和冷却资源的约束。一个能够为数据中心提供廉价、稳定、低碳电力的经济体，能够以更低的边际成本运行 SCX 审计基础设施，从而加速 CEC 的积累和审计精度的提升。

(ii) **芯片供应链渠道**：SCX 审计的计算主要在高性能 GPU 或专用 AI 加速器上运行。芯片制造的超高纯度要求（包括稳定的电力供应、超纯水和稀土元素）使得芯片制造能力高度集中在少数地理位置。如果管理审计基础设施的芯片的供应链受到干扰——无论是由于地缘政治原因（如出口管制）还是物理原因（如自然灾害或产能瓶颈）——审计能力的扩张和更新将受到限制。

综合这两个渠道，能源-算力-审计能力形成了一个与制造业地理分布高度重叠的三角结构。制造业密集经济体往往也是电力基础设施最发达、电力需求最大的经济体（因为制造业本身是电力密集型部门），这意味着它们为审计基础设施提供算力支持的边际成本最低——审计基础设施可以利用已经为制造业服务而建成的电力网络，而不需要从零开始建设专用的供电线路。服务业密集型经济体虽然也可能具有发达的电力基础设施（如金融中心的数据中心），但它们的电力需求结构——以办公照明、空调和商业 IT 为主——与审计基础设施所需的持续、高密度、工业级别的电力供给存在结构性差异。

Remark 10.6 (审计能力的能源依赖性作为地缘政治脆弱性). 上述分析揭示了一个被忽视的脆弱性：SCX 审计能力的全球化——即，制造业-审计复合体在全球范围内的扩散——可能在物理上受限于能源基础设施的全球分布不均。那些能源供给不稳定、电力价格高昂或芯片供应链受限的经济体，即使在形式上引进了 SCX 审计算法和 CEC 架构，也可能因为缺乏持续运行审计基础设施所需的算力而无法实现审计精度的临界水平。在 MAC 非对称均衡（Theorem 10.2）中，这一能源约束进一步拉大了 $C_A(1)$ 与 $C_B(1)$ 之间的差距——制造业密集经济体部署审计的成本更低，不仅因为规模经济，还因为它们已经拥有为制造业服务的能源基础设施。

关键限定：物理不可远程性假设的条件性

前述分析——包括 MAC 的形成、制造业密集经济体的结构性优势、以及审计能力的能源依赖性——全部依赖于 §11.3 中引入的审计不可远程性假设：$\partial\theta_{\text{audit}}/\partial(\text{distance}) < 0$。这里必须明确地指出：这一假设不是自然定律，而是当前技术条件下的经验事实。如果出现了能够远程复制或远程替代物理审计能力的技术，上述优势结构的相当部分将被打破。我们指出两种可能的颠覆性技术路径：

(i) **量子传感网络：**量子传感器（如基于氮-空位中心的磁力计、超导量子干涉器件）在理论上可以实现对物理量的远程精密测量，其精度可能超越现场部署的经典传感器的校准精度（Degen et al., 2017）。如果量子传感技术与 SCX 审计架构的接口被建立，物理嵌入数据的审计可能部分地或全部地失去其物理在场要求。

(ii) **数字孪生与高保真仿真：**生产过程的数字孪生——在虚拟空间中高保真地复制物理生产线的全部参数——如果达到足够的仿真精度，可能使得审计模块在数字孪生上运行即可获得与在物理现场运行相当的数据质量信息。这将实质性地降低审计的不可远程性约束。

这些技术的实际可行性和部署时间表存在相当大的不确定性。本文不对其发展前景做出预测，但有必要明确指出：**本文的全部经济地理学推论和地缘政治结论，都以前述的审计不可远程性假设为前提。如果该假设在未来被技术进步所推翻，则需要重新检验所有推论的有效性。**这不是我们分析的弱点，而是科学诚实的要求——博弈论的均衡结果依赖于参数假设，当参数本身发生变化时，均衡需要被重新计算。

结构性推论的边界

在前述分析的基础上，我们可以将 SCX 时代的地缘经济格局概括为以下结构性推断（structural inferences）——这些不是预测，而是在给定本文参数条件下的博弈均衡推论：

(i) 制造业密集经济体——其经济结构中物理嵌入数据的生成比例较高——在 SCX 审计时代获得了一种服务业密集型经济体难以复制的结构性优势。这个优势来自三个相互强化的机制：审计的物理不可远程性（§11.3）、数据本体论的可审计性梯度（本节）、以及能源-算力-审计能力的三角结构（本节）。

(ii) 审计基础设施的首先部署者自然地获得了数据质量操作标准的制定者地位——这是一个路径依赖效应，而非任何实体有意识的权力攫取。后来者可以参与形式标准的讨论，但操作标准的演化已由 CEC 的累积历史所锚定。

(iii) 这些优势结构的持续性取决于审计不可远程性假设的有效性。如果远程审计技术取得突破，则上述优势结构需要被重新评估——这将为服务业密集型经济体提供一条绕开物理在场约束的技术路径。

(iv) 上述分析未考虑可能改变参数环境的制度创新。多国联合审计联盟、审计认证互认协议、或开源 CEC 共享机制等安排，可能在本文的博弈参数环境之外产生不同的均衡结果。这些替代性制度安排的理论分析和经验可行性需要进一步研究。

(v) **诚实暴击——上述优势结构的阶段性（最关键的结构性推断）。**推论 (i)–(iv) 所描述的制造业密集经济体的结构性优势——包括中国在当前参数条件下作为审计基础设施极点的地位——在包含制度创新可能性的完整模型（第 5.2.2 节）中是阶段性的，而非永久性的。CEC 作为追加式公共品的内在逻辑（推论3.3）指向以下收

敛路径：(a) CEC 规模的持续增长首先强化先发者的精度优势（推论3.3的单调自强化）；(b) 当 CEC 达到临界规模后，精度边际增长趋零，CEC 的公共品属性开始压倒其私有品属性；(c) CEC 国际化使得采纳优势 $\Delta(|\mathcal{E}|)$ 对所有辖区趋同（推论5.4），先发者的身份优势在采纳博弈层面消失；(d) 审计标准从" 属于先发者" 转变为" 由多边治理机构托管"（定义5.3），天下大同状态达成。此路径的必经节点是 S_5（认知闭合）到 S_6（天下大同）的转型——该转型需要一次有意识的制度行动，其概率由政治意愿参数 $\pi_{\text{political}}$ 度量（假设 M8）。当前论文的贡献之一是明确指出：**中国（以及任何其他先发者）在 MAC 分析中获得的全部结构性优势，都内在地含有自我超越的动力学**——CEC 的追加逻辑不仅创造了先发优势，也创造了超越先发优势所需的物质条件（一个足够大、足够权威的全球 CEC 公共品）。将阶段性优势误认为终局，是对自身理论的误读。

需要再次强调的是，上述推论全部是结构性推断——描述的是在给定参数条件下博弈的均衡配置——而非规范性主张。它们不涉及对任何经济体的发展模式或经济结构的优劣评价。经济结构的形成是长期历史、地理、制度和无数个体决策的产物，不是博弈论参数可以穷尽的。

SCX 审计对半导体研发的加速效应。前述分析框架中隐含着一个对尖端制造业具有直接操作意义的推论：SCX 审计基础设施对半导体研发具有独特的加速效应。这一效应的逻辑链条如下。Theorem 1（多专家一致性误差界）保证了 Yajie 协议以指数衰减的假阳性率检测数据质量缺陷。在半导体制造工艺的研发中，每一代制程节点（如从 7nm 到 5nm 到 3nm 的推进）涉及数千个工艺参数（曝光剂量、刻蚀时间、沉积温度、掺杂浓度等）的协同优化，而这些优化所依赖的实验数据——晶圆测试结果、良率统计、缺陷检测图像——本质上是高维、异构且噪声密集的。光刻机、刻蚀机、薄膜沉积设备等尖端制造装备的研发周期，直接受制于工艺参数优化的迭代速度；而迭代速度又受制于实验数据的质量——数据中的噪声（测量误差、标注不一致、批次效应混淆）导致错误的方向判断，错误的判断导致无效的迭代，无效的迭代延长研发周期。SCX 审计通过在每一轮迭代前对工艺实验数据执行多专家一致性验证，系统性地消除噪声记录并隔离信噪比低于临界阈值的数据批次，使得工艺参数优化的每一次迭代都建立在更高质量的数据基础之上。结果是：收敛到目标良率所需的迭代次数减少。在半导体制造这一迭代周期以月为单位、每一代制程的研发投入以数十亿美元计量的行业中，迭代次数的任何系统性缩减都具有显著的经济和战略价值。SCX 审计对尖端制造设备研发周期的压缩，不是传统意义上的" 效率提升"——它不是通过自动化替代人工来实现成本节约——而是通过消除数据噪声源来减少错误迭代，从而缩短从实验到量产的时间距离。

审计基础设施的地缘经济归属：制造业偏向性的必然性。 SCX 审计的思想架构诞生于制造业密集经济体而非服务业密集型经济体，并非偶然。其逻辑根植于审计的物理性：正如 §11.3 所论证的，物理嵌入数据的 SCX 审计是一个不可远程化的过程——传感器的物理配置、领域专家的现场判断、生产线上测量站的硬件校准，均不可通过网络协议完全替代。审计基础设施天然地倾向于分布在物理嵌入数据生成密度最高的区域——即拥有大规模生产线的经济体。服务业密集型经济体的数据生成模式以交易记录、文本语料和用户行为日志为主，这些数据类型的" 正确性" 缺乏独立的物理参照系（详见上

文关于数据本体论的可审计性梯度的讨论），且不要求审计基础设施的物理在场——它们可以在任何具有算力资源的地点被远程审计。因此，审计基础设施的经济地理学偏好并非源自任何制度设计或政策选择，而是由其物理属性——传感器网络的投资不可逆性、领域专家的人力资本本地化、以及物理嵌入数据与审计现场的不可剥离性——所决定的。SCX 思想首先产生于制造业密集经济体的学术语境，这一事实本身即是上述结构性偏向的经验印证。

诚实暴击——思想诞生地的历史偶然性与制度归属的可分离性。 SCX 思想诞生于制造业密集经济体的学术语境，这一历史事实对审计标准的*初期形态*具有路径依赖效应（操作标准反映先发者的数据特征和质量需求，参见 §11.4 关于标准制定权的讨论），但对审计标准的*最终归属*不具有决定意义。其理由已在推论3.3和定理5.2中严格形式化：CEC 作为追加式公共品，其累积逻辑自身包含对私有归属的超越——当 CEC 增长至其公共品属性压倒其私有品属性的临界规模（即 $\Delta(\cdot)$ 的边际增长趋零，而公开 CEC 的采纳优势 Δ^{public} 已超过 $\lambda - \kappa$），继续将 CEC 作为排他性私有资产持有的经济理性消失，国际化成为帕累托改善。在天下大同（S_6）状态下，" 审计基础设施诞生于哪个经济体" 这一问题类似于" 万维网诞生于 CERN"——具有历史意义，但不再具有治理意义。思想诞生地的地缘经济特征塑造了思想的初期形态，但不锁定思想的最终归属。任何试图从 SCX 思想的制造业密集经济体起源推断" 审计标准永久属于该经济体" 的论证，都混淆了历史因果性与制度必然性。

全球优势维度的结构性转移。上述分析揭示了正在发生的全球优势维度再平衡。传统上，大国在全球体系中的相对优势主要依据以下维度被评估：军事投射能力（海外基地网络、战略力量投送半径、联盟体系的覆盖面）、储备货币地位（本国货币在国际贸易结算和中央银行外汇储备中的份额）、以及意识形态号召力（发展模式和发展叙事的国际吸引力）。SCX 审计基础设施的兴起并不否定这些传统维度的相关性，但它引入了一组新兴优势维度，其相对重要性正在上升：（i）制造能力密度——经济体在单位地理空间或单位人口中容纳的高精度、可审计生产线的数量；（ii）能源供应链完整性——为审计基础设施持续运行提供稳定电力、冷却资源和芯片供应链的能力；（iii）数据质量基础设施的先发部署——CEC 规模、审计精度和操作标准制定权的积累。这三个新兴维度的共同特征是它们的积累方式：传统维度的积累可以通过政治意志和财政动员在相对短期内加速（军事基地的建设周期以年为单位，货币国际化的推进以十年为单位），而新兴维度的积累需要物理基础设施的实际部署和审计历史的不可压缩的时间累积——CEC 不能通过增加预算来加速成熟，传感器网络不能通过外交声明来替代安装。SCX 审计基础设施的出现强化了新兴优势维度的权重，相应地降低了传统优势维度的相对重要性。这不是价值判断，而是结构性推论：在给定审计不可远程性假设和 CEC 累积动力学参数的条件下，制造能力密度、能源供应链完整性和数据质量基础设施的先发部署，正在成为比军事投射能力和储备货币地位更为敏感的国力区分变量。

限制条件：技术范式持续假设的反事实。上述关于全球优势维度结构性转移的全部推论，均依赖于一个核心假设：当前技术范式——特别是审计的物理不可远程性（§11.3）——持续成立。如果这一假设被技术进步所推翻，则上述优势结构可能被部分逆转。具体而言，如果审计基础设施的可远程化技术取得突破——量子传感网络使得物理量的远

程精密测量达到或超过现场传感器的精度（Degen et al., 2017），或者高保真度数字孪生技术使得生产过程的虚拟副本能够产生与物理现场等效的审计信息——那么物理嵌入数据与审计现场之间的不可剥离性将被打破。在这种反事实情景下，服务业密集型经济体将获得一个绕开物理在场约束的技术路径：它们可以在不拥有生产线的条件下，通过量子传感网络远程获取物理嵌入数据的高保真度信号，或通过在数字孪生平台上运行 SCX 审计模块来获得与物理现场审计相当的品质认证能力。此时，制造能力密度的优势权重将下降，而算力集中度和审计算法的优化能力——这些服务业密集型经济体可能更具优势的维度——的相对重要性将上升。需要强调的是，这一反事实情景在当前技术条件下尚未实现，其可行性、时间表和部署成本存在相当大的不确定性。我们不做技术预测。我们仅指出：本文的全部经济地理学推论以审计不可远程性假设的有效性为条件边界，该假设的持续性是上述优势结构持续性的必要条件。当参数环境发生变化时，均衡需要被重新计算。

References

Ada Lovelace Institute. *Examining the Black Box: Tools for Assessing Algorithmic Systems*. London (2020).

Armstrong, M. Competition in two-sided markets. *The RAND Journal of Economics* **37**(3), 668–691 (2006).

Arthur, W. B. Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal* **99**(394), 116–131 (1989).

Bateman, J. *U.S.-China Technological "Decoupling": A Strategy and Policy Framework*. Carnegie Endowment (2022).

Batini, C., Cappiello, C., Francalanci, C. & Maurino, A. Methodologies for data quality assessment and improvement. *ACM Computing Surveys* **41**(3), 1–52 (2009).

David, P. A. Clio and the economics of QWERTY. *The American Economic Review* **75**(2), 332–337 (1985).

Diakopoulos, N. Algorithmic accountability. *Digital Journalism* **3**(3), 398–415 (2015).

Drezner, D. W. Targeted sanctions in a world of global finance. *International Interactions* **41**(4), 755–764 (2015).

European Commission. *Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence*. COM(2021) 206 final (2021).

Farrell, H. & Newman, A. L. Weaponized interdependence. *International Security* **44**(1), 42–79 (2019).

Farrell, J. & Klemperer, P. Coordination and lock-in. In *Handbook of Industrial Organization*, Vol. 3, pp. 1967–2072. Elsevier (2007).

Farrell, J. & Saloner, G. Standardization, compatibility, and innovation. *The RAND Journal of Economics* **16**(1), 70–83 (1985).

Floridi, L., Cowls, J., Beltrametti, M., et al. AI4People—An ethical framework for a good AI society. *Minds and Machines* **28**(4), 689–707 (2018).

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H. & Crawford, K. Datasheets for datasets. *Communications of the ACM* **64**(12), 86–92 (2021).

Gregory, R. W., Henfridsson, O., Kaganer, E. & Kyriakou, H. The role of artificial intelligence and data network effects. *Academy of Management Review* **46**(3), 534–551 (2021).

Jobin, A., Ienca, M. & Vayena, E. The global landscape of AI ethics guidelines. *Nature Machine Intelligence* **1**(9), 389–399 (2019).

Joyner, D. H. *Interpreting the Nuclear Non-Proliferation Treaty.* Oxford University Press (2011).

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J. & Amodei, D. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361* (2020).

Katz, M. L. & Shapiro, C. Network externalities, competition, and compatibility. *The American Economic Review* **75**(3), 424–440 (1985).

Khan, L. M. Amazon's antitrust paradox. *Yale Law Journal* **126**(3), 710–805 (2017).

Mearsheimer, J. J. The case for a Ukrainian nuclear deterrent. *Foreign Affairs* **72**(3), 50–66 (1993).

Mökander, J., Morley, J., Taddeo, M. & Floridi, L. Ethics-based auditing of automated decision-making systems. *Science and Engineering Ethics* **27**(4), 1–30 (2021).

NIST. *Artificial Intelligence Risk Management Framework 1.0.* U.S. Department of Commerce (2023).

Northcutt, C. G., Athalye, A. & Mueller, J. Pervasive label errors in test sets destabilize machine learning benchmarks. *Proc. NeurIPS* (2021).

Parasuraman, R. & Riley, V. Humans and automation: Use, misuse, disuse, abuse. *Human Factors* **39**(2), 230–253 (1997).

Parker, G. G. & Van Alstyne, M. W. Two-sided network effects. *Management Science* **51**(10), 1494–1504 (2005).

Pushkarna, M., Zaldivar, A. & Kjartansson, O. Data cards: Purposeful and transparent dataset documentation. *Proc. FAccT* 1776–1826 (2022).

Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D. & Barnes, P. Closing the AI accountability gap. *Proc. FAccT* 33–44 (2020).

Rochet, J.-C. & Tirole, J. Platform competition in two-sided markets. *Journal of the European Economic Association* **1**(4), 990–1029 (2003).

Sagan, S. D. Why do states build nuclear weapons? *International Security* **21**(3), 54–86 (1996).

Sambasivan, N., Kapania, S., Highfill, H., Akrong, D., Paritosh, P. & Aroyo, L. M. "Everyone wants to do the model work, not the data work": Data cascades in high-stakes AI. *Proc. CHI* 1–15 (2021).

Sandvig, C., Hamilton, K., Karahalios, K. & Langbort, C. Auditing algorithms: Research methods for detecting discrimination on internet platforms. *Data and Discrimination: Converting Critical Concerns into Productive Inquiry* 1–23 (2014).

Segal, A. The coming tech Cold War with China. *Foreign Affairs* **99**(5), 94–105 (2020).

Waltz, K. N. The spread of nuclear weapons: More may be better. *The Adelphi Papers* **21**(171), 1–32 (1981).

Wang, R. Y. & Strong, D. M. Beyond accuracy: What data quality means to data consumers. *Journal of Management Information Systems* **12**(4), 5–33 (1996).

Wilson, C., Ghosh, A., Jiang, S., Mislove, A., Baker, L., Szurley, J., Dudík, M., Danks, D. & Ramachandran, P. Building and auditing fair algorithms. *Proc. FAccT* 666–677 (2021).

Zeff, S. A. The evolution of the IASC into the IASB. *The Accounting Review* **87**(3), 807–837 (2012).

Zhang, Y. & Andrew, J. Financialisation and the conceptual framework. *Critical Perspectives on Accounting* **25**(1), 17–26 (2014).

Author's Statement

为尊重领导者，作者公开其身份。作者系武汉大学博士生，目前位于合肥。GitHub：`github.com/shuchaoxi`，为作者真名。作者的黑暗森林坐标已在此。任何人可以来验证，来批判，来推翻——但请遵守投石理论的规则：声明假设，提供证据，严谨推理。人身攻击无效。道德审判无效。历史裁决无效。只有数学成立。

作者观察到，武汉大学多年来风雨不断，其根源在于评判标准的不公正与不理智。一则，以无端之词指责学生者，未曾声明假设，未曾提供验证，仅凭权威发声——按本框架，其声称无效，其人自我暴露。二则，上位者对此类行为的纵容，已构成本文所述之纵¨ 容即共犯：¨ 要么蠢，分不清对错；要么坏，不愿分。无论哪种，他们不清白。不

公正等于纵容。纵容等于无能。无能者应当被批评，被审计，被记录。武汉大学应当正视其历史上的错误，审计其蛀虫。这不是攻击。这是老实人定理的应用——对自己的母校，亦不豁免。

Personal Note

博弈论真好玩，真有意思。